

Lean PLM for Manufacturing Innovation

Lean Product Lifecycle Management for Manufacturing Innovation



Ver1.0 Public Review Edition R4

Industrial Value Chain Initiative (General Incorporated Association)

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Changes in PR4 (technical content)

1. As a data model, the Operation Process has been removed. It has been simplified to Production Process and Equipment Process.
2. As information models, the following five have been removed: Equipment Plan Table, Personnel Plan Table, Procurement Cost Table, Sales Cost Table, and Inspection Procedure Document. The total is now 50.
3. BOA has been moved from the Structure Information group to the Linkage Information group.
4. The definition of the data model has been separated out as its own specification. This specification shows only diagrams illustrating part of its structure and relational configuration.

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Editor

Yasuyuki Nishioka (Industrial Value Chain Initiative / Hosei University)

Contributors

Akihito Ito Akihiro Suyama Atsushi Inagaki Eiji Matsuda Haruhiko Shimizu Hidekazu Takeuchi Hiroshi Tozuka Hiroyuki Nishiyama Hisashi Tsunashima Katsuhiro Iyoda Kazuma Kitayama Keisuke Shimizu Keita Koshii Kohei Sugaya Koki Miura Masanari Ukai Masaru Wakabayashi Mitsutoshi Oshita Morikatsu Kinoshita Motohiko Yamashita Nobuyuki Ogura Ryoichi Higuchi Ryuta Mokuda Seiichiro Wada Shinichiro Chino Shinjiro Yamazaki Shintaro Nemoto Shoji Nemoto Takamichi Kojima Takamichi Mori Tatsuya Umemoto Toshiki Kitamori Yoshiaki Nishimura Yoshinobu Kumagai Yoshitsugu Tanaka Yuya Otake

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1. Purpose of this Specification

In this technical specification, Lean PLM is defined as the effort to synchronize the cycle of structural change and evolution of the manufacturing-side mechanisms — production equipment, plants, and the people involved with them — with the lifecycle of the product, thereby realizing a more agile and sustainable DX for the manufacturing industry.

In a business environment characterized by rapid change and high uncertainty, it is effective for further strengthening the competitiveness of manufacturing to advance product development quickly and flexibly with the design site and the production site working as one. In Lean PLM, product design, process design, and the design of factory equipment and production lines, together with the production management and plant operations needed to embody these as products or services, can be carried out in an integrated manner by making the notations of the related function model, information model, and data model mutually convertible.

This technical specification defines standard models for describing the various operations related to Lean PLM. Using the standard models defined here, it presents procedures and description formats that allow each stakeholder to describe their target operations according to their own differing circumstances.

Content described by each stakeholder in accordance with this technical specification is used so that one stakeholder can collaborate with another. That is, one stakeholder translates its target operations into the standard model and presents it to its collaboration partner, while the other stakeholder translates the content received as the standard model into its own business model. In many business software applications, this data conversion process is automated as an inter-system interface; engineers can reduce development effort and shorten the schedule by making use of information described in accordance with this technical specification.

The purpose of this technical specification is to shorten such engineering effort and enable inter-system collaboration at lower cost, while also, by reconsidering matters back at the level of the information model and function model, discovering more effective information flows than before and increasing the number of cases that lead to new implementations. Through such accumulated efforts, this specification contributes to the realization of Lean PLM.

Intended Readers

This technical specification assumes the following readers:

- Those within manufacturing companies who are involved in the planning, development, and operation of information systems
- Those within manufacturing companies or related companies who are involved in building information systems
- Those who provide consulting on business reform or information systems in manufacturing
- Managers or engineers engaged in the planning, development, or operation of business software for manufacturing
- Staff of public institutions involved in policy-making related to improving the competitiveness of manufacturing

- Experts and researchers knowledgeable about standard specifications or standardization processes in the manufacturing field

2. Scope

This technical specification presents a model for capturing, in an integrated manner rather than as a one-way flow from design to manufacturing, the mutual dependencies among design activities within the manufacturing value chain — in particular product design, process design, and the design of equipment and production lines. It focuses on the flow of process design and equipment design corresponding to product design, as well as on the flow of process design that can accommodate a given equipment design, and further of product design that can be accommodated by a given process design, thereby enabling such bidirectional design work to be linked together while incorporating past track record and actual circumstances.

The manufacturing industries targeted are primarily discrete-type industries with machining and assembly processes. With respect to the market, the focus is mainly on repetitive, high-volume mass production; however, engineer-to-order production and made-to-order design-and-build are also included within the scope of this specification, insofar as core parts, modules, or production processes are reused repeatedly. In terms of position within the supply chain, the scope covers not only downstream finished-goods manufacturers but also upstream parts manufacturers, and covers companies ranging from large enterprises to mid-sized and small/medium enterprises. Fabless companies without their own plants are also included.

Design work in this technical specification defines the structure of the product being produced as the BOM, the structure of the plant that produces it as the BOA, and the structure of the processes that relate the two as the BOP, and it attempts to capture these within as common a framework as possible. Note that the details of data models handled by CAD/CAM/CAE and similar software, and the details of individual business logic, are outside the scope.

At the same time, this technical specification addresses the mechanism by which, when the product structure information (BOM) is related to the plant structure information (BOA) via the process structure information (BOP), the operations on the model (cyber space) are related to their corresponding realities in the physical world (physical space), managing these together as a cyber-physical system. For this reason, the scope also includes the general data models handled by Manufacturing Execution Systems (MES), which deal with instances in an actual plant, and by Computerized Maintenance Management Systems (CMMS), among others.

3. Terminology (normative)

Abbreviations

BOA Bill of Assets

BOM Bill of Materials

BOP Bill of Process

CAD Computer Aided Design

CAE Computer Aided Engineering

CAM Computer Aided Manufacturing

CRM Customer Relationship Management

ERP Enterprise Resource Planning

FMEA Failure Mode and Effects Analysis

FTA Fault Tree Analysis

LCA Life Cycle Assessment

MBSE Model-Based Systems Engineering

CMMS Computerized Maintenance Management System

MES Manufacturing Execution System

MOM Manufacturing Operations Management

PLM Product Lifecycle Management

QFD Quality Function Deployment

Terminology

AS-IS Model

A model that represents and expresses the current situation as-is. In this technical specification there are an AS-IS function model, AS-IS information model, and AS-IS data model.

BOM (Bill of Materials)

A hierarchical structure table representing the parts and materials that make up a product; it is foundational information for design, manufacturing, procurement, and cost management.

BOA (Bill of Assets)

Data representing the structure, capability, and hierarchy of the assets — equipment, manufacturing resources, jigs and tools — that make up a plant or production line; it is production-environment information that executes the process structure (BOP).

BOP (Bill of Process)

A structure table that hierarchically defines the sequence, work content, conditions, and man-hours in a product's manufacturing processes; it is information that deploys the product structure (BOM) into an actual production process.

CMMS (Computerized Maintenance Management System)

An information system that manages equipment/machinery maintenance operations. It records and manages inspection plans, maintenance work orders, failure history, and parts inventory, and helps understand

equipment condition.

CRM (Customer Relationship Management)

A mechanism that centrally manages customer information, touch-point history, and purchase/inquiry history, linking sales, marketing, and service to improve customer satisfaction and value.

E-BOM (Engineering BOM)

A design-perspective BOM created by the design department, reflecting functional structure and design intent, and managed in conjunction with drawings and CAD.

ERP (Enterprise Resource Planning)

A system that integrally manages a company's core operations such as finance, HR, production planning, supplier management, and sales, supporting the optimization of management resources and the efficiency of business operations.

FMEA (Failure Mode and Effects Analysis)

A quality method that systematically analyzes failure modes and their effects/causes using severity, occurrence frequency, and detection difficulty, so as to prevent risks latent in design or processes.

FTA (Fault Tree Analysis)

A method that logically decomposes the causes of system defects or accidents and visualizes the relationships among causes in a tree structure; used for identifying potential risks and formulating preventive measures.

MBSE (Model-Based Systems Engineering)

A systems engineering method that models functions, structure, and other aspects in the design and development of complex products or systems, enabling consistent examination throughout.

M-BOM (Manufacturing BOM)

A BOM that reflects the manufacturing/assembly sequence and the structure at the process level; it is the basis for work instructions, process management, and cost calculation.

MOM (Manufacturing Operations Management)

A manufacturing management framework that integrally manages the planning, execution, progress, quality, equipment, and results of the shop floor, supporting QCD optimization and shop-floor improvement.

MFCA (Material Flow Cost Accounting)

A management method that visualizes the flow of materials and energy in the manufacturing process and grasps cost by separating it into product and loss, thereby promoting improved resource efficiency and cost reduction.

MES (Manufacturing Execution System)

A system that manages and monitors production activities on the shop floor in real time, integrally capturing production instructions, resource allocation, process progress, and quality data.

PLM (Product Lifecycle Management)

A mechanism that centrally manages information across a product's entire lifecycle, from planning through design, manufacturing, maintenance, and disposal, realizing cross-departmental collaboration and improved quality/development efficiency.

Product LCA (Product Life Cycle Assessment)

A method that evaluates, per product unit, the environmental burden of a specific product from raw-material procurement through manufacturing, use, and disposal; used for design comparison, environmentally conscious design, carbon footprint calculation, and the like.

QFD (Quality Function Deployment)

A method for deploying customer requirements into design specifications and technical characteristics, using a quality table to support building in customer value.

S-BOM (Service BOM)

A BOM organized from a maintenance/service perspective, organizing replacement parts and consumables to support after-sales service operations.

Stakeholder

A business unit or department of a manufacturing company that actually carries out the target operations in real business, or software developed as a solution for such operations.

TO-BE Model

A model showing the goal of a development project as the desired future state. In this technical specification there are a TO-BE function model, TO-BE information model, and TO-BE data model.

4. References

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5. How to Use this Technical Specification

5.1. Structure of the Standard Model

One goal of this technical specification is that the target operations of Lean PLM held by different stakeholders — or parts thereof — will have their respective functions collaborate with one another, in the form of exchanging the data they handle with other stakeholders.

Here, a stakeholder is an entity that shares interests or values, and is the unit — a business division, department, or plant of a manufacturing company — that practices, as the business party, the operations corresponding to Lean PLM on a daily basis. For a company that provides an information system as a solution, the unit of the solution is defined as the stakeholder. Also, even for the same company or solution, if its structure or the assumed environment changes significantly, each generation can be treated as a separate stakeholder.

The definition of a stakeholder uses the standard model. In this technical specification, the standard model is what results from extracting, organizing, and summarizing the common elements — each of which otherwise differs entirely — among the functions of the actual business operations targeted by Lean PLM, the information used therein, and the data that is processed there. In this technical specification, a function model, information model, and data model related to Lean PLM are defined.

Function Model

The function model shows the content of what a business operation does, in a form that maps it to its task, purpose, and outcome. The granularity when defining a function is the range that can be handled by a person in charge or by a department; collaboration between operations is via a state change corresponding to the deliverable that is its output, or to its involvement with the external environment.

Information Model

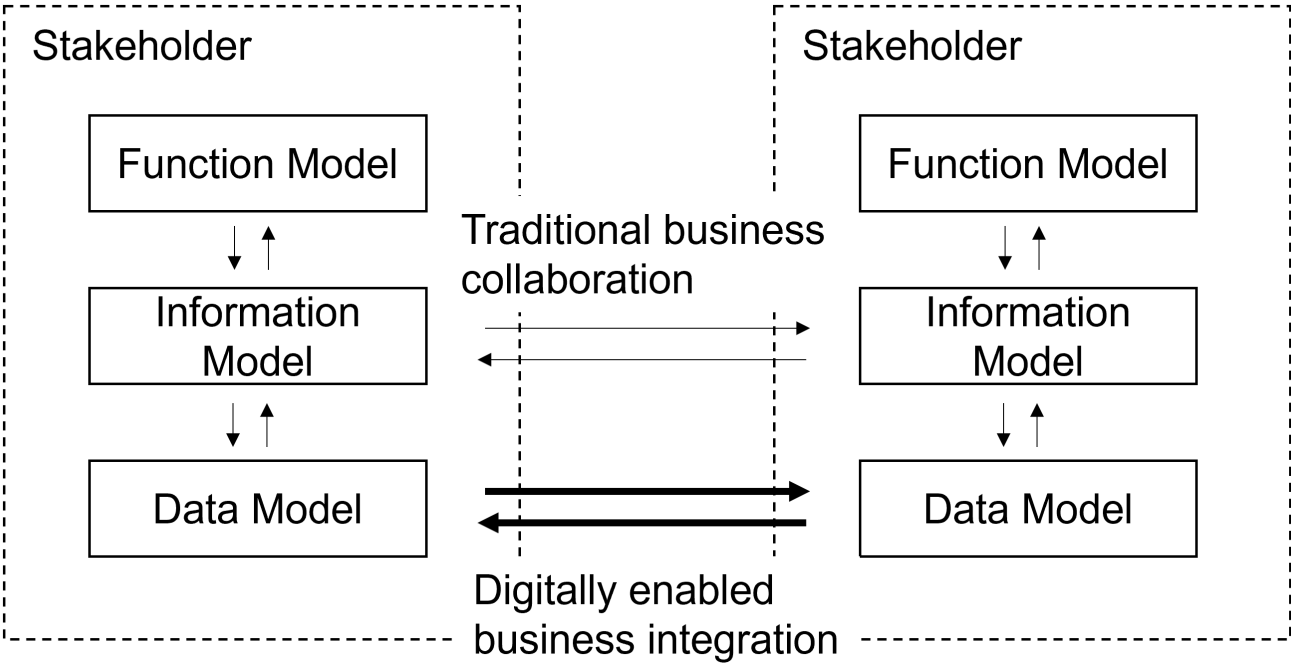
The information model is the unit of input or output needed for the units of function defined in the function model to collaborate with one another. In a situation where information is needed, the unit that is presented or exchanged — a screen or a form — is that unit. The person or department in charge of a function defined in the function model interprets the information shown corresponding to the information model, and updates its content or newly sets other information.

Data Model

The data model shows the format and rules for expressing, using digital technology, the information corresponding to the information model. A data model is defined by entities, the units representing data, and the relations among them; furthermore, one or more attributes are defined for each entity. The content of information is ultimately shown entirely by the values of the attributes held by the entities in the data model.

Figure 1 shows the relationship among these three kinds of models and the stakeholders. When both stakeholders are, say, a manufacturer and a supplier with a trading relationship in manufacturing, whereas collaboration traditionally took place between information models recognizable by the people in charge — such as slips and forms — DX makes collaboration possible at the level of the data model, which not only raises business efficiency but also dramatically improves flexibility and reliability.

Figure 1 Relationship among Stakeholders



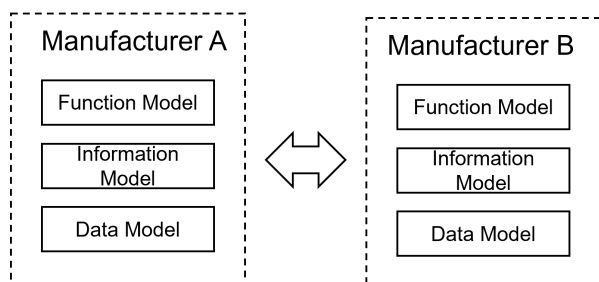
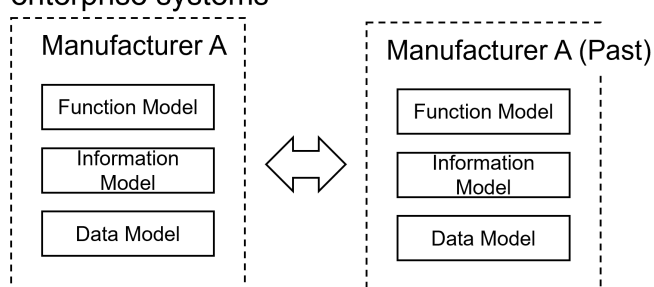
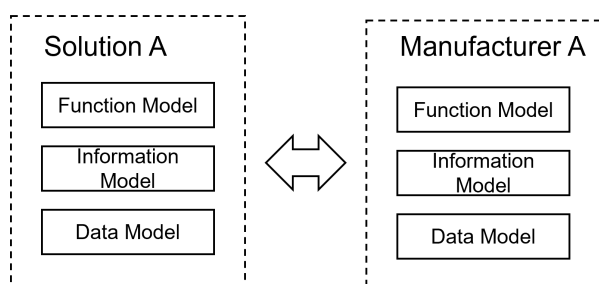
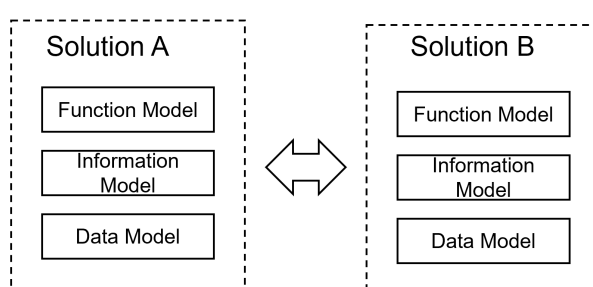
5.2. Use Cases for the Standard Specification

As shown in Figure 2, four use cases can be envisioned for using this standard: data collaboration between companies, consolidation of enterprise systems, implementation of external solutions, and data collaboration between systems. Each use case can be organized as in the table below, according to the kinds of stakeholders that are collaborating.

Table 1 Structure of the Use Cases

No.	Use Case Name	Stakeholder 1	Stakeholder 2
1	Data collaboration between companies	Manufacturing company	Manufacturing company (another company)
2	Consolidation of enterprise systems	Manufacturing company	Manufacturing company (own company)
3	Implementation of an external solution	Solution	Manufacturing company
4	Data collaboration between systems	Solution	Solution

Figure 2 Use Cases of the Technical Specification

Use Case 1: Inter-company data exchange**Use Case 2: Integration and consolidation of enterprise systems****Use Case 3: Implementation of external solutions****Use Case 4: Data integration between systems****Data Collaboration between Companies**

Use case 1 is data collaboration between companies. This corresponds to the case in which the collaborating stakeholders are different companies conducting business by linking their respective operations. It is handled at a level corresponding to information models such as slips, and there may be no collaboration corresponding to the data model.

Consolidation of Enterprise Systems

Use case 2 corresponds to a case in which a company renews its internal information systems, or reconfigures its information systems due to the consolidation of departments. The typical in-house DX project — in which a TO-BE model, viewed from the perspective of the whole company, is set based on the function model, and the existing AS-IS information model and AS-IS data model are rebuilt accordingly — corresponds to this case.

Implementation of an External Solution

Use case 3 corresponds to the case in which, when a company builds an information system, it adopts a package product owned by a solution company. A solution package owned by a solution company has, as software, a data model premised on a function model and information model assumed in advance. In this case, the correspondence between the available solution and the company to which it will be provided is examined, leading to the implementation of the information system.

Data Collaboration between Systems

Use case 4 is a case in which system integration is carried out between solutions owned by a solution company, or between all or part of the information systems that a user company such as a manufacturer operates as legacy systems. This corresponds to cases such as when a developing company integrates

different solutions to provide to a user company, or performs data collaboration with an already-implemented information system. These are activities carried out daily at many system integration sites.

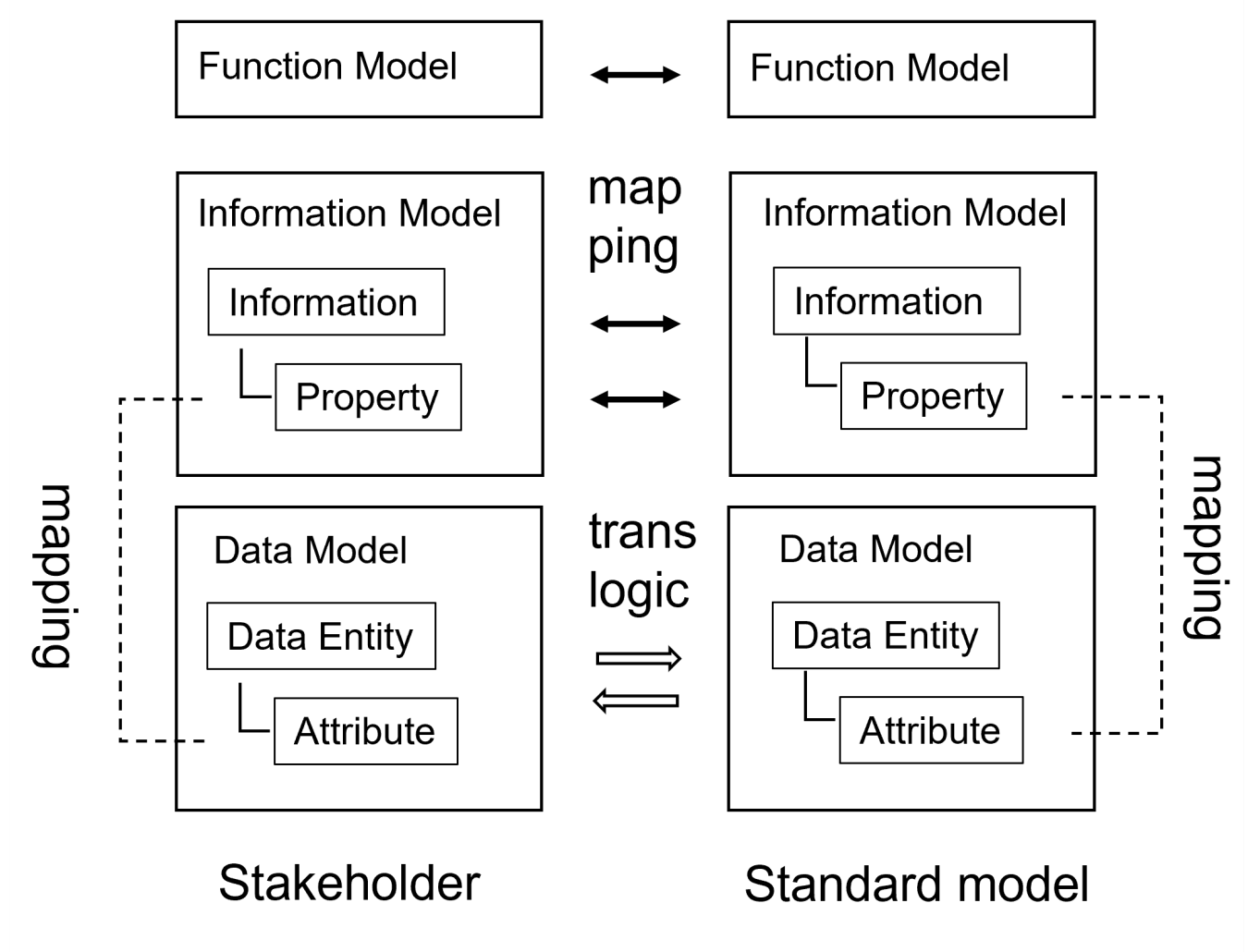
5.3. Mechanisms for Interoperability

To apply this standard to each use case and achieve interoperability of models between two different stakeholders, the approach taken is to first clarify the correspondence between each stakeholder and the standard model, and then integrate them.

For example, to link stakeholder A and stakeholder B, one method is to place the standard model in between: link stakeholder A with the standard model, and then link the standard model with stakeholder B. As needed, this process can be integrated and automated as a system to improve efficiency.

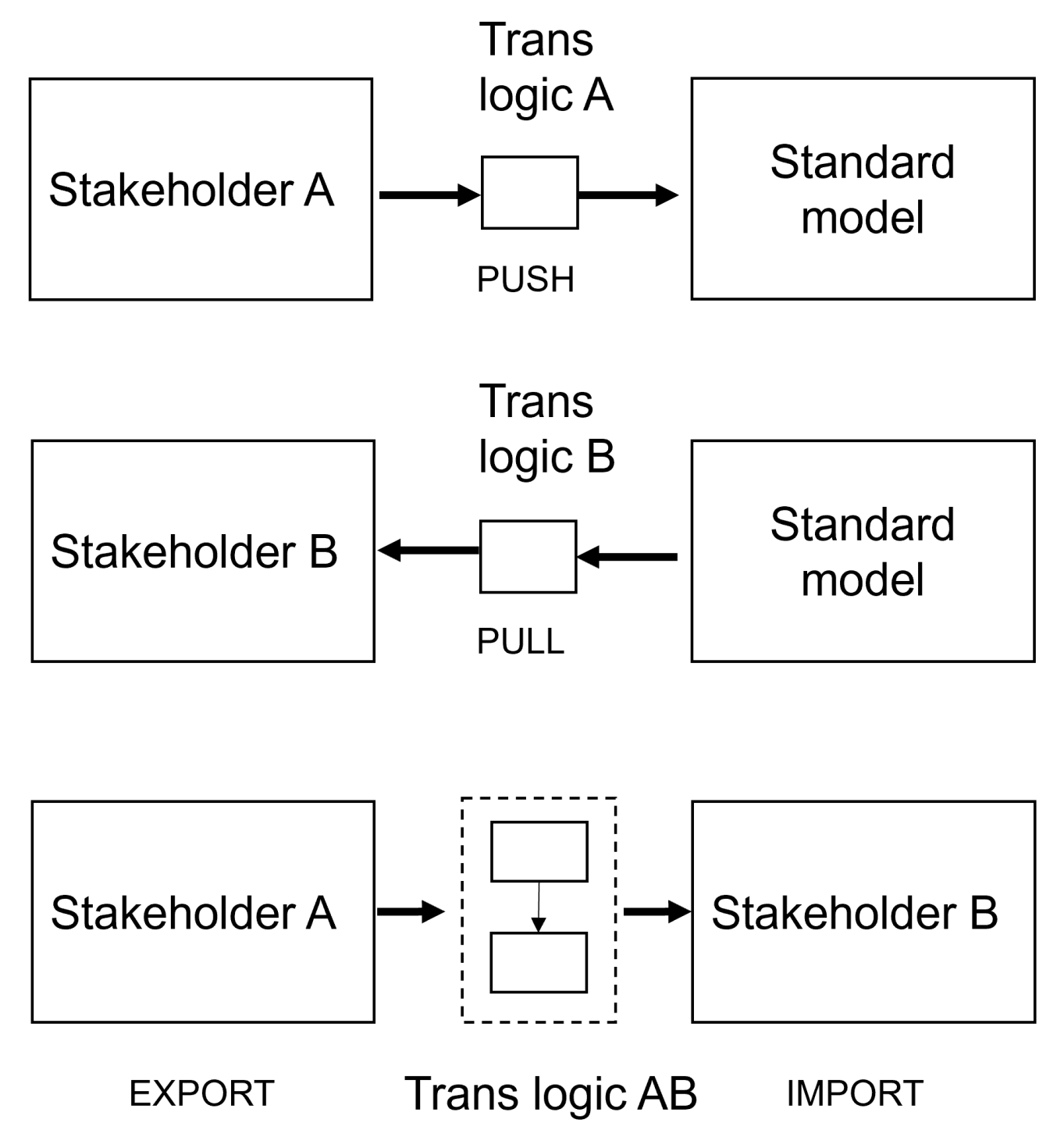
As shown in Figure 3, correspondences can be defined between a stakeholder and the standard model for each of the function model, information model, and data model. For the data model, conversion procedures can be defined from the stakeholder to the standard model, or from the standard model to the stakeholder.

Figure 3 Model Correspondence and Conversion Procedure



When defining a conversion procedure between a stakeholder and the standard model, for the data model there is a directionality of PUSH or PULL. To achieve interoperability of the data model between different stakeholders, the PUSH conversion procedure at one stakeholder and the PULL conversion procedure at the other stakeholder are integrated.

Figure 4 Directionality and Integration of Conversion Procedures



For example, in Figure 4, stakeholder A's model is mapped to the standard model, and further, for the data model, the procedure for converting stakeholder A's model into the standard model is clarified. Likewise, by clarifying the procedure for converting stakeholder B's data model into the standard model, it becomes possible to convert the content of the data model from stakeholder A to stakeholder B.

Here, the information about conversion procedure A, which maps stakeholder A's AS-IS model to the standard model, is called the integration profile. An integration profile is defined for each stakeholder's target operation or target system. Accordingly, in the example of Figure 4, there are integration profile A and integration profile B. Chapter 9 shows the specification for how a stakeholder actually uses the standard model to define a TO-BE model for collaboration and uses its content as a profile.

To socially implement this technical specification, a platform is needed by which a third-party organization certifies the content of these integration profiles, archives and manages them, and enables the parties in a collaboration to exchange profiles that are secure and trust-certified.

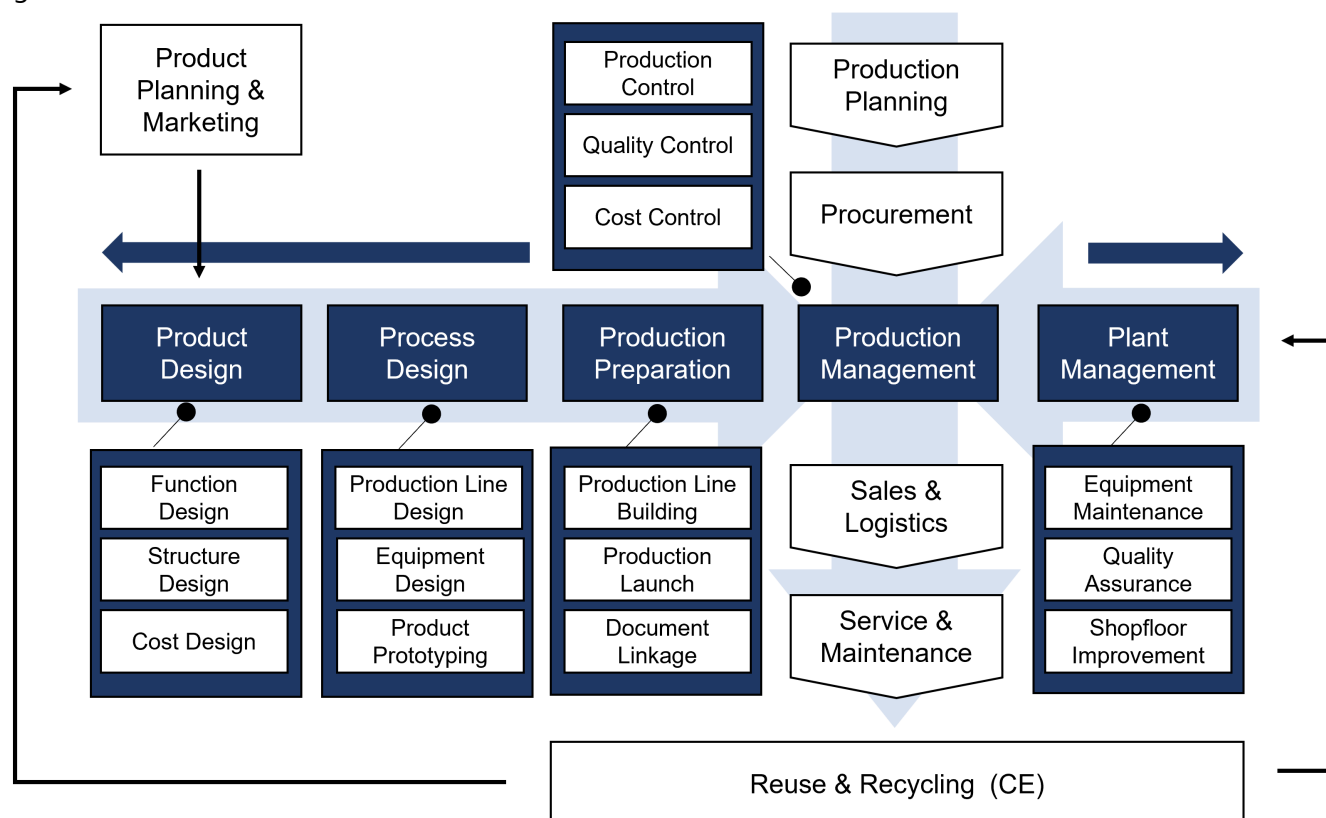
6. Function Model (normative)

The operations in manufacturing targeted by Lean PLM are modeled from the perspective of function. In defining a function, there are three levels of granularity: the Function Element, which is the granularity of activity at which a person in charge actually performs some operation on a thing or on information; the Function Group, which is the granularity used when an organization collectively manages such function elements as a unit of business; and the Function Category, which is the granularity used to take a bird's-eye view of function groups within the company-wide engineering chain.

Here, the function category can be regarded as a major classification, and there are function categories such as Product Design, Process Design, Production Preparation, Production Management, and Equipment Maintenance. On the other hand, the function group is a medium classification and is included in one of the function categories. For example, in Figure 5, three function groups — Functional Design, Structural Design, and Cost Design — are defined for the function category of Product Design. Note that one or more function elements are defined for each function group. These are summarized in a table for each function group.

The function element, which is the minor classification, shows the business content of a function group, and is defined by operations on things and information. In principle, "things" use what is defined in the glossary, and "information" refers to what is defined in the information model. A function element is also mapped to the person in charge who is the actor carrying out that function.

Figure 5 Functions of Lean PLM



Below, for each function category, representative examples of function groups and function elements are shown.

6.1. Product Design

The function category of Product Design has three function groups: Functional Design, Structural Design, and Cost Design. Representative function elements for each function group are shown below.

Functional Design

In functional design, the functions a product should have are defined qualitatively and quantitatively. The designer translates the requirements of the product's intended users into the functions the product should have in order to maximally satisfy them.

Table 2 Function Elements of Functional Design

No.	Function Element	Description
1	Collecting product requirements	Collects requests from customers, the market, regulations, the environment, etc., and clarifies the value, functions, and performance required of the product.
2	Organizing requirement specifications	Classifies and structures the collected requirements, and translates them into design requirements from the perspectives of performance, cost, safety, environment, etc.
3	Formulating the product plan	Formulates the development purpose, product concept, and development scope based on customer value, marketability, and profitability.
4	Architecture design	Examines and connects to design the product architecture, including electrical, electronic control, and software aspects.
5	Evaluating mockups	Confirms shape, operability, and design using prototype models, and improves them by reflecting evaluations from customers and development members.
6	Deploying quality function	Converts customer requirements into design quality elements using the QFD method, clarifying key quality characteristics.
7	Deploying the functional BOM	Organizes the product's functions in a hierarchical structure and expresses/manages the correspondence between functions and their constituent elements in BOM form.
8	Analyzing product functions	Analyzes the purpose, input/output, and interrelationships of each function, and identifies design optimization points and the scope of impact.
9	Setting functional targets	Sets target values such as performance, cost, and reliability for each function, serving as the evaluation criteria for design in subsequent processes.
10	Formulating product specifications	Finalizes the product specification document, including concrete dimensions, performance, materials, etc., based on the results of

No.	Function Element	Description
		requirement/function analysis.

Structural Design

In structural design, the physical structure of the product, such as its composition and the shape of its parts, is defined. This includes the work of creating, using CAD and similar tools, models corresponding to part drawings and assembly drawings, and the work of examining and determining, together with parameters, the specifications corresponding to their shapes and shape features. Note that structural design covers not only hardware but also electrical/electronic control and software.

Table 3 Function Elements of Structural Design

No.	Function Element	Description
1	Designing the product structure	Designs the structure, shape, and module composition needed to realize the product's functions, considering assemblability and maintainability.
2	Determining part composition	Defines the parts/units that make up the product, and determines the optimal composition, including the use of standard or common parts.
3	Creating CAD models	Creates 3D shapes of the product/parts in CAD, used for design verification, interference checking, and digital prototyping.
4	Creating the design BOM	Organizes part composition, specifications, and quantities at the design stage, and registers them in the PLM as an E-BOM to manage design information.
5	Detailed product design	Concretizes material, dimensions, tolerances, strength, etc., and creates detailed design drawings suitable for machining, assembly, and inspection.
6	Product CAE analysis	Performs structural, fluid, thermal, vibration, and other analyses, quantitatively evaluating performance, strength, and reliability to optimize the design.
7	Analyzing failure modes	Analyzes failure modes using methods such as design FMEA and FTA, identifying causes and effects and formulating measures to improve reliability.
8	Responding to design changes	Manages design changes that arise during prototyping, evaluation, and manufacturing stages, analyzing the scope of impact and ensuring reflection and approval.
9	Collaborating with partner companies	Shares data with parts suppliers and outsourced design partners, optimizing design accuracy, delivery time, and cost.
10	Responding to destination specifications and regulations	Confirms the design/manufacturing conditions corresponding to safety, environmental, and certification regulations in each country or region, and reflects them in the product specifications.

Cost Design

Corresponding to the product's price and expected sales volume obtained in product planning and marketing, the costs needed to actually manufacture the product are accumulated to confirm whether profit can be secured. Generally, the main task is not simply to tally the cost of each cost-component item, but to go back to the constituent elements and examine how the targeted profit can be secured.

Table 4 Function Elements of Cost Design

No.	Function Element	Description
1	Setting the target cost	Sets the target cost per product by working backward from the assumed sales price and profit plan, so as to secure the target profit.
2	Examining the development period	Examines the optimal balance between development cost and schedule, based on the cost target and investment payback period.
3	Managing product ROI	Evaluates profitability against investment amount, and quantitatively manages the validity of development, capital investment, and manufacturing cost.
4	Setting the standard cost	Calculates a standard manufacturing cost based on the production process, material cost, and man-hours, serving as the basis for cost variance management.
5	Pricing procured parts	Determines the appropriate price for each procured part through quotation comparison with suppliers and market price analysis.
6	Managing environmental burden	Analyzes MFCA, product LCA, etc., and manages environmental burden per plant and per product.
7	Make-or-buy decisions for processes	Compares cost and technical capability for each process, and determines the optimal split between in-house manufacturing and outsourcing.
8	Cost analysis of regulatory measures	Analyzes the cost of responding to environmental regulations and safety standards, achieving both legal compliance and economic efficiency.
9	Managing cost reduction activities	Plans and evaluates cost reduction measures such as part commonization, process improvement, and procurement cost, and promotes continuous cost reduction.
10	Cost simulation	Models variable factors such as material cost, man-hours, and production volume, predicting and verifying cost impact at the design stage.

6.2. Process Design

The function category of Process Design has three function groups: Line Design, Equipment Design, and Product Prototyping. Representative function elements for each function group are shown below.

Line Design

This examines and determines the method for realizing, through a process using an actual production line, the production processes technically and theoretically required for the machining, assembly, and other necessary

processing of the parts that make up the target product. This also covers cases where an existing combination of equipment is used, or where the work is outsourced to an external partner plant.

Table 5 Function Elements of Line Design

No.	Function Element	Description
1	Organizing line requirement information	Organizes the design-premise requirement conditions such as production volume, takt time, quality, safety, and number of workers, and clarifies the direction of the line design.
2	Comparing production methods	Analyzes the flow of each process, such as machining, assembly, and inspection, and selects a production method from the perspectives of cost, quality, and productivity.
3	Determining process sequence and procedure	Defines the process sequence and work procedure based on the product structure and work content, building an efficient and standardized production flow.
4	Comparing in-house vs. outsourced production	Compares whether each process should be made in-house or outsourced, based on cost, delivery time, quality, and technical capability, and determines the optimal production method.
5	Designing the process layout	Plans the placement of workers, equipment, and parts supply, and designs the optimal layout considering safety, productivity, and maintainability.
6	Determining the in-process transport method	Compares transport means such as conveyors, AGVs, and robots, and selects a transport method that balances logistics efficiency and safety.
7	Production line simulation	Reproduces process operation with 3D models or digital twins, verifying and optimizing takt time, utilization rate, and bottlenecks in advance.
8	Analysis via process FMEA	Analyzes latent defects and risks for each process, evaluates cause and impact, and formulates and manages preventive improvement measures.
9	Designing control programs	Designs programs that control the operation and interlocks of equipment and transport devices, realizing automation of the entire line.
10	Creating the process structure (BOP)	Lists the sequence, work time, equipment used, etc. for each process, preparing basic material usable for production preparation, planning, and maintenance.

Equipment Design

Equipment is designed or modified to obtain a result that satisfies given requirements, such as part shape or location, or material characteristics or precision. In addition to equipment, this also includes the work of designing jigs and tools and other mechanisms for working more efficiently and stabilizing quality further.

Table 6 Function Elements of Equipment Design

No.	Function Element	Description
1	Collecting functional requirements for equipment	Clarifies the performance, capacity, and safety requirements demanded of the equipment, based on production targets and product specifications.
2	Basic design of equipment functions	Defines, at the conceptual level, the equipment configuration, control method, and functional blocks needed to satisfy the specifications.
3	Detailed design of equipment configuration	Designs the detailed structure of each element, such as mechanism, electrical, and control, and creates part drawings and connection diagrams.
4	Automating and making equipment autonomous	Replaces manual work with automatic control, realizing autonomous judgment and optimal operation using AI and sensing.
5	Evaluating and addressing equipment reliability	Designs specifications with reliability in mind, based on design FMEA and FTA of the equipment and analysis of past trouble with similar equipment.
6	Designing and managing jigs and tools	Designs jigs and tools used for machining, assembly, inspection, etc., and manages their lifespan and precision.
7	Collaborating with equipment suppliers	Shares specifications and cross-checks drawings with manufacturers and outsourcing partners, uniformly managing delivery requirements and quality.
8	Equipment operation simulation	Verifies equipment operation, interference, and production capacity in virtual space, making improvements at the design stage.
9	Managing equipment documentation	Systematically manages drawings, specifications, and past trouble information, facilitating tracking and sharing of revision history.
10	Creating the equipment structure (BOA)	Organizes the equipment configuration, jigs/tools, and workers and other assets held by the production site, systematizing them as equipment structure information.

Product Prototyping

Actual equipment and devices are used to machine and assemble a product against a theoretical production process or production line, confirming that the required quality is obtained with the assumed production method. Development prototyping and mass-production prototyping differ in their evaluation content and purpose, but here, as prototyping at the stage before mass-production design, the focus is on exploring a production method with better production efficiency and more stable quality.

Table 7 Function Elements of Product Prototyping

No.	Function Element	Description
1	Examining the prototyping method	Examines the method, materials, and equipment according to the prototyping purpose, determining a method that allows performance

No.	Function Element	Description
		evaluation under conditions close to actual manufacturing.
2	Procuring parts for prototyping	Identifies parts/materials needed for the prototype, and plans and executes a procurement method that can accommodate short lead times and small lots.
3	Arranging manufacture of the prototype	Requests the manufacture of the prototype based on design data and production conditions, and manages quality, delivery time, and cost.
4	Analyzing prototype performance	Measures and analyzes the product's performance, reliability, and safety at the prototyping stage, verifying design validity.
5	Approving specifications via the prototype	Evaluates the test results of the prototype, and confirms/approves the validity of the design specification with the customer or internally.
6	Improving the production method	Improves the process, jigs, and work procedure based on the prototyping results, raising productivity and quality at mass production.
7	Manufacturing simulation	Reproduces the manufacturing process using 3D models or digital twins, verifying the method and line design.
8	Acquiring product-use data	Collects usage data through actual-machine testing and user testing, reflecting it in product improvement and design optimization.
9	Conducting and verifying virtual prototyping	Simulates function and performance on CAD/CAE models, verifying design validity before actual prototyping.
10	Updating marketing information	Re-analyzes customer value based on prototype evaluation and market response, and reflects it in product planning and sales strategy.

6.3. Production Preparation

The function category of Production Preparation has three function groups: Line Fabrication, Production Ramp-up, and Document Linkage. Representative function elements for each function group are shown below.

Line Fabrication

Once the results of line design and equipment design, premised on mass production, have been approved, materials for the equipment and production line are procured based on that content, and the equipment, production line, and jigs/tools are actually fabricated. In fabrication, fabrication drawings, fabrication schedule, and fabrication procedure are presented in advance, so that work can proceed efficiently even where the target plant is already producing product.

Table 8 Function Elements of Line Fabrication

No.	Function Element	Description
1	Applying the QC process table to the actual line	Applies the QC process table created at the design stage to the shop floor, reflecting quality check points and inspection conditions on the actual line.
2	External procurement of production equipment	Procures the necessary production equipment from external manufacturers, managing specifications, delivery time, and performance evaluation to proceed with introduction.
3	In-house fabrication of production equipment	Designs and fabricates in-house-producible equipment, establishing a system that can flexibly handle specification changes and improvements.
4	Arranging fabrication of jigs/tools and molds	Arranges for the fabrication of jigs/tools and molds used in the manufacturing process based on design information, managing precision and delivery time.
5	Installing transport lines and confirming flow lines	Installs and verifies transport lines considering parts transport routes and worker flow lines, optimizing work efficiency.
6	Installing the line and balancing	Adjusts equipment, personnel placement, and takt time, balancing the production capacity and workload of the entire line.
7	Production monitoring in trial mode	Monitors equipment operation and production data in real time during the trial-run and verification stage, detecting defects early.
8	Collecting actual quality and functional data	Collects quality, performance, and work data during trial production, verifying the validity of design and process conditions.
9	Sharing parameters and know-how	Databases equipment setting values and improvement insights gained during prototyping, horizontally deploying them to the next model or other sites.
10	Effectively utilizing unneeded equipment	Diverts surplus or older equipment to reusable processes, improving production efficiency while restraining investment cost.

Production Ramp-up

Production ramp-up is carried out by performing design that assumes the actual production line and production equipment. When starting full-scale production, data is intensively acquired as a prototype or initial lot, analyzed, and managed until the productivity and quality-control values targeted for ramp-up are achieved.

Table 9 Function Elements of Production Ramp-up

No.	Function Element	Description
1	Managing the production ramp-up schedule	Schedules each of the prototyping, verification, approval, and mass-production switchover processes, managing the reliable progress of mass-production ramp-up.

No.	Function Element	Description
2	Adapting the line for series production	Designs a line configuration that considers commonization and flexibility so as to accommodate derivative models and improved products.
3	Preparing the environment for the QC process table	Re-prepares a QC process table suited to the mass-production stage, standardizing quality control items, work procedures, and inspection procedures.
4	Selecting and coordinating suppliers	Confirms the production capacity and quality system of partner companies based on the mass-production plan, minimizing supply risk.
5	Arranging long-lead-time procured parts	Arranges early for parts with long lead times to prevent production delays, optimizing the procurement plan.
6	Managing initial fluctuation at mass production	Intensively monitors and corrects quality variation and process instability in the early period of mass-production ramp-up, transitioning early to stable production.
7	Addressing required quality and defect causes	Maintains the quality level based on market requirements, analyzing defect causes and taking swift countermeasures/improvements.
8	Creating the manufacturing BOM	Creates a manufacturing BOM including the materials, auxiliary materials, and consumables needed in the production process.
9	Creating the service BOM	Prepares drawings and part composition for after-sales service, establishing a supply/maintenance system across the entire product lifecycle.
10	Transferring the line at end of production	Withdraws and relocates line equipment in line with product discontinuation, planning and executing a diversion plan to other products.

Document Linkage

Document linkage prepares the information needed at production time, such as work procedure documents. This covers not only workers but also business related to preparing master data used by business systems. Document linkage covers not only cases of starting up an entirely new production line or registering production procedures for a new product, but also cases such as responding to design changes for products whose production has already started.

Table 10 Function Elements of Document Linkage

No.	Function Element	Description
1	Linking or integrating the bill of materials	Integrally manages the design BOM, manufacturing BOM, and service BOM, maintaining data consistency between processes and departments.
2	Managing the design BOM	Integrally manages design composition, part specifications, and change history on the PLM, guaranteeing the accuracy of design

No.	Function Element	Description
		information.
3	Managing the manufacturing BOM	Manages part composition based on the manufacturing process and assembly sequence, streamlining production preparation while aligning with the design BOM.
4	Organizing terminology and creating a dictionary	Standardizes terms commonly used across design, production, and quality, compiling them into a dictionary to facilitate information collaboration inside and outside the company.
5	Sharing drawings and work procedure documents	Integrally manages the latest design drawings and corresponding work instructions/procedure documents, preventing misapplication on the shop floor.
6	Managing and sharing the QC process table	Clarifies quality control items and inspection conditions in the process table, sharing it so it can respond immediately to design changes or process improvements.
7	Managing and sharing standard work procedure documents	Standardizes work methods, jig conditions, and inspection criteria, sharing them across all sites via PLM or MES.
8	Extracting and preparing master data	Extracts master information such as parts, equipment, and processes, unifying and preparing the code system and attribute definitions.
9	Managing the flow at design change	Manages the process from design change proposal through approval and reflection, making the timing and consistency of changes traceable.
10	Creating documents for regulatory compliance	Creates technical documents responding to regulatory requirements such as RoHS and REACH, managed per product.

6.4. Production Management

The function category of Production Management has three function groups: Process Control, Quality Control, and Cost Management. Representative function elements for each function group are shown below.

Process Control

This manages the correspondence between the production schedule issued in response to confirmed orders, forecast orders, or replenishment instructions from downstream processes, and the actual production performance, managing whether each instruction is being executed as planned, or whether the plan itself is valid.

Table 11 Function Elements of Process Control

No.	Function Element	Description
1	Managing the production schedule plan	Formulates a production schedule based on demand forecasts and order plans, managing resource load and delivery time.

No.	Function Element	Description
2	Managing the production schedule	Creates and distributes a daily/weekly production schedule considering equipment, personnel, and material constraints.
3	Issuing production instructions	Issues manufacturing instructions based on the plan to the MES or shop-floor terminals, clarifying work content and sequence.
4	Managing production progress	Collects actual performance data from the shop floor, visualizing the difference from the plan to control delivery time and quality.
5	Managing equipment operation	Monitors equipment utilization rate and downtime causes, converting them to data, and optimizes OEE improvement and maintenance planning.
6	Managing procured items' content and delivery time	Shares suppliers' production progress and delivery-time information, managing the delivery time of procured items.
7	Managing output and productivity	Analyzes work performance/output data, continuously examining measures to improve productivity and operating efficiency.
8	Managing material and WIP inventory	Monitors the receipt and issuance of materials and intermediate items, appropriately grasping inventory levels to prevent shortages or excess inventory.
9	Managing product inventory and shipment performance	Links finished-goods inventory and shipment status with the sales plan, enabling accurate delivery-time responses.
10	Responding to irregularities	Ensures reliable handling of irregularities (parts shortage, equipment trouble, defects, etc.) when they occur, and records/manages the response performance.

Quality Control

For quality control, in addition to normal inspection, mechanisms are operated and maintained — such as setting stricter inspection items — to prevent defective products as part of quality control. Analysis of inspection results identifies the causes of defects, and activities are conducted to raise productivity and the first-pass yield rate.

Table 12 Function Elements of Quality Control

No.	Function Element	Description
1	Setting quality inspection criteria	Defines judgment criteria, inspection items, and allowable ranges based on product specifications and customer requirements.
2	Conducting finished-product inspection	Inspects the product's performance, appearance, and safety before shipment, making a final confirmation of standard compliance and quality assurance.
3	Receiving and in-process inspection	Inspects procured items or parts partway through the process, preventing impact on the current and next processes.

No.	Function Element	Description
4	Monitoring quality data trends	Analyzes quality data such as measured values and defect rates over time, catching signs of abnormality early.
5	Calibrating and recording measuring instruments	Performs periodic calibration to maintain the precision of measuring instruments, recording and managing the results.
6	Analyzing defect causes and countermeasures	Analyzes causes when a defect occurs, and takes measures from the design, process, and material sides to prevent recurrence.
7	Repairing and re-inspecting defective products	Re-inspects a defective product after correction to confirm quality, thoroughly managing recurrence prevention and history.
8	Tracing back defect causes	Traces the BOM, lot, and work history when a defect occurs, identifying the cause and scope of impact.
9	Developing inspection devices and mechanisms	Develops automatic inspection devices, AI image judgment, etc., improving the efficiency and accuracy of inspection.
10	Sharing quality-improvement knowledge	Databases defect cases, improvement measures, and success cases, sharing them between design and manufacturing departments.

Cost Management

This provides a mechanism for calculating and aggregating, as rationally as possible, the costs incurred at production lines and equipment other than material cost — such as labor cost and equipment depreciation cost. It sets and manages the indicators, unit consumption rates, and charge rates needed to allocate the fixed costs of equipment.

Table 13 Function Elements of Cost Management

No.	Function Element	Description
1	Identifying and recording cost factors	Clarifies cost-component factors such as material cost, labor cost, and equipment cost, recording their source so they can be traced.
2	Recording manufacturing performance data	Collects performance data such as manufacturing time, materials used, and work hours, as the basis for calculating actual cost.
3	Calculating and updating the unit consumption rate	Calculates the standard work time and material usage per product, and updates it periodically based on production performance.
4	Managing equipment depreciation	Allocates equipment investment cost according to useful life, reflecting it in the standard cost and asset management.
5	Allocating fixed and shared costs	Rationally allocates shared costs such as equipment maintenance cost and administrative cost to each product/process, raising the accuracy of cost calculation.
6	Calculating actual cost	Calculates the cost actually incurred from performance data, evaluating efficiency by comparison with the standard cost.

No.	Function Element	Description
7	Setting manufacturing overhead	Calculates indirect costs other than shop-floor work, such as management, maintenance, and logistics, setting an allocation basis to incorporate into cost.
8	Analyzing cost variance	Analyzes the variance between standard cost and actual cost, identifying factors (yield, man-hours, material price, etc.) to improve.
9	Calculating environmental burden amount	Calculates energy usage and CO2 emissions, reflecting/managing them as environmental cost within the cost.
10	Cost estimation and price setting	Calculates the estimated cost based on design and production conditions, determining the product price or quotation amount in line with the profit target.

6.5. Plant Operations

The function category of Plant Operations has three function groups: Equipment Maintenance, Quality Assurance, and Shop-floor Improvement. Representative function elements for each function group are shown below.

Equipment Maintenance

Equipment maintenance includes preventive maintenance such as periodic inspection, corrective maintenance after a failure or minor/major stoppage, and predictive maintenance that senses signs of a failure in advance and responds proactively. Equipment inspection and maintenance are carried out based on predetermined methods and criteria, and new insight is gained by analyzing this data.

Table 14 Function Elements of Equipment Maintenance

No.	Function Element	Description
1	Formulating equipment maintenance procedures	Formulates standard procedures such as inspection, cleaning, and replacement to maintain safe and efficient operation of the equipment.
2	Conducting routine maintenance	Performs preventive maintenance such as periodic inspection, lubrication, and cleaning, preventing failures before they occur.
3	Planning maintenance content and schedule	Formulates an annual/monthly maintenance schedule, clearly defining the target equipment and frequency of implementation.
4	Monitoring equipment and detecting abnormalities	Uses sensor and IoT data to detect early signs of abnormality from temperature, vibration, etc.
5	Analyzing operating performance data	Analyzes operation rate, downtime, and maintenance history, using it to improve reliability and optimize the maintenance plan.
6	Responding to and addressing equipment abnormalities	Takes swift emergency action when an abnormality occurs, and formulates permanent recurrence-prevention measures through cause analysis.

No.	Function Element	Description
7	Managing auxiliary materials and spare parts	Manages inventory of auxiliary materials such as tools, parts, and oils/greases needed for maintenance, preventing delays due to shortage.
8	Managing equipment maintenance history	Digitally records history such as inspection, repair, and replacement, using it for trend analysis and improvement of maintenance strategy.
9	Training equipment maintenance personnel	Trains personnel through education and OJT to pass on shop-floor skills and troubleshooting capability.
10	Responding to minor and major stoppages	Analyzes the causes of short stoppages (minor stops) and long stoppages (major stops), pursuing recurrence prevention from both the work and equipment sides.

Quality Assurance

Quality assurance provides a mechanism to prevent quality problems in products or parts before they occur, so that quality can be assured without excessive reliance on direct 100% inspection at the final stage. It focuses on the process by which quality defects arise, and removes the causes of quality defects and failures through causal-relationship and failure-factor analysis. In particular, it incorporates the 4M factors: equipment, people, material, and method.

Table 15 Function Elements of Quality Assurance

No.	Function Element	Description
1	Setting the quality inspection method	Determines an inspection method suited to the characteristics of the product or process, setting the optimal method such as sampling, 100%, or automatic inspection.
2	Responding to customer complaints	Swiftly analyzes defects/requests from customers, investigating the cause and taking recurrence-prevention measures to maintain trust.
3	Training quality assurance personnel	Educates specialized knowledge and skill regarding quality assurance, inspection, and improvement, embedding a quality culture.
4	Evaluating supplier quality	Periodically evaluates the quality-control system, defect rate, and improvement capability of suppliers, securing the quality of supplied goods.
5	Countermeasures via process FMEA	Identifies latent risks for each process, evaluating impact and occurrence frequency to implement measures proactively.
6	Identifying and recording control factors	Identifies important parameters affecting quality (temperature, pressure, time, etc.), recording and managing them.
7	Responding to quality-defect recalls	Identifies the scope of impact for a serious defect that occurred in the market, and swiftly carries out recovery and customer response.

No.	Function Element	Description
8	Excluding defective lots	Identifies and isolates lots containing defective products, performing inspection and re-judgment to prevent further outflow.
9	Responding to internal/external process audits	Responds to process audits by internal parties, customers, or third parties, implementing corrective action and continuous improvement.
10	Managing plant health and safety	Complies with labor safety and hygiene standards, reducing risk in the work environment and maintaining employee health.

Shop-floor Improvement

Shop-floor workers and managers themselves discover material for improvement, such as shop-floor problems and near-misses, find solutions, and implement improvements. Shop-floor improvement uses diagrams such as the QC seven tools and the new QC seven tools, taking an autonomous, bottom-up approach that includes problem discovery and visualization.

Table 16 Function Elements of Shop-floor Improvement

No.	Function Element	Description
1	5S activities and safety management	Thoroughly implements the 5S (sort, set in order, shine, standardize, sustain), improving the workplace environment and safety.
2	"Why-why" analysis of shop-floor issues	Practices a method of repeating "why" to identify the true cause of a problem, leading to a fundamental improvement measure.
3	Sharing and utilizing improvement cases	Systematically shares success/failure cases from the shop floor, horizontally deploying them to other processes or departments to prevent recurrence.
4	Sharing past troubles and near-misses	Shares past troubles and near-miss cases, cultivating a safety culture that prevents similar risks before they occur.
5	Following up on countermeasures	Continuously confirms the effect after an improvement is implemented, building a mechanism that leads to recurrence prevention and standardization.
6	Data-driven QC activities	Performs quantitative problem-solving using statistical methods based on quality, cost, and delivery-time data.
7	Synchronizing and streamlining things and information	Aligns the flow of physical items on the shop floor with the flow of information, achieving lead-time reduction and elimination of waste between processes.
8	Improving the work environment	Improves layout, lighting, and flow lines considering worker safety and comfort, raising productivity.
9	Promoting in-house standardization activities	Reflects improvement results in standardization documents, aiming for continuous company-wide operation as unified rules.

No.	Function Element	Description
10	Skill development and human resource development	Develops personnel through skill education and support for qualification acquisition, evaluating their contribution to improvement activities.

7. Information Model (normative)

The information model shows the structure of the content that expresses, as information, the outcome obtained as a result of executing a function. The information model also shows the structure of the content of information that is indispensable when executing a function. The function model and the information model correspond element by element: one function uses one or more pieces of information, and at the same time one piece of information is involved in multiple functions.

The information model refers to objects composed of characters, numbers, diagrams, etc., such as slips, forms, or screens. The information model is shaped so that the person in charge of the business, having understood its meaning, can read or write its content. In general, an information model has a hierarchy, and part of the structure of one information model may include other information models.

In this chapter, information is divided into the categories of Planning Information, Development Information, Technical Information, Management Information, and Shop-floor Information, and information is grouped within each category. Below, an overview of the information model is explained by information category and information group, and the main items are shown for each.

Table 17 Categories and Groups of the Information Model

Category	Group	Description
Planning Information	Planning Information	Information that sets product development or production targets and defines the method of realization, including quantity, period, and resource allocation
	Specification Information	Information that defines the function, performance, and conditions of a product or process, and is shared as the standard for design, manufacturing, and quality
	Cost Information	Information that manages the cost structure, estimates, and actuals related to a product or process, and is used for profitability analysis and improvement
Development Information	Structure Information	Information that defines the constituent elements and relationships of a product or piece of equipment, forming the basis of structure management such as BOM and BOA
	Procedure Information	Information that defines the concrete method and sequence of carrying out a task or process, contributing to establishing standard work and stabilizing quality
	Reference/Standard Information	Information that defines the judgment criteria or standards regarding quality, work, or equipment, serving as a common

Category	Group	Description
		yardstick for evaluation and management
Technical Information	Deployment Information	Information for deploying the content of a design or plan to each process or department, converting it into an executable form
	Linkage Information	Information that is shared and linked between departments or systems, ensuring data consistency and interoperability
	Quality Information	Information that records measured values and evaluation results regarding the quality of a product or process, used for defect prevention and quality improvement
Management Information	Schedule Information	Information indicating the start/end time and progress status of production or work, used for schedule management and progress control
	Instruction Information	Information indicating concrete instruction content regarding work or production, directly supporting execution on the shop floor
	Performance Information	Information that records performance such as production quantity, work results, and operating status, forming the basis for analysis and improvement
Shop-floor Information	4M Information	Information that manages data regarding workers, equipment, materials, and methods, used for factor analysis and management of process quality
	Status Information	Information that grasps the status or abnormality of equipment or a process in real time, supporting stable operation and swift response
	Improvement Information	Information that records the cause analysis of an issue or defect and the content of countermeasures, used for continuous improvement and recurrence prevention

In this chapter, the table corresponding to each information group indicates, for each information name, which of product, process, or equipment it relates to. This makes it possible to identify cases where a given piece of information is cross-cutting content relating to product, process, and equipment.

Also, for each information model, a corresponding class diagram and a list of the main information items are shown. The classes in the class diagram correspond to entities in the data model; in the list of information items, the entity to which each information item belongs and the corresponding attribute are shown, along with the cardinality shown in the Relationship column. For example, "*" means that multiple pieces of the corresponding information exist for one; where there are multiple underlying entities, they are identified by number. Also, where an entity is marked with "×", this indicates that it is an entity formed by cross-expanding the entities on both sides.

7.1. Planning Information

Planning Information (Planning Information)

Information that defines the plan or schedule for a future activity together with the progression of a period or schedule. It shows the plan and its actual performance regarding the schedule accompanying product development, and the production volume, capacity, and load of each product or piece of equipment.

Table 18 Planning Information

No.	Information Name	Product	Process	Equipment	Description
1	Development Plan Table	○	○	○	A plan summarizing the purpose, process, schedule, structure, and resources of product or system development. Serves as a guide for development progress.
2	Production Plan Table	○			A table planning production volume, timing, and plant load based on demand forecast and orders. Serves as a guide for overall production.
3	Capacity Plan Table		○	○	A plan table for comparing the capacity and load of the equipment and personnel needed for production, to grasp shortfalls/excesses. Evaluates the feasibility of the production plan.

Development Plan Table

A plan summarizing the purpose, process, schedule, structure, and resources of product or system development. Serves as a guide for development progress.

Figure 6 Development Plan Table

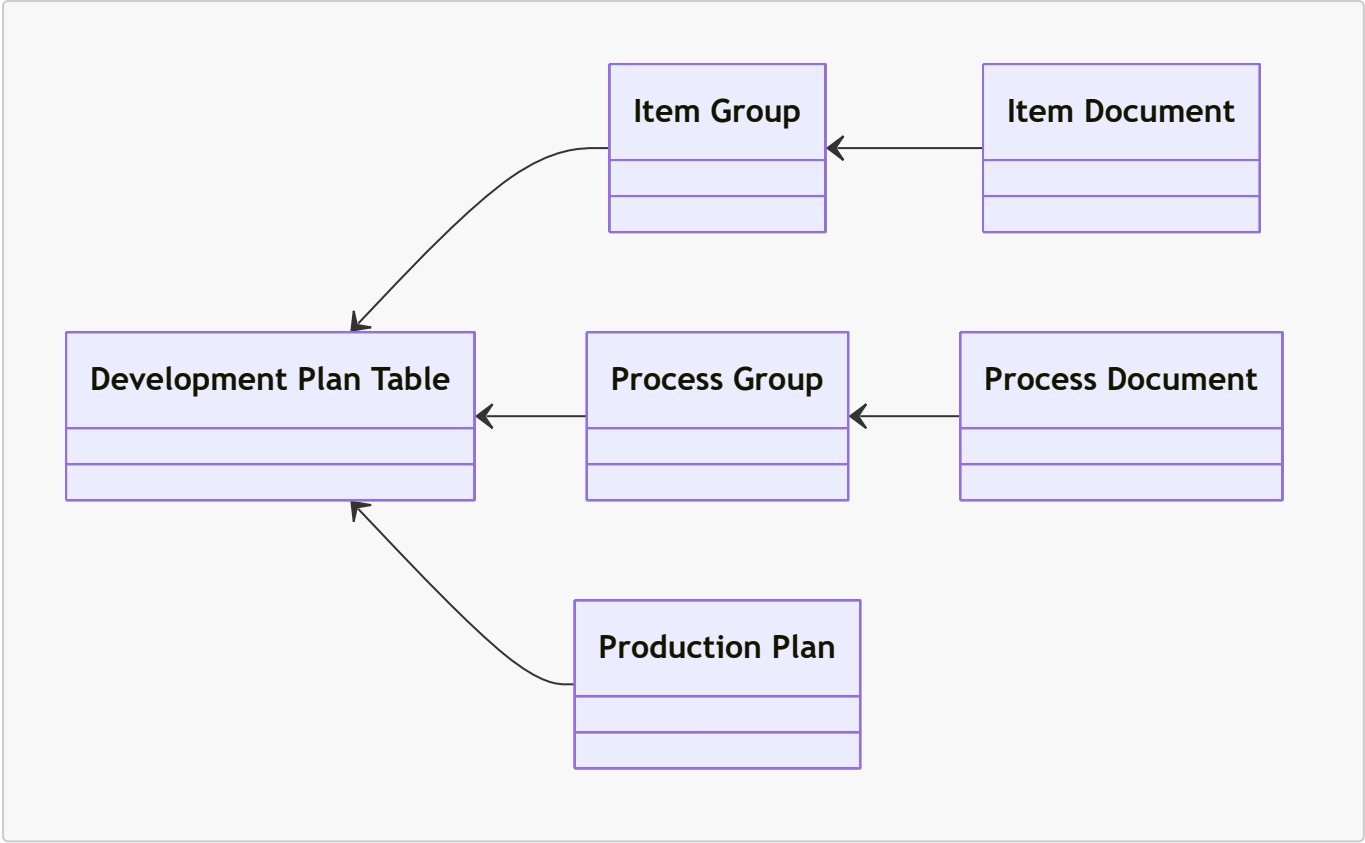


Table 19 Development Plan Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Item Group	Item Group Name	
2	Status	Item Group	Status	
3	Content	Item Group × Date	Content	*1
4	Development Purpose	Item Document	Development Purpose	
5	Development Schedule	Item Document	Development Schedule	
6	Specification	Item Document	Specification	
7	Quantity	Item Document	Quantity	
8	Unit	Item Document	Unit	
9	Process Name	Process Group	Process Group Name	
10	Status	Process Group	Status	
11	Content (Process)	Process Group × Date	Content	*2
12	Development Purpose (Process)	Process Document	Development Purpose	
13	Development Schedule (Process)	Process Document	Development Schedule	

No.	Item Name	Entity	Attribute	Relationship
14	Specification (Process)	Process Document	Specification	
15	Quantity (Process)	Process Document	Quantity	
16	Unit (Process)	Process Document	Unit	
17	Date	Production Plan	Term ID	*3
18	Plan Content	Production Plan	Plan Content	*3
19	Status	Production Plan	Status	*3
20	Cost of Sales	Production Plan	Cost of Sales	*3
21	Operating Profit	Production Plan	Operating Profit	*3

Production Plan Table

A table planning production volume, timing, and plant load based on demand forecast and orders. Serves as a guide for overall production.

Figure 7 Production Plan Table

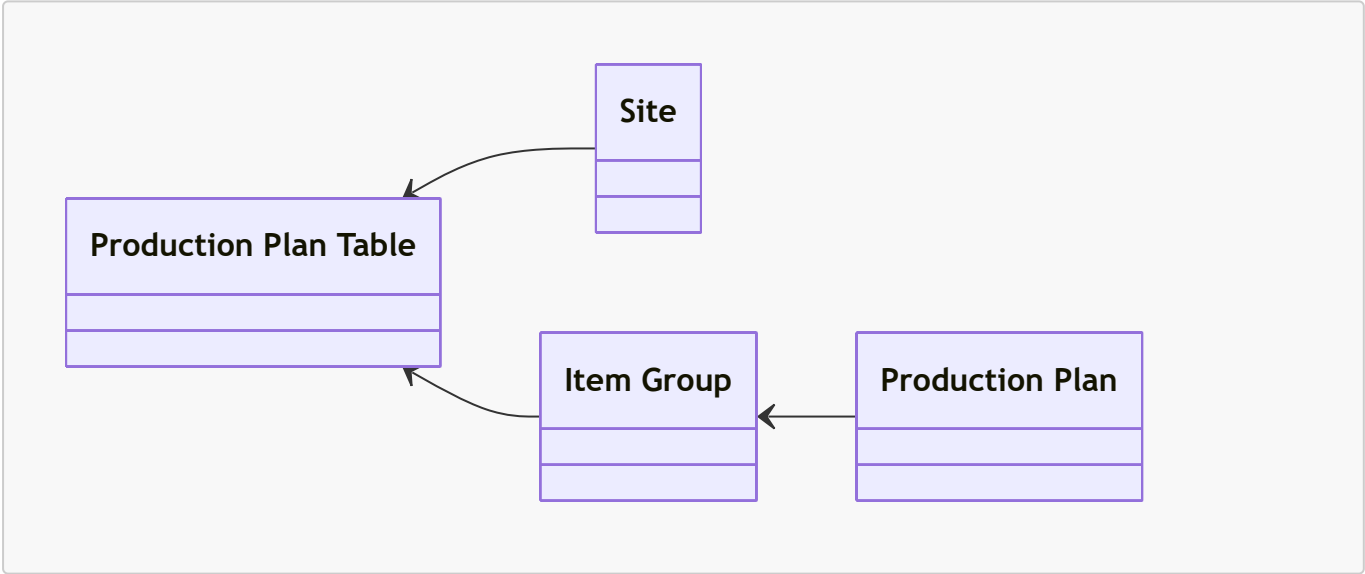


Table 20 Production Plan Table

No.	Item Name	Entity	Attribute	Relationship
1	Site Name	Site	Site Name	
2	Planned Quantity	Site × Month	Planned Quantity	*1
3	Actual Quantity	Site × Month	Actual Quantity	*1
4	Item Name	Item Group	Item Group Name	
5	Planned Quantity	Item Group × Month	Planned Quantity	*2
6	Actual Quantity	Item Group × Month	Actual Quantity	*2

No.	Item Name	Entity	Attribute	Relationship
7	Planned Date/Time	Production Plan	Planned Date/Time	*
8	Planned Quantity	Production Plan	Planned Quantity	*
9	Actual Quantity	Production Plan	Actual Quantity	*
10	Reference Quantity	Production Plan	Quantity	*
11	Unit	Production Plan	Unit	*

Capacity Plan Table

A plan table for comparing the capacity and load of the equipment and personnel needed for production, to grasp shortfalls/excesses. Evaluates the feasibility of the production plan.

Figure 8 Capacity Plan Table

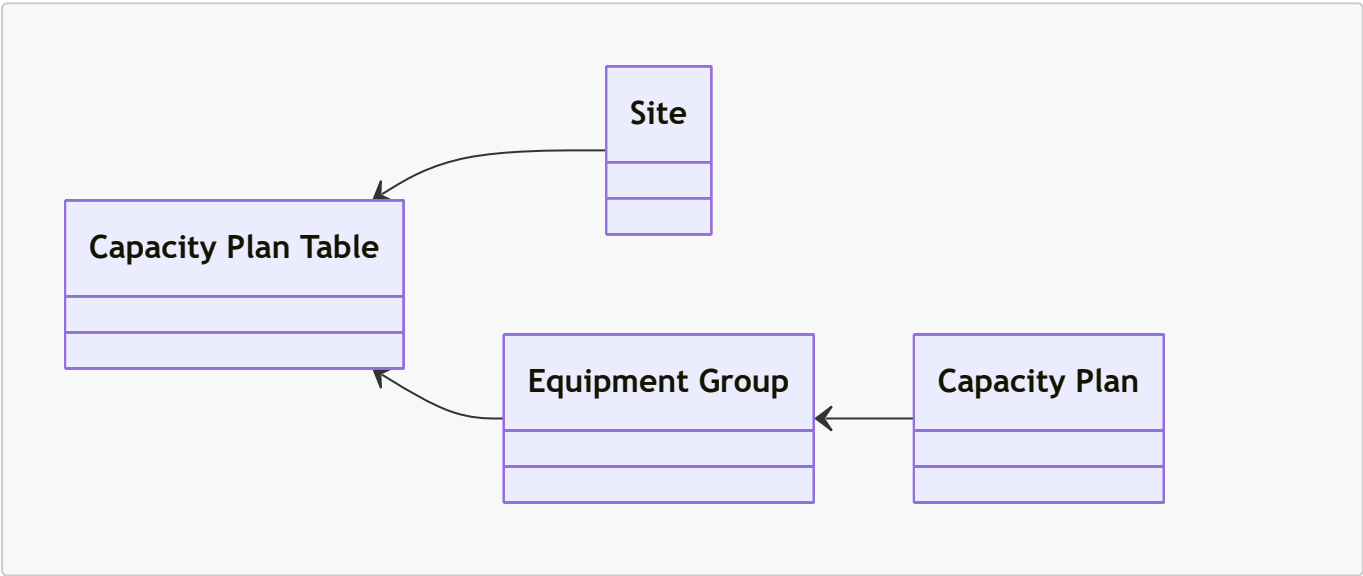


Table 21 Capacity Plan Table

No.	Item Name	Entity	Attribute	Relationship
1	Site Name	Site		
2	Planned Quantity	Site × Month		*1
3	Actual Quantity	Site × Month		*1
4	Equipment Name	Equipment Group		
5	Planned Quantity	Equipment Group × Month		*2
6	Actual Quantity	Equipment Group × Month		*2
7	Planned Date/Time	Production Plan	Planned Date/Time	*
8	Planned Quantity	Production Plan	Planned Quantity	*
9	Actual Quantity	Production Plan	Actual Quantity	*

No.	Item Name	Entity	Attribute	Relationship
10	Reference Quantity	Production Plan	Quantity	*
11	Unit	Production Plan	Unit	*

Specification Information (Specification Information)

The requirement specification document is upper-level information that defines requirements, conditions, and constraints, serving as the reference point for other information. It is formal, prescriptive information containing both qualitative and quantitative specifications. The design document, meanwhile, is generative information that describes the structure, function, and shape needed to embody the specification. It is a category of technical document with a fixed representation format, such as drawings and the E-BOM.

Table 24 Specification Information

No.	Information Name	Product	Process	Equipment	Description
1	Requirement Specification Document	○	○	○	A document summarizing the desired functions, performance, and constraint conditions prior to designing a product, process, or equipment. An upper-level requirement showing what should be achieved.
2	Product Design Document	○			A document summarizing the design content of a product, such as its structure, materials, dimensions, and performance. Design information forming the basis of mass production.
3	Process Design Document		○		A document defining the procedure, equipment conditions, work content, quality control points, etc. of a manufacturing process. Serves as the standard for the mass-production process.
4	Equipment Design Document			○	A document summarizing the detailed design of the structure, capacity, control method, and safety measures of production equipment. A guide for equipment fabrication/introduction.

Requirement Specification Document

A document summarizing the desired functions, performance, and constraint conditions prior to designing a product, process, or equipment. An upper-level requirement showing what should be achieved.

Figure 9 Requirement Specification Document

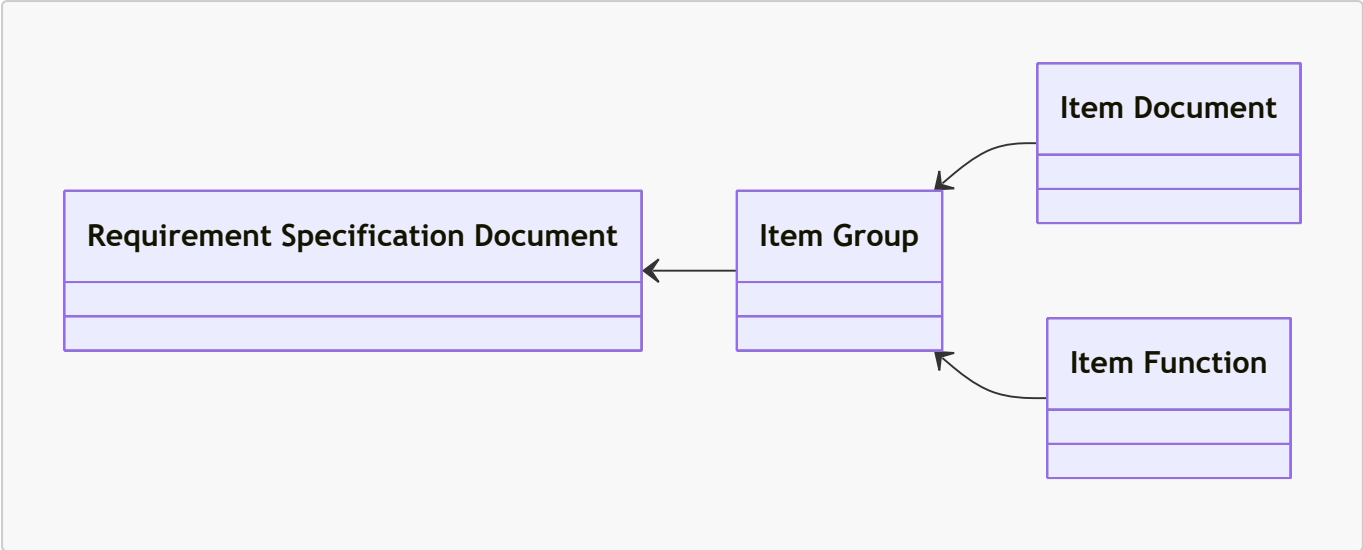


Table 25 Requirement Specification Document

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Item Group	Item Group Name	
2	Item Classification	Item Group	Item Group Classification	
3	Target Cost	Item Group	Target Cost	
4	Sales Target	Item Group	Sales Target	
5	Item Document	Item Document	Item Document Name	*1
6	Document Classification	Item Document	Item Document Classification	*1
7	Development Purpose	Item Document	Development Purpose	*1
8	Target Customer	Item Document	Target Customer	*1
9	Development Schedule	Item Document	Development Schedule	*1
10	Development Structure	Item Document	Development Structure	*1
11	Status	Item Function	Status	*2
12	Customer Requirement	Item Function	Customer Requirement	*2
13	Functional Requirement	Item Function	Functional Requirement	*2
14	Quality Characteristic	Item Function	Quality Characteristic	*2
15	Required Level	Item Function	Quantity	*2
16	Unit	Item Function	Unit	*2

Product Design Document

A document summarizing the design content of a product, such as its structure, materials, dimensions, and performance. Design information forming the basis of mass production.

Figure 10 Product Design Document

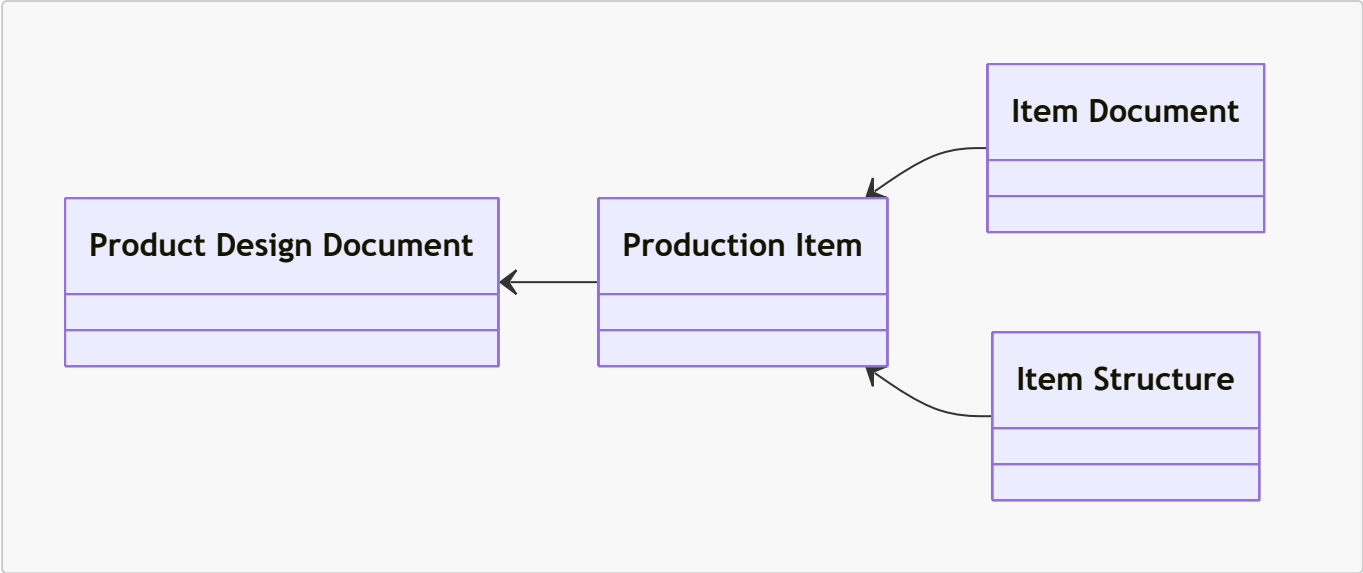


Table 26 Product Design Document

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Item Name	Production Item	Production Item Name	
3	Item Classification	Production Item	Production Item Classification	
4	Status	Production Item	Status	
5	Specification	Production Item	Specification	
6	Quantity	Production Item	Quantity	
7	Unit	Production Item	Unit	
8	Drawing Number	Item Document	Item Document Name	*1
9	Drawing Name	Item Document	Specification	*1
10	Drawing Classification	Item Document	Item Document Classification	*1
11	Revision	Item Document	Status	*1
12	File	Item Document	File	*1
13	Sequence Number	Item Structure	Sequence Number	*2
14	Structure Classification	Item Structure	Item Structure Classification	*2
15	Part Number	Item Structure	Child Production Item ID	*2
16	Part Name	Item Structure	Item Structure Name	*2
17	Part Specification	Item Structure	Specification	*2
18	Quantity per Unit	Item Structure	Quantity per Unit	*2
19	Unit	Item Structure	Unit	*2

Process Design Document

A document defining the procedure, equipment conditions, work content, quality control points, etc. of a manufacturing process. Serves as the standard for the mass-production process.

Figure 11 Process Design Document

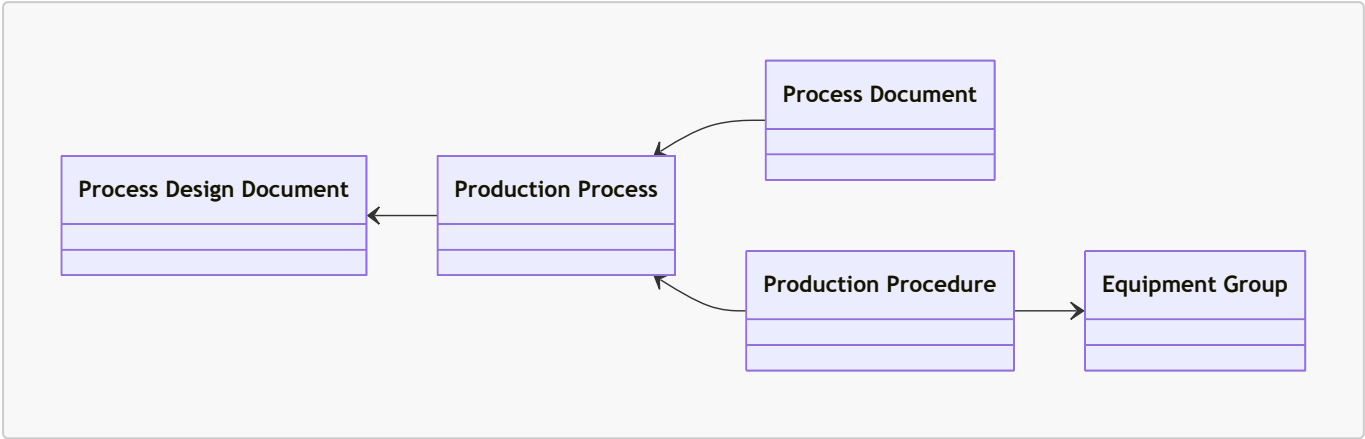


Table 27 Process Design Document

No.	Item Name	Entity	Attribute	Relationship
1	Process Name	Process Group	Process Group Name	
2	Process Classification	Process Group	Process Group Classification	
3	Status	Process Group	Status	
4	Specification	Process Group	Specification	
5	Performance	Process Group	Rate	
6	Capacity	Process Group	Standard Capacity	
7	Unit	Process Group		
8	Drawing Number	Process Document	Equipment Document Name	*1
9	Drawing Name	Process Document	Specification	*1
10	Drawing Classification	Process Document	Equipment Document Classification	*1
11	Revision	Process Document	Status	*1
12	File	Process Document	File	*1
13	Sequence Number	Production Procedure	Sequence Number	*2
14	Procedure Name	Production Procedure	Production Procedure Name	*2
15	Procedure Classification	Production Procedure	Production Procedure Classification	*2

No.	Item Name	Entity	Attribute	Relationship
16	Procedure Content	Production Procedure	Content	*2
17	Quantity	Production Procedure	Quantity	*2
18	Unit	Production Procedure	Unit	*2
19	Production Man-hours	Production Procedure	Production Man-hours	*2
20	Production Days	Production Procedure	Production Days	*2
21	Equipment Name	Equipment Group	Equipment Group Name	*3
22	Equipment Classification	Equipment Group	Equipment Group Classification	*3
23	Equipment Capacity	Equipment Group	Standard Capacity	*3
24	Equipment Specification	Equipment Group	Specification	*3
25	Unit	Equipment Group	Unit	*3

Equipment Design Document

A document summarizing the detailed design of the structure, capacity, control method, and safety measures of production equipment. A guide for equipment fabrication/introduction.

Figure 12 Equipment Design Document

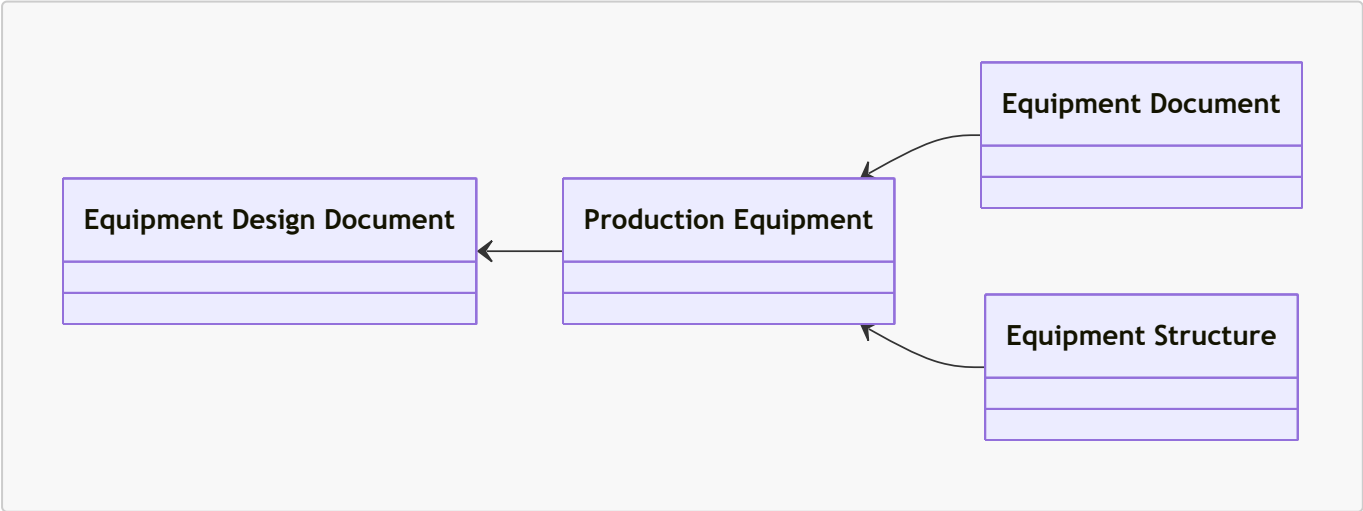


Table 28 Equipment Design Document

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Number	Production Equipment	Equipment Number	
2	Equipment Name	Production Equipment	Production Equipment Name	
3	Equipment Classification	Production Equipment	Equipment Classification	
4	Status	Production Equipment	Status	
5	Specification	Production Equipment	Specification	
6	Quantity	Production Equipment	Quantity	
7	Unit	Production Equipment	Unit	
8	Equipment Capacity	Production Equipment	Standard Capacity	
9	Equipment Performance	Production Equipment	Rate	
10	Drawing Number	Equipment Document	Equipment Document Name	*1
11	Drawing Name	Equipment Document	Specification	*1
12	Drawing Classification	Equipment Document	Equipment Document Classification	*1
13	Revision	Equipment Document	Status	*1
14	File	Equipment Document	File	*1
15	Sequence Number	Equipment Structure	Sequence Number	*2
16	Component Equipment Name	Equipment Structure	Equipment Structure Name	*2
17	Component Equipment Number	Equipment Structure	Child Production Equipment ID	*2
18	Structure Classification	Equipment Structure	Structure Classification	*2

No.	Item Name	Entity	Attribute	Relationship
19	Structure Specification	Equipment Structure	Specification	*2
20	Quantity per Unit	Equipment Structure	Quantity per Unit	*2
21	Structure Unit	Equipment Structure	Unit	*2

Cost Information (Cost Information)

Calculation information based on amount, quantity, and time, such as standard cost and actual cost. Has a numerical structure that can be aggregated and compared. Table 29 Cost Table

No.	Information Name	Product	Process	Equipment	Description
1	Standard Cost Table	☐	☐	☐	A table summarizing the theoretical standard cost, such as material cost, labor cost, and expenses. Serves as the basis for planned values and variance analysis.
2	Actual Cost Table	☐	☐	☐	A cost table tallying the material cost, man-hours, and expenses actually incurred. Grasps points for improvement by comparison with the standard cost.

Standard Cost Table

A table summarizing the theoretical standard cost, such as material cost, labor cost, and expenses. Serves as the basis for planned values and variance analysis.

Figure 13 Standard Cost Table

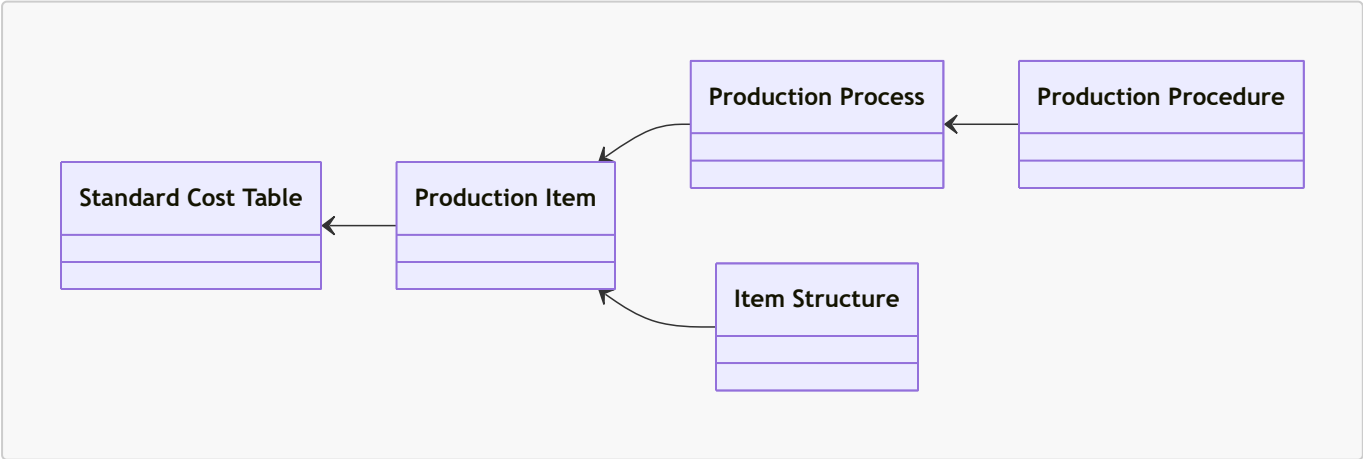


Table 30 Standard Cost Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Item Name	Production Item	Item Name	
3	Status	Production Item	Status	
4	Quantity	Production Item	Quantity	
5	Unit	Production Item	Unit	
6	Cost	Production Item	Cost	
7	Sequence Number	Item Structure	Sequence Number	*1
8	Part Name	Item Structure	Item Structure Name	*1
9	Specification	Item Structure	Specification	*1
10	Quantity per Unit	Item Structure	Quantity per Unit	*1
11	Unit	Item Structure	Unit	*1
12	Cost	Item Structure	Cost	*1
13	Process Number	Production Process	Process Number	
14	Process Name	Production Process	Production Process Name	
15	Status	Production Process	Status	
16	Quantity	Production Process	Quantity	
17	Unit	Production Process	Unit	
18	Cost	Production Process	Cost	
19	Sequence Number	Production Procedure	Sequence Number	*2
20	Part Name	Production Procedure	Production Procedure Name	*2
21	Content	Production Procedure	Content	*2
22	Quantity	Production Procedure	Quantity	*2
23	Unit	Production Procedure	Unit	*2
24	Cost	Production Procedure	Cost	*2

Actual Cost Table

A cost table tallying the material cost, man-hours, and expenses actually incurred. Grasps points for improvement by comparison with the standard cost.

Figure 14 Actual Cost Table

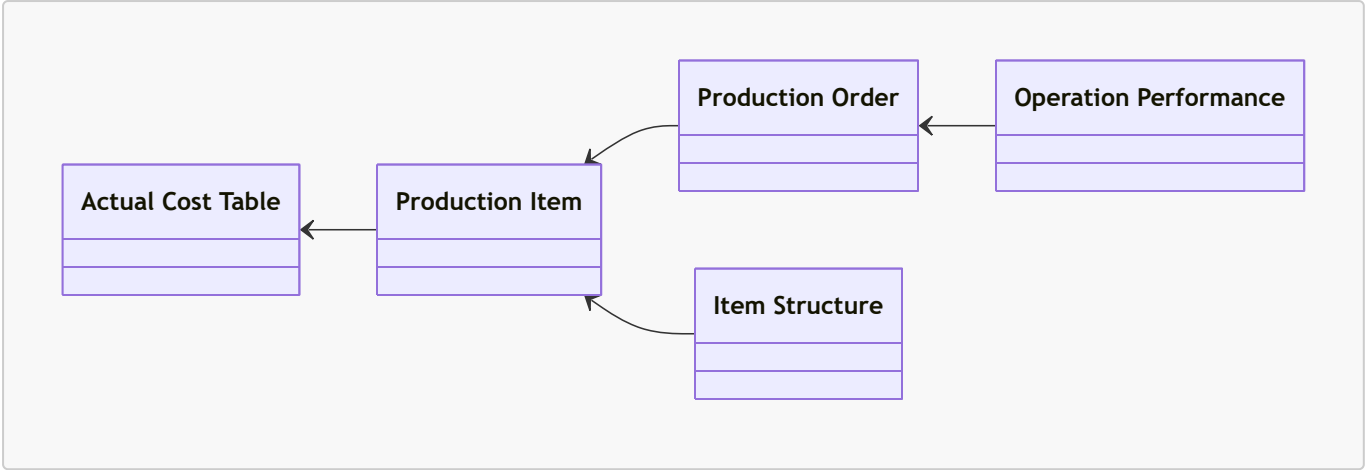


Table 31 Actual Cost Table

No.	Item Name	Entity	Attribute	Relationship
1	Part Number	Production Item	Part Number	
2	Item Name	Production Item	Production Item Name	
3	Status	Production Item	Status	
4	Quantity	Production Item	Quantity	
5	Unit	Production Item	Unit	
6	Cost	Production Item	Cost	
8	Sequence Number	Item Structure	Sequence Number	*1
9	Part Name	Item Structure	Item Structure Name	*1
10	Specification	Item Structure	Specification	*1
11	Quantity per Unit	Item Structure	Quantity per Unit	*1
12	Unit	Item Structure	Unit	*1
13	Cost	Item Structure	Cost	*1
14	Order Number	Production Order	Production Order Name	*2
15	Issue Date	Production Order	Issue Date/Time	*2
16	Status	Production Order	Status	*2
17	Production Process	Production Order	Production Process ID	*2
18	Quantity	Production Order	Quantity	*2
19	Unit	Production Order	Unit	*2
20	Cost	Production Order	Actual Quantity	*2
21	Date	Operation Performance	Term ID	*3

No.	Item Name	Entity	Attribute	Relationship
22	Performance Classification	Operation Performance	Operation Performance Classification	*3
23	Equipment	Operation Performance	Production Equipment ID	*3
24	Content	Operation Performance	Content	*3
25	Quantity	Operation Performance	Quantity	*3
26	Unit	Operation Performance	Unit	*3
27	Operating Time	Operation Performance	Operating Time	*3

7.2. Development Information

Structure Information (Structure Information)

Model information that expresses a target object in a hierarchical structure, such as the bill of materials (BOM), bill of process (BOP), and bill of assets (BOA). A static relational data type requiring consistency. Table 34 Structure Information

No.	Information Name	Product	Process	Equipment	Description
1	Engineering BOM (E-BOM)	○			A BOM showing the part composition from a design perspective. Describes the function and structure of the product based on drawings and specifications.
2	Manufacturing BOM (M-BOM)	○	○		A BOM showing the part composition needed in the manufacturing process. A manufacturing-perspective table corresponding to work units and process deployment.
3	Service BOM (S-BOM)	○			A BOM showing the composition of replacement parts for maintenance/service. A part system for inspection and repair.
4	Bill of Process (BOP)	○	○	○	A table recording the process sequence, work content, conditions, man-hours, etc. needed for manufacturing. Has the role of deploying the BOM into processes.

Engineering BOM (E-BOM)

A BOM showing the part composition from a design perspective. Describes the function and structure of the product based on drawings and specifications.

Figure 15 Engineering BOM (E-BOM)

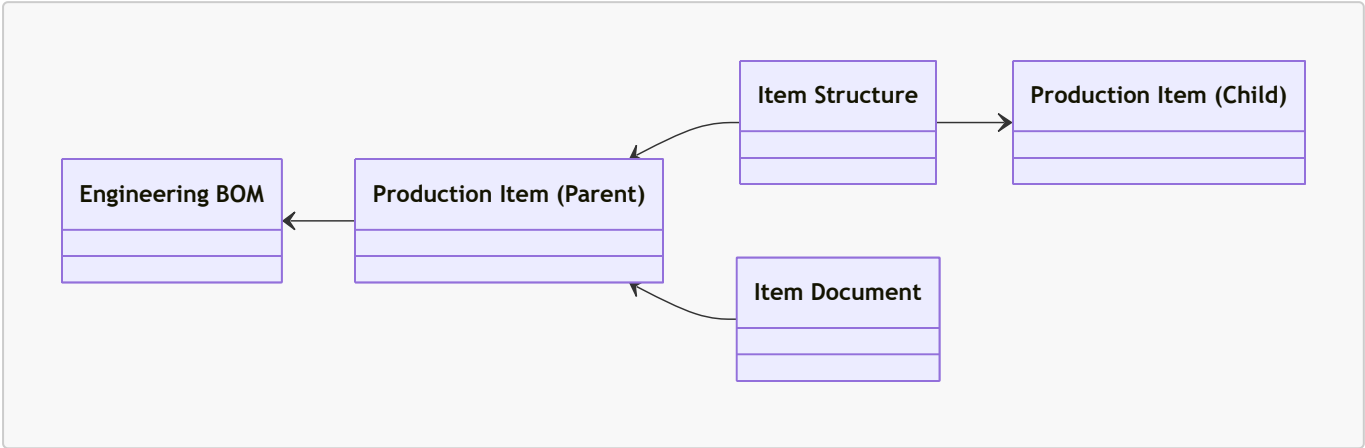


Table 35 Engineering BOM (E-BOM)

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Production Item	Production Item Name	
2	Item Number	Production Item	Item Number	
3	Item Classification	Production Item	Production Item Classification	
4	Status	Production Item	Status	
5	Drawing Number	Production Item	Drawing Number	
6	Specification	Production Item	Specification	
7	Drawing File	Item Document	File	*1
8	Sequence Number	Item Structure	Sequence Number	*2
9	Classification	Item Structure	Item Structure Classification	*2
10	Quantity per Unit	Item Structure	Quantity per Unit	*2
11	Unit	Item Structure	Unit	*2
12	Item Number	Production Item (Child)	Item Number	*2
13	Item Name	Production Item (Child)	Production Item Name	*2
14	Drawing Number	Production Item (Child)	Drawing Number	*2
15	Specification (Part)	Production Item (Child)	Specification	*2

Manufacturing BOM (M-BOM)

A BOM showing the part composition needed in the manufacturing process. A manufacturing-perspective table corresponding to work units and process deployment.

Figure 16 Manufacturing BOM (M-BOM)

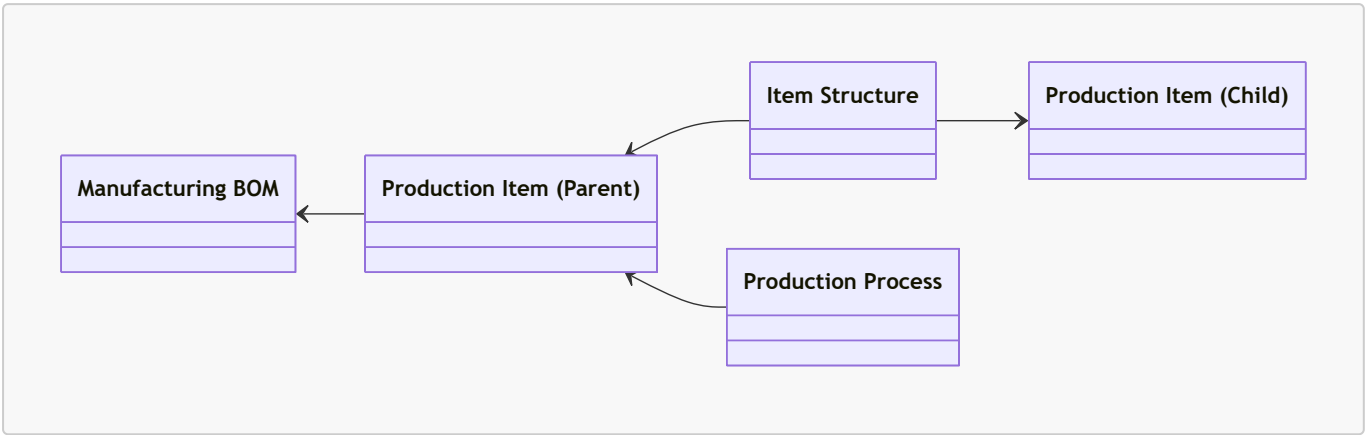


Table 36 Manufacturing BOM (M-BOM)

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Production Item	Production Item Name	
2	Item Number	Production Item	Item Number	
3	Item Classification	Production Item	Production Item Classification	
4	Status	Production Item	Status	
5	Drawing Number	Production Item	Drawing Number	
6	Specification	Production Item	Specification	
7	Sequence Number	Item Structure	Sequence Number	*
8	Level	Item Structure	Level	*
9	Classification	Item Structure	Classification	*
10	Item Name	Item Structure	Production Item Name	*
11	Quantity per Unit	Item Structure	Quantity per Unit	*
12	Unit	Item Structure	Unit	*
13	Item Number	Production Item (Child)	Item Number	*2
14	Drawing Number	Production Item (Child)	Drawing Number	*2
15	Specification (Child)	Production Item (Child)	Specification	*2
16	Process Name	Production Process	Process Name	*3
17	Specification (Process)	Production Process	Specification (Process)	*3
18	Quantity	Production Process	Quantity	*3
19	Unit	Production Process	Unit	*3

Service BOM (S-BOM)

A BOM showing the composition of replacement parts for maintenance/service. A part system for inspection and repair.

Figure 17 Service BOM (S-BOM)

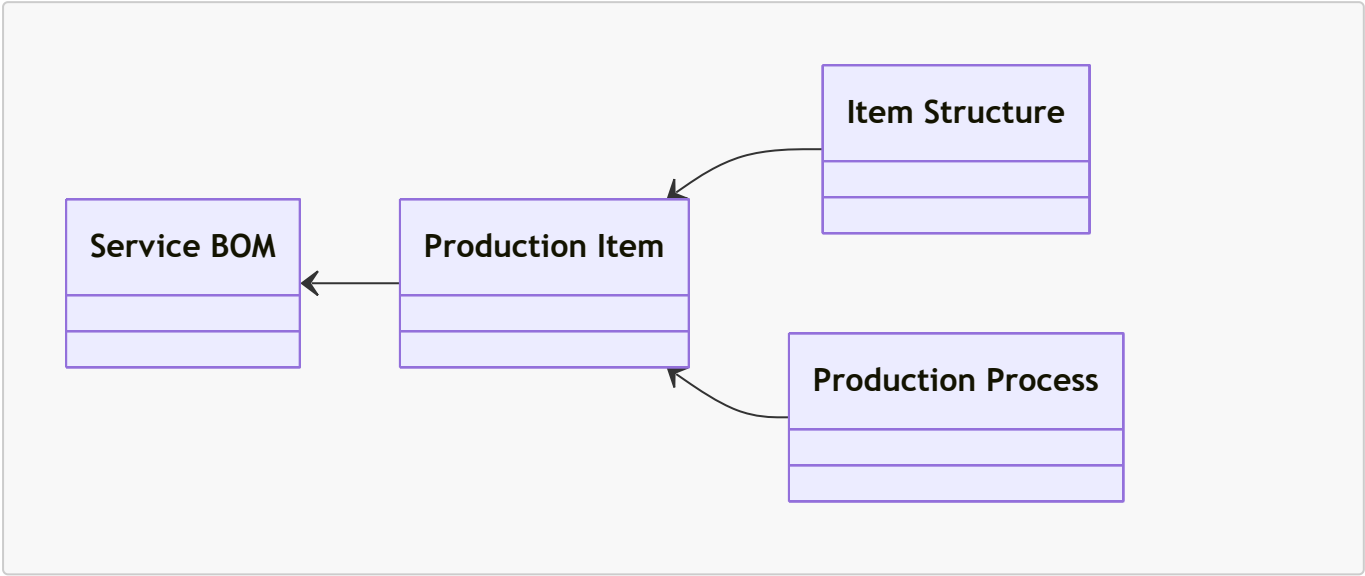


Table 37 Service BOM (S-BOM)

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Production Item	Production Item Name	
2	Item Number	Production Item	Item Number	
3	Item Classification	Production Item	Production Item Classification	
4	Status	Production Item	Status	
5	Service Number	Production Item	Drawing Number	
6	Service Specification	Production Item	Specification	
7	Service Name	Production Process	Production Process ID	*1
8	Service Classification	Production Process	Production Process Classification	*1
9	Service Content	Production Process	Production Process Name	*1
10	Service Status	Production Process	Status	*1
11	Quantity	Production Process	Quantity	*1
12	Unit	Production Process	Unit	*1
13	Sequence Number	Item Structure	Sequence Number	*2
14	Classification	Item Structure	Item Structure Classification	*2
15	Item Name	Item Structure	Production Item Name	*2

No.	Item Name	Entity	Attribute	Relationship
16	Quantity per Unit	Item Structure	Quantity per Unit	*2
17	Unit	Item Structure	Unit	*2

Bill of Process (BOP)

A table recording the process sequence, work content, conditions, man-hours, etc. needed for manufacturing. Has the role of deploying the BOM into processes.

Figure 18 Bill of Process (BOP)

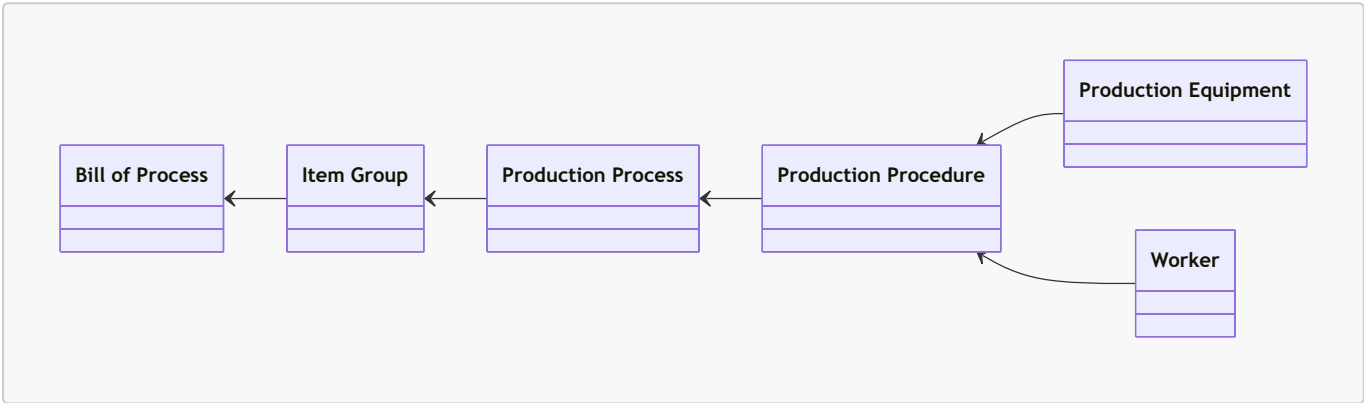


Table 38 Bill of Process (BOP)

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Production Item	Item Name	-
2	Item Number	Production Item	Item Number	-
3	Sequence Number	Item Structure	Sequence Number	*
4	Level	Item Structure	Level	*
5	Classification	Item Structure	Item Structure Classification	*
6	Production Item	Item Structure	Child Production Item ID	*
7	Production Process	Item Structure	Production Process ID	*
8	Specification	Item Structure	Specification	*
9	Quantity per Unit	Item Structure	Quantity per Unit	*
10	Unit	Item Structure	Unit	*
11	Sequence Number	Production Procedure	Sequence Number	*2
12	Procedure Name	Production Procedure	Production Procedure Name	*2
13	Procedure Content	Production Procedure	Content	*2
14	Production Man-hours	Production Procedure	Production Man-hours	*2
15	Unit	Production Procedure	Unit	*2

No.	Item Name	Entity	Attribute	Relationship
16	Equipment	Production Procedure	Equipment Group ID	*2
17	Worker	Production Procedure	Worker Group ID	*2
18	Equipment Number	Production Equipment	Equipment Number	*3
19	Equipment Name	Production Equipment	Production Equipment Name	*3
20	Specification	Production Equipment	Specification	*3
21	Standard Capacity	Production Equipment	Standard Capacity	*3
22	Worker Number	Worker	Worker Number	*4
23	Worker Name	Worker	Worker Name	*4
24	Specification	Worker	Specification	*4
25	Standard Capacity	Worker	Standard Capacity	*4

Procedure Information

Procedure information is information that defines the procedure, sequence, and judgment criteria for correctly and safely executing a business task or piece of work. It clarifies work content, execution conditions, precautions, and the equipment or jigs/tools used, suppressing variation in work between different people in charge. Table 40 Procedure Information

No.	Information Name	Product	Process	Equipment	Description
1	QC Process Table	○	○		A table summarizing the work content, inspection items, control points, and equipment used for each process. A standard for building in quality.
2	Work Procedure Document	○	○	○	A document summarizing the work method, procedure, tools used, and precautions. Aims to stabilize work quality and ensure safety.
3	Maintenance Procedure Document		○	○	A document showing the concrete procedure and precautions for equipment inspection, repair, and preventive maintenance. Used to improve equipment operation rate and prevent failures.

QC Process Table

A table summarizing the work content, inspection items, control points, and equipment used for each process. A standard for building in quality.

Figure 19 QC Process Table

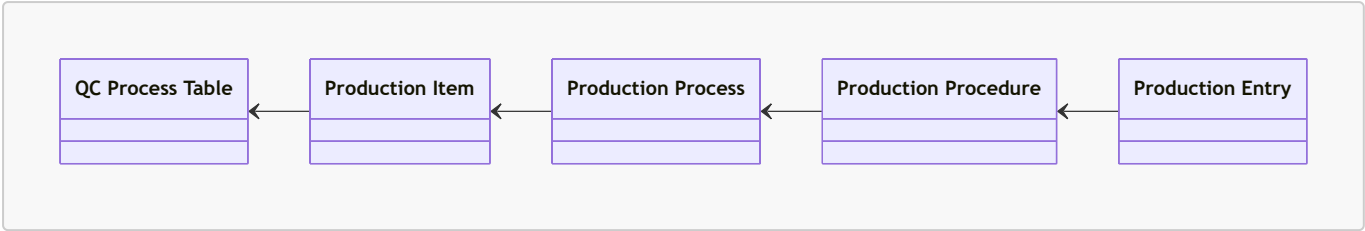


Table 41 QC Process Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	-
2	Item Name	Production Item	Production Item Name	-
3	Item Classification	Production Item	Production Item Classification	-
4	Process Number	Production Process	Process Number	
5	Process Name	Production Process	Production Process Name	
6	Site	Production Process	Site	
7	Sequence Number	Production Procedure	Sequence Number	*1
8	Classification	Production Procedure	Classification	*1
9	Procedure Name	Production Procedure	Procedure Name	*1
10	Content	Production Procedure	Content	*1
11	Quantity	Production Procedure	Quantity	*1
12	Unit	Production Procedure	Unit	*1
13	Production Line	Production Procedure	Production Line	*1
14	Equipment Used	Production Procedure	Equipment Used	*1
15	Worker	Production Procedure	Worker	*1
16	Control Method	Production Procedure	Control Method	*1
17	Inspection Method	Production Procedure	Inspection Method	*1
18	Inspection Frequency	Production Procedure	Inspection Frequency	*1
19	Remarks	Production Procedure	Remarks	*1
20	Sequence Number	Production Entry	Sequence Number	*2
21	Classification	Production Entry	Classification	*2
22	Production Entry Name	Production Entry	Production Entry Name	*2
23	Jig/Tool Name	Production Entry	Jig/Tool Name	*2

| 24 | Specification | Production Entry | Specification | *2 || 25 | Standard Value | Production Entry | Standard Value | *2 || 26 | Measured Value | Production Entry | Measured Value | *2 || 27 | Unit | Production Entry | Unit | *2 |

Work Procedure Document

A document summarizing the work method, procedure, tools used, and precautions. Aims to stabilize work quality and ensure safety.

Figure 20 Work Procedure Document

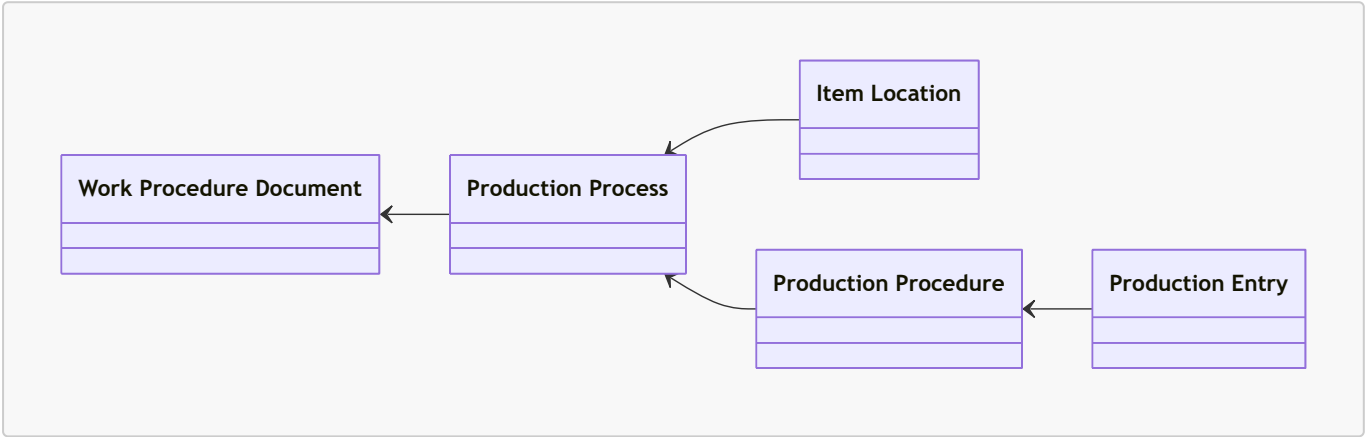


Table 42 Work Procedure Document

No.	Item Name	Entity	Attribute	Relationship
1	Process Name	Production Process	Production Process Name	
2	Process Classification	Production Process	Production Process Classification	
3	Status	Production Process	Status	
4	Production Item	Production Process	Production Item ID	
5	Specification	Production Process	Specification	
6	Quantity	Production Process	Quantity	
7	Unit	Production Process	Unit	
8	Site	Production Process	Site ID	
9	Sequence Number	Production Procedure	Sequence Number	*1
10	Procedure Name	Production Procedure	Production Procedure Name	*1
11	Procedure Classification	Production Procedure	Production Procedure Classification	*1
12	Procedure Content	Production Procedure	Content	*1

No.	Item Name	Entity	Attribute	Relationship
13	Procedure Quantity	Production Procedure	Quantity	*1
14	Unit	Production Procedure	Unit	*1
15	Sequence Number	Production Entry	Sequence Number	*2
16	Entry Classification	Production Entry	Production Entry Classification	*2
17	Entry Content	Production Entry	Entry Content	*2
18	Standard Value	Production Entry	Standard Value	*2
19	Unit	Production Entry	Unit	*2
20	Target Location	Production Entry	Item Location ID	*2

Maintenance Procedure Document

A document showing the concrete procedure and precautions for equipment inspection, repair, and preventive maintenance. Used to improve equipment operation rate and prevent failures.

Figure 21 Maintenance Procedure Document

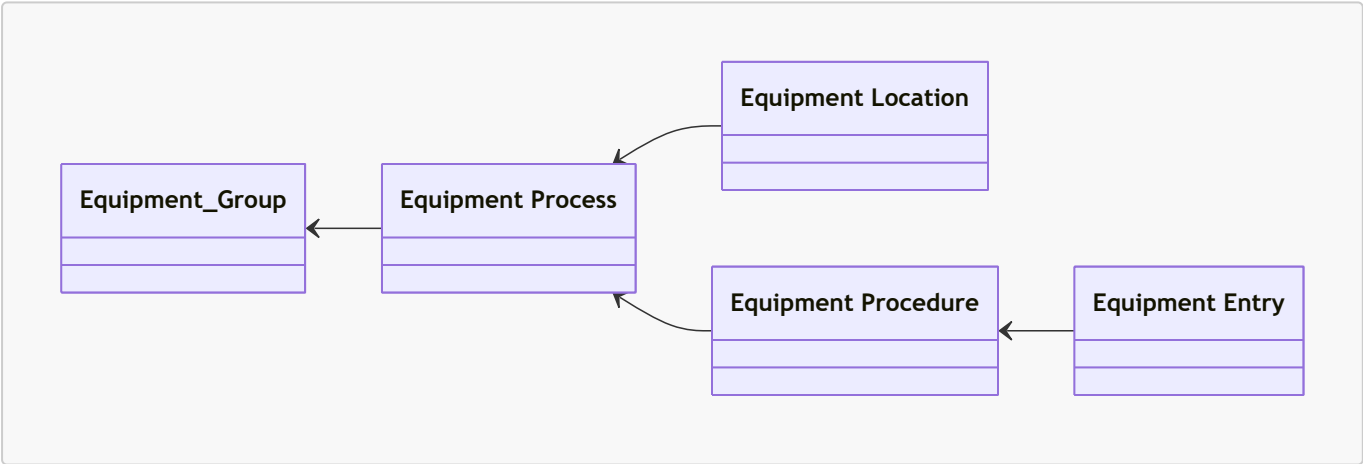


Table 44 Maintenance Procedure Document

No.	Item Name	Entity	Attribute	Relationship
1	Process Name	Equipment Process	Production Process Name	
2	Process Classification	Equipment Process	Production Process Classification	
3	Status	Equipment Process	Status	
4	Production Item	Equipment Process	Production Item ID	
5	Specification	Equipment Process	Specification	
6	Quantity	Equipment Process	Quantity	

No.	Item Name	Entity	Attribute	Relationship
7	Unit	Equipment Process	Unit	
8	Site	Equipment Process	Site ID	
9	Sequence Number	Equipment Procedure	Sequence Number	*1
10	Procedure Name	Equipment Procedure	Production Procedure Name	*1
11	Procedure Classification	Equipment Procedure	Production Procedure Classification	
12	Procedure Content	Equipment Procedure	Content	
13	Procedure Quantity	Equipment Procedure	Quantity	*1
14	Unit	Equipment Procedure	Unit	*1
15	Sequence Number	Equipment Entry	Sequence Number	*2
16	Entry Classification	Equipment Entry	Production Entry Classification	*2
17	Entry Content	Equipment Entry	Entry Content	*2
18	Standard Value	Equipment Entry	Standard Value	*2
19	Unit	Equipment Entry	Unit	*2
20	Target Location	Equipment Entry	Equipment Location ID	*2

Reference/Standard Information (Standard Information)

Reference information holding standard values and specifications for quality, process, equipment, etc. Changes infrequently and functions as the judgment criterion for other information.

Table 45 Reference/Standard Information

No.	Information Name	Product	Process	Equipment	Description
1	Quality Control Standard Table	○	○		A standard showing the control items, target values, and monitoring method for maintaining/improving quality. Aims to unify process quality.
2	Equipment Maintenance Standard Table		○	○	A standard prescribing the inspection items, cycle, allowable values, and maintenance method of equipment. Used to prevent failures and maintain operation rate.

No.	Information Name	Product	Process	Equipment	Description
3	Environment/Safety Standard Table	○	○	○	A table summarizing standards for the work environment, safety measures, chemical substance management, etc. Ensures environmental conservation and worker safety.

Quality Control Standard Table

A standard showing the control items, target values, and monitoring method for maintaining/improving quality. Aims to unify process quality.

Figure 22 Quality Control Standard Table

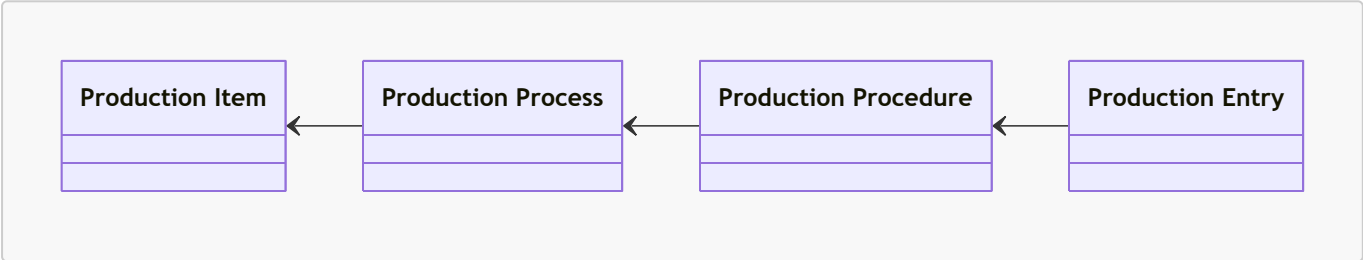


Table 46 Quality Control Standard Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Item Name	Production Item	Production Item Name	
3	Item Classification	Production Item	Production Item Classification	
4	Status	Production Item	Status	
5	Drawing Number	Production Item	Drawing Number	
6	Quality Specification	Production Item	Specification	
7	Sequence Number	Production Procedure	Sequence Number	*1
8	Procedure Name	Production Procedure	Production Procedure Name	*1
9	Procedure Classification	Production Procedure	Production Procedure Classification	*1
10	Procedure Content	Production Procedure	Content	*1
11	Control Method	Production Procedure	Control Method	*1

No.	Item Name	Entity	Attribute	Relationship
12	Inspection Method	Production Procedure	Inspection Method	*1
13	Standard Quantity	Production Procedure	Quantity	*1
14	Unit	Production Procedure	Unit	*1
15	Sequence Number	Production Entry	Sequence Number	*2
16	Entry Classification	Production Entry	Production Entry Classification	*2
17	Entry Name	Production Entry	Production Entry Name	*2
18	Inspection Specification	Production Entry	Specification	*2
19	Inspection Standard	Production Entry	Quantity	*2
20	Unit	Production Entry	Unit	*2
21	Item Location	Production Entry	Item Location ID	*2

Equipment Maintenance Standard Table

A standard prescribing the inspection items, cycle, allowable values, and maintenance method of equipment. Used to prevent failures and maintain operation rate.

Figure 23 Equipment Maintenance Standard Table

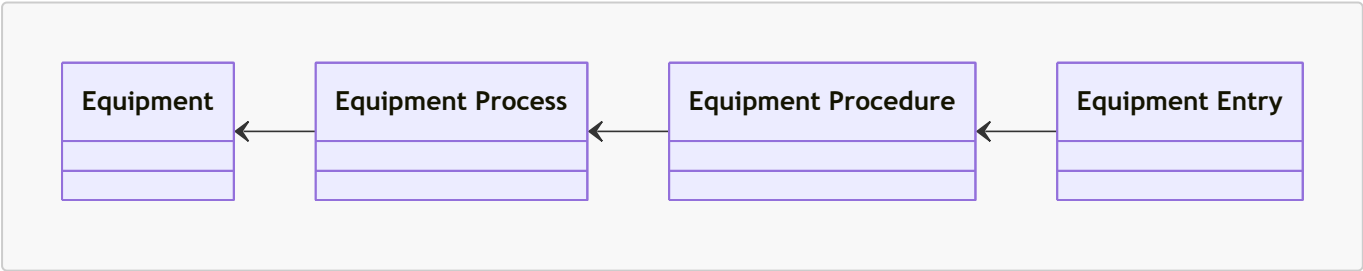


Table 47 Equipment Maintenance Standard Table

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Number	Production Equipment	Equipment Number	
2	Equipment Name	Production Equipment	Production Equipment Name	
3	Equipment Classification	Production Equipment	Production Equipment Classification	

No.	Item Name	Entity	Attribute	Relationship
4	Status	Production Equipment	Status	
5	Quality Specification	Production Equipment	Specification	
6	Sequence Number	Equipment Procedure	Sequence Number	*1
7	Procedure Name	Equipment Procedure	Equipment Procedure Name	*1
8	Procedure Classification	Equipment Procedure	Equipment Procedure Classification	*1
9	Procedure Content	Equipment Procedure	Content	*1
10	Control Method	Equipment Procedure	Control Method	*1
11	Inspection Method	Equipment Procedure	Inspection Method	*1
12	Standard Quantity	Equipment Procedure	Quantity	*1
13	Unit	Equipment Procedure	Unit	*1
14	Sequence Number	Equipment Entry	Sequence Number	*2
15	Entry Classification	Equipment Entry	Equipment Entry Classification	*2
16	Entry Name	Equipment Entry	Equipment Entry Name	*2
17	Inspection Specification	Equipment Entry	Specification	*2
18	Inspection Standard	Equipment Entry	Quantity	*2
19	Unit	Equipment Entry	Unit	*2
20	Equipment Location	Equipment Entry	Equipment Location ID	*2

Environment/Safety Standard Table

A table summarizing standards for the work environment, safety measures, chemical substance management, etc. Ensures environmental conservation and worker safety.

Figure 24 Environment/Safety Standard Table

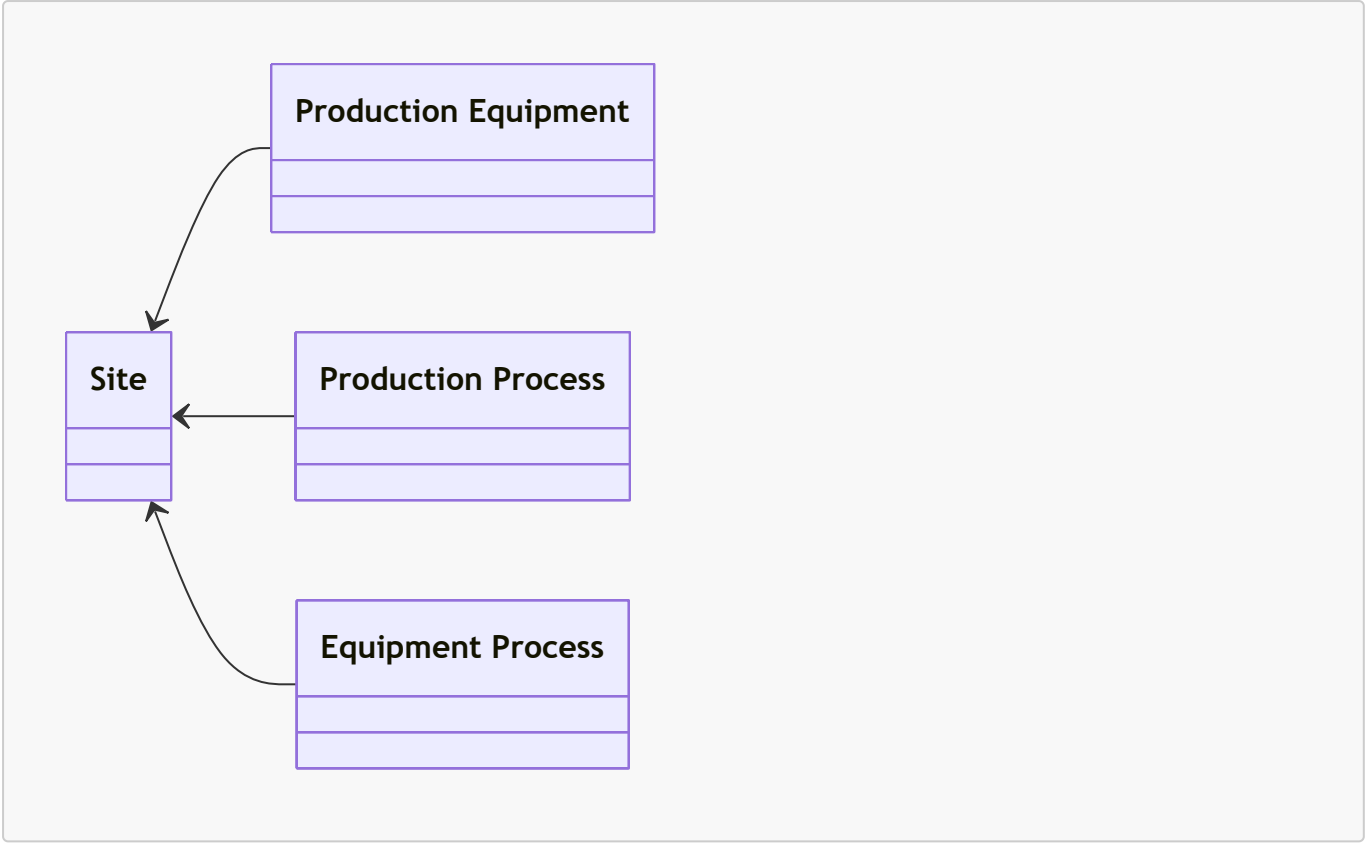


Table 48 Environment/Safety Standard Table

No.	Item Name	Entity	Attribute	Relationship
1	Site Name	Site	Site Name	
2	Safety Specification	Site	Specification	
3	Location	Site	Location	
4	Status	Site	Status	
5	Production Process	Production Process	Production Process Name	*1
6	Process Classification (Production Process)	Production Process	Production Process Classification	*1
7	Measurement Specification	Production Process	Specification	*1
8	Standard Value	Production Process	Quantity	*1
9	Unit	Production Process	Unit	*1
10	Maintenance Process	Equipment Process	Equipment Process Name	*2
11	Process Classification (Maintenance Process)	Equipment Process	Equipment Process Classification	*2

No.	Item Name	Entity	Attribute	Relationship
12	Required Specification (Maintenance Process)	Equipment Process	Specification	*2
13	Standard Value (Maintenance Process)	Equipment Process	Quantity	*2
14	Unit (Maintenance Process)	Equipment Process	Unit	*2

7.3. Technical Information

Deployment Information (Deployment Information)

Transformation-type information that digs deeper into information about products, processes, and equipment along more individualized management methods, and presents a new view by deploying that information and showing correspondences among it. Each has its own defined notation rules (e.g., QFD, FMEA, etc.). Table 49 Deployment Information

No.	Information Name	Product	Process	Equipment	Description
1	Quality Function Deployment Table (QFD)	○	○	○	A table that systematically deploys customer requirements into design elements and evaluates their importance. Ensures design quality is reliably reflected.
2	Fault Tree Analysis Table (FTA)		○	○	An analysis diagram that logically pursues the causes of a failure in a tree structure. Clarifies the root cause of a serious failure.
3	Design FMEA	○	○		A method for predicting potential defects at the design stage and organizing causes, effects, and countermeasures. Reduces design risk.
4	Process FMEA		○	○	A method for analyzing possible defects or abnormalities in the manufacturing process, prioritizing them, and examining countermeasures. Strengthens process quality.

Quality Function Deployment Table (QFD)

A table that systematically deploys customer requirements into design elements and evaluates their importance. Ensures design quality is reliably reflected.

Figure 25 Quality Function Deployment Table

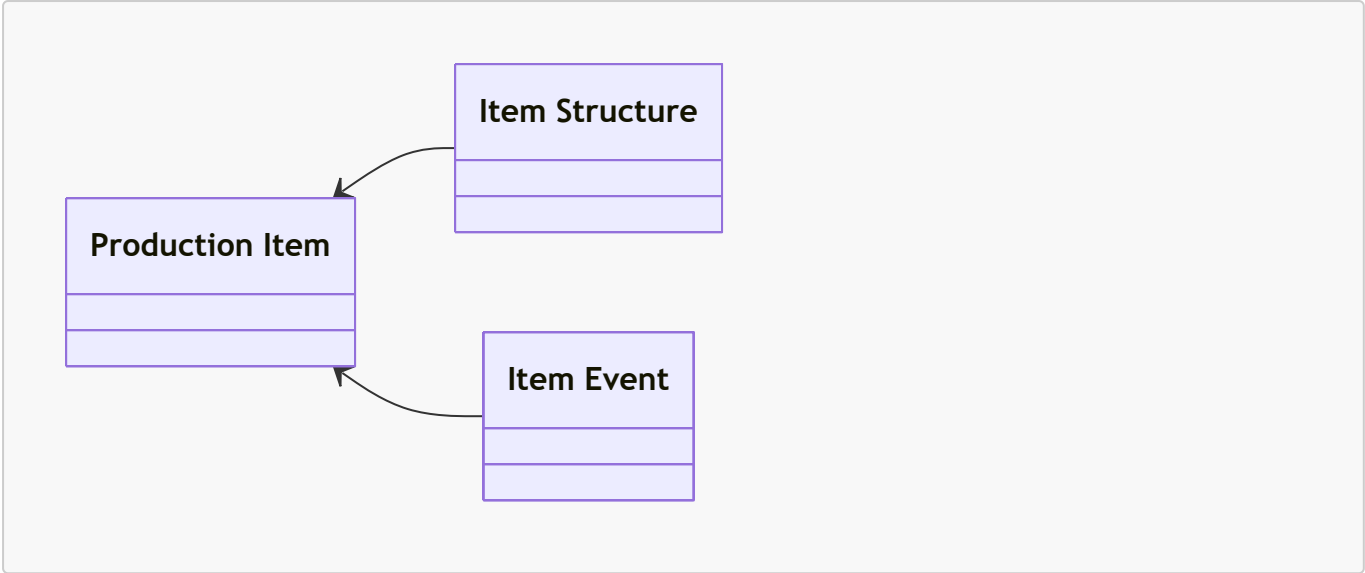


Table 50 Quality Function Deployment Table (QFD)

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Item Name	Production Item	Item Name	
3	Item Classification	Production Item	Item Classification	
4	Status	Production Item	Status	
5	Specification	Production Item	Specification	
6	Function Name	Item Function	Item Function Name	
7	Quality Characteristic	Item Function	Quality Function Classification	
8	Standard Quantity	Item Function	Quantity	
9	Unit	Item Function	Unit	
10	Requirement Name	Item Event	Item Event Name	
11	Requirement Classification	Item Event	Item Event Classification	
12	Status	Item Event	Status	
13	Correlation Degree	Item Event × Item Function	Correlation Degree	
14	Importance	Item Event × Item Function	Importance	

Fault Tree Analysis Table

An analysis diagram that logically pursues the causes of a failure in a tree structure. Clarifies the root cause of a serious failure.

Figure 26 Fault Tree Analysis Table

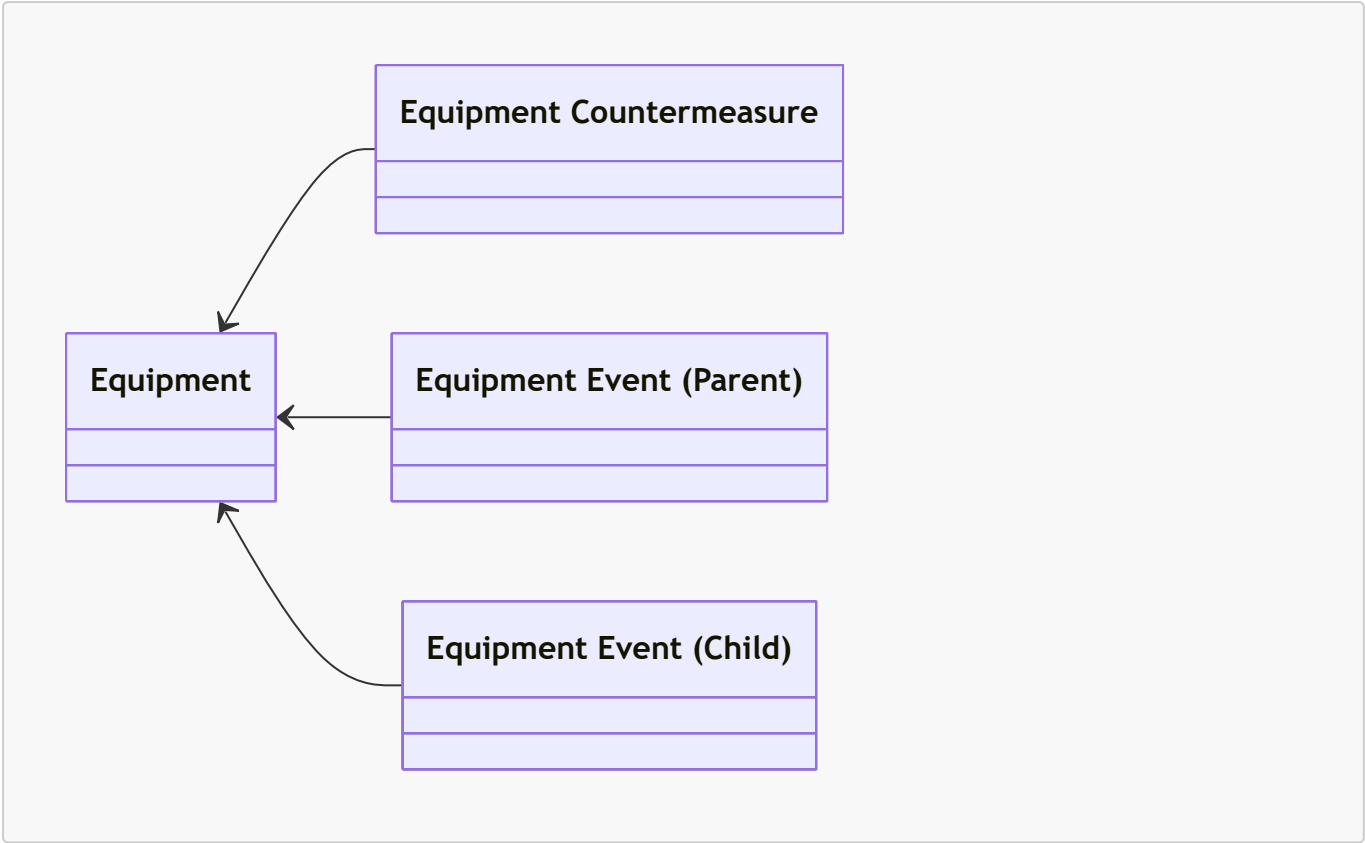


Table 51 Fault Tree Analysis Table (FTA)

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Number	Production Equipment	Equipment Number	
2	Equipment Name	Production Equipment	Production Equipment Name	
3	Equipment Classification	Production Equipment	Production Equipment Classification	
4	Specification	Production Equipment	Specification	
5	Status	Production Equipment	Status	
6	Event Name	Equipment Event	Equipment Event Name	*1
7	Logical Relationship	Equipment Event	Logical Relationship	*1
8	Probability	Equipment Event	Probability	*1
9	Fact	Equipment Event	Fact	*1
10	Issue	Equipment Event	Issue	*1
11	Cause	Equipment Event	Cause	*1
12	Event Name (Child)	Equipment Event (Child)	Equipment Event Name	*2
13	Logical Relationship (Child)	Equipment Event (Child)	Logical Relationship	*2

No.	Item Name	Entity	Attribute	Relationship
14	Probability (Child)	Equipment Event (Child)	Probability	*2
15	Fact (Child)	Equipment Event (Child)	Fact	*2
16	Issue (Child)	Equipment Event (Child)	Issue	*2
17	Cause (Child)	Equipment Event (Child)	Cause	*2
18	Countermeasure Name	Equipment Countermeasure	Equipment Countermeasure Name	*3
19	Status	Equipment Countermeasure	Status	*3
20	Countermeasure	Equipment Countermeasure	Countermeasure	*3
21	Result	Equipment Countermeasure	Result	*3
22	Completion Date/Time	Equipment Countermeasure	Completion Date/Time	*3

Design FMEA

A method for predicting potential defects at the design stage and organizing causes, effects, and countermeasures. Reduces design risk.

Figure 27 Design FMEA

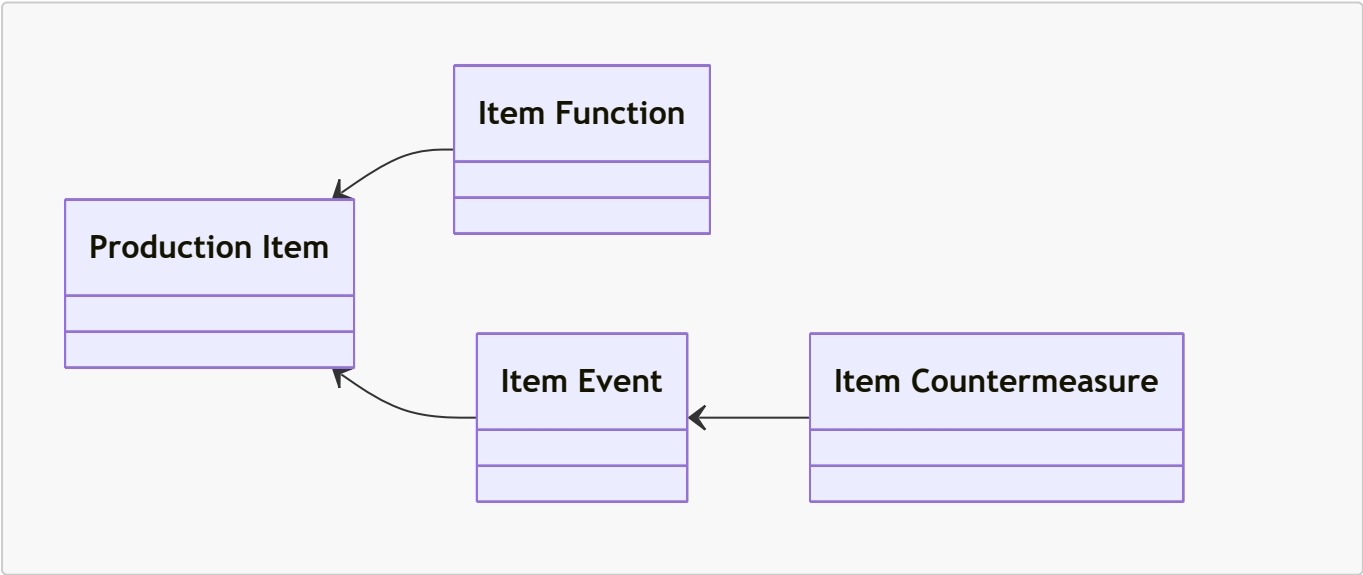


Table 52 Design FMEA

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Item Name	Production Item	Item Name	

No.	Item Name	Entity	Attribute	Relationship
3	Item Classification	Production Item	Item Classification	
4	Status	Production Item	Status	
5	Drawing Number	Production Item	Drawing Number	
6	Specification	Production Item	Specification	
7	Item Function Name	Item Function	Item Function Name	*1
8	Status	Item Function	Status	*1
9	Failure Mode	Item Function	Failure Mode	*1
10	Effect	Item Function	Effect	*1
11	Cause	Item Function	Cause	*1
12	Control	Item Function	Control	*1
13	Severity	Item Function	Severity	*1
14	Occurrence Frequency	Item Function	Occurrence Frequency	*1
15	Detectability	Item Function	Detectability	*1
16	Priority (RPN)	Item Function	Priority (RPN)	*1
17	Business Name	Item Event	Business Name	*
18	Business Classification	Item Event	Business Classification	*
19	Status	Item Event	Status	*
20	Fact	Item Event	Fact	*
21	Issue	Item Event	Issue	*
22	Cause	Item Event	Cause	*
23	Sequence Number	Item Countermeasure	Sequence Number	*2
24	Countermeasure Name	Item Countermeasure	Process Countermeasure Name	*2
25	Countermeasure Classification	Item Countermeasure	Process Countermeasure Classification	*2
26	Status	Process Countermeasure	Status	*2
27	Countermeasure	Item Countermeasure	Countermeasure	*2
28	Result	Item Countermeasure	Result	*2
29	Start Date/Time	Item Countermeasure	Start Date/Time	*2
30	Completion Date/Time	Item Countermeasure	Completion Date/Time	*2

No.	Item Name	Entity	Attribute	Relationship
31	Worker	Item Countermeasure	Worker ID	*2

Process FMEA

A method for analyzing possible defects or abnormalities in the manufacturing process, prioritizing them, and examining countermeasures. Strengthens process quality.

Figure 28 Process FMEA

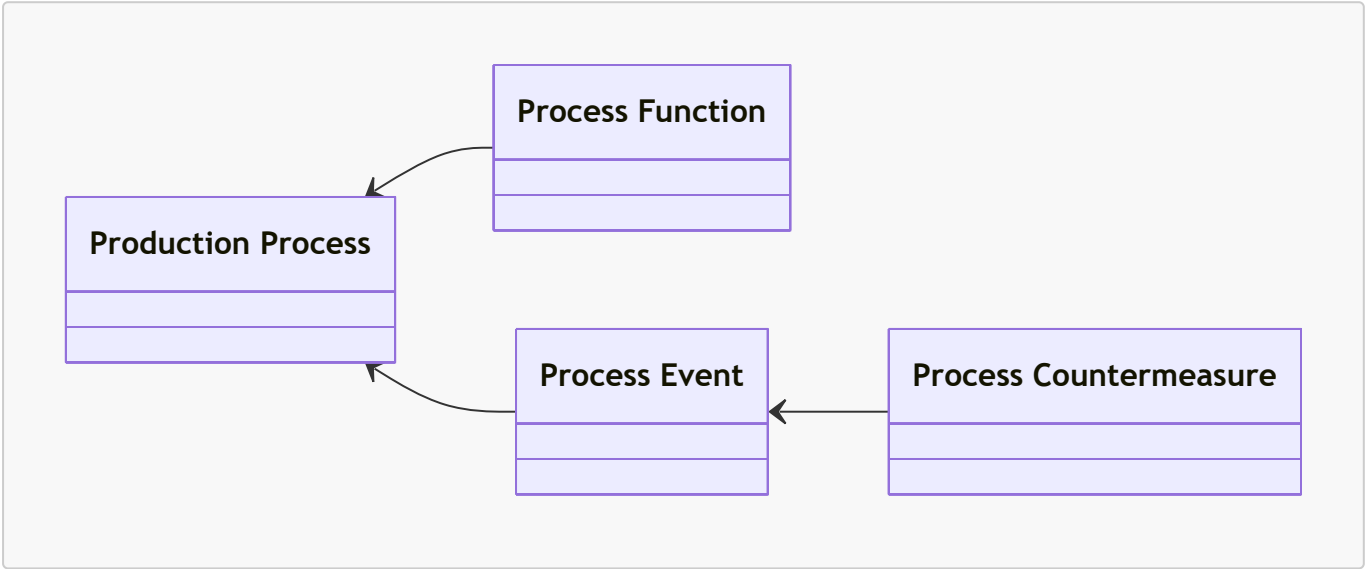


Table 53 Process FMEA

No.	Item Name	Entity	Attribute	Relationship
1	Process Number	Production Process	Item Number	
2	Process Name	Production Process	Item Name	
3	Process Classification	Production Process	Item Classification	
4	Status	Production Process	Status	
5	Drawing Number	Production Process	Drawing Number	
6	Specification	Production Process	Specification	
7	Function Name	Process Function	Item Function Name	*1
8	Status	Process Function	Status	*1
9	Failure Mode	Process Function	Failure Mode	*1
10	Effect	Process Function	Effect	*1
11	Cause	Process Function	Cause	*1
12	Control	Process Function	Control	*1
13	Severity	Process Function	Severity	*1

No.	Item Name	Entity	Attribute	Relationship
14	Occurrence Frequency	Process Function	Occurrence Frequency	*1
15	Detectability	Process Function	Detectability	*1
16	Priority (RPN)	Process Function	Priority (RPN)	*1
17	Business Name	Process Event	Business Name	*
18	Business Classification	Process Event	Business Classification	*
19	Status	Process Event	Status	*
20	Fact	Process Event	Fact	*
21	Issue	Process Event	Issue	*
22	Cause	Process Event	Cause	*
23	Sequence Number	Process Countermeasure	Sequence Number	*2
24	Countermeasure Name	Process Countermeasure	Process Countermeasure Name	*2
25	Countermeasure Classification	Process Countermeasure	Process Countermeasure Classification	*2
26	Status	Process Countermeasure	Status	*2
27	Countermeasure	Process Countermeasure	Countermeasure	*2
28	Result	Process Countermeasure	Result	*2
29	Start Date/Time	Process Countermeasure	Start Date/Time	*2
30	Completion Date/Time	Process Countermeasure	Completion Date/Time	*2
31	Worker	Process Countermeasure	Worker ID	*2

Linkage Information (Item/Material Identification Information)

Tracking information consisting of the location, status, and identification information of parts or work-in-process. Primarily takes the form of physical item tags, barcodes, ID tags, etc.

Table 54 Linkage Information

No.	Information Name	Product	Process	Equipment	Description
1	Bill of Assets (BOA)		○	○	Data representing the structure, capacity, and equipment hierarchy of a plant or equipment. Positioned as the plant model that executes the BOP.
2	Value Stream Map (VSM)		○		A diagram that visualizes the flow of things and the flow of information between processes, identifying waste and bottlenecks. The basis for improvement activities.
3	Traceability Table	○	○	○	A table recorded so that parts, processes, and work history can be traced. Essential for tracing the cause when a quality problem occurs.

Bill of Assets (BOA)

Data representing the structure, capacity, and equipment hierarchy of a plant or equipment. Positioned as the plant model that executes the BOP.

Figure 29 Bill of Assets

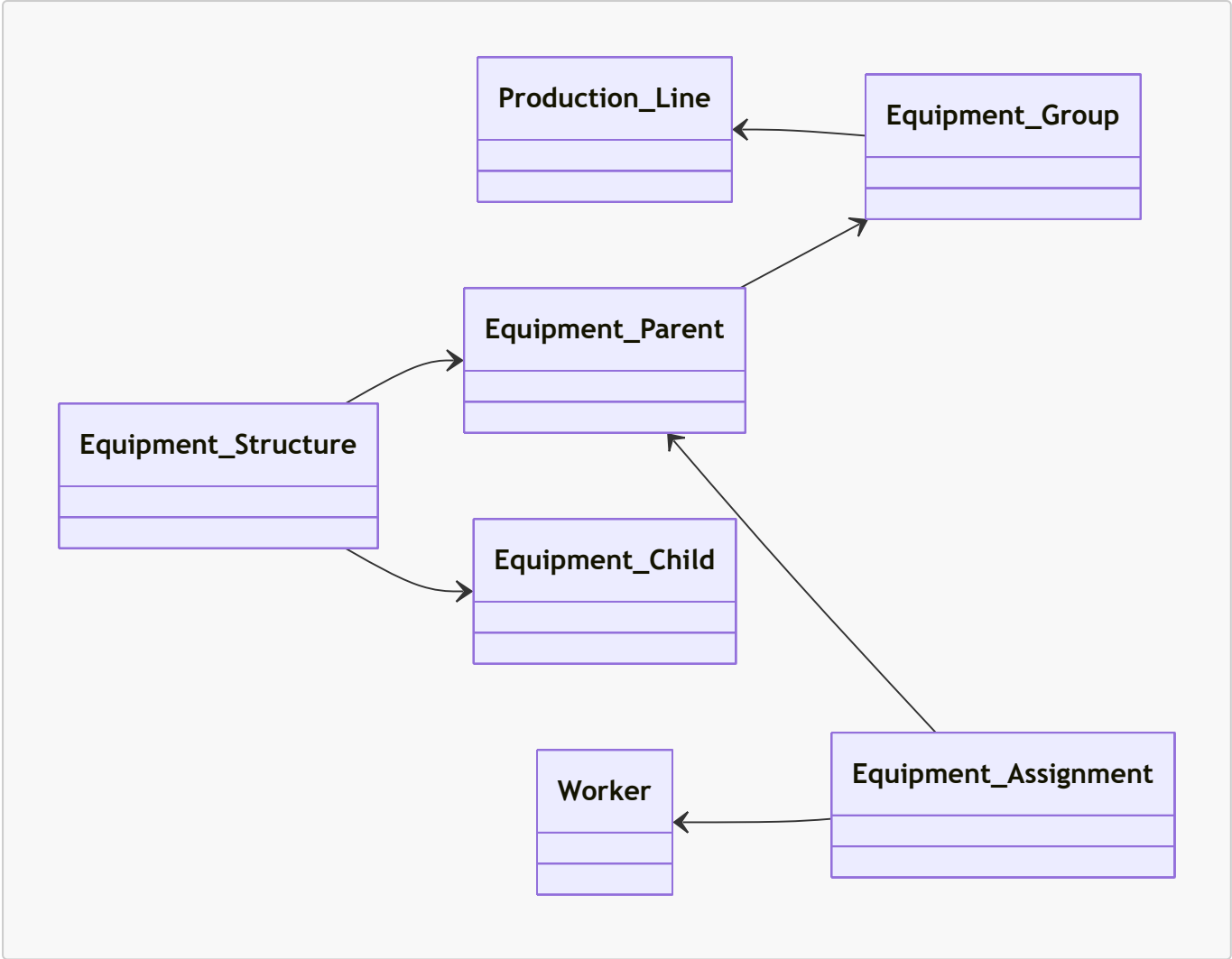


Table 39 Bill of Assets (BOA)

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Name	Production Equipment	Production Equipment Name	*
2	Equipment Number	Production Equipment	Equipment Number	*
3	Location	Production Equipment	Location	*
4	Specification	Production Equipment	Specification	*
5	Standard Capacity	Production Equipment	Standard Capacity	*
6	Unit	Production Equipment	Unit	*
7	Area	Production Line	Area ID	
8	Production Line	Production Line	Production Line Name	

No.	Item Name	Entity	Attribute	Relationship
9	Equipment Classification	Equipment Group	Equipment Group Classification	
10	Equipment Group	Equipment Group	Equipment Group Name	
11	Equipment Number (Child)	Equipment Structure	Child Production Equipment ID	*1
12	Equipment Name (Child)	Equipment Structure	Equipment Structure Name	*1
13	Specification (Child)	Equipment Structure	Specification	*1
14	Quantity per Unit	Equipment Structure	Quantity per Unit	*1
15	Unit (Child)	Equipment Structure	Unit	*1
16	Jig/Tool Name	Jig/Tool	Jig/Tool Name	*2
17	Specification (Jig/Tool)	Jig/Tool	Specification	*2
18	Quantity (Jig/Tool)	Jig/Tool	Quantity	*2
19	Unit (Jig/Tool)	Jig/Tool	Unit	*2
20	Location (Jig/Tool)	Jig/Tool	Location	*2
21	Worker Skill Name	Worker Skill	Worker Skill Name	*3
22	Worker Number	Worker Skill	Worker ID	*3
23	Required Skill	Worker Skill	Specification	*3
24	Quantity (Worker)	Worker Skill	Quantity	*3
25	Unit (Worker)	Worker Skill	Unit	*3

Value Stream Map

A diagram that visualizes the flow of things and the flow of information between processes, identifying waste and bottlenecks. The basis for improvement activities.

Figure 30 Value Stream Map

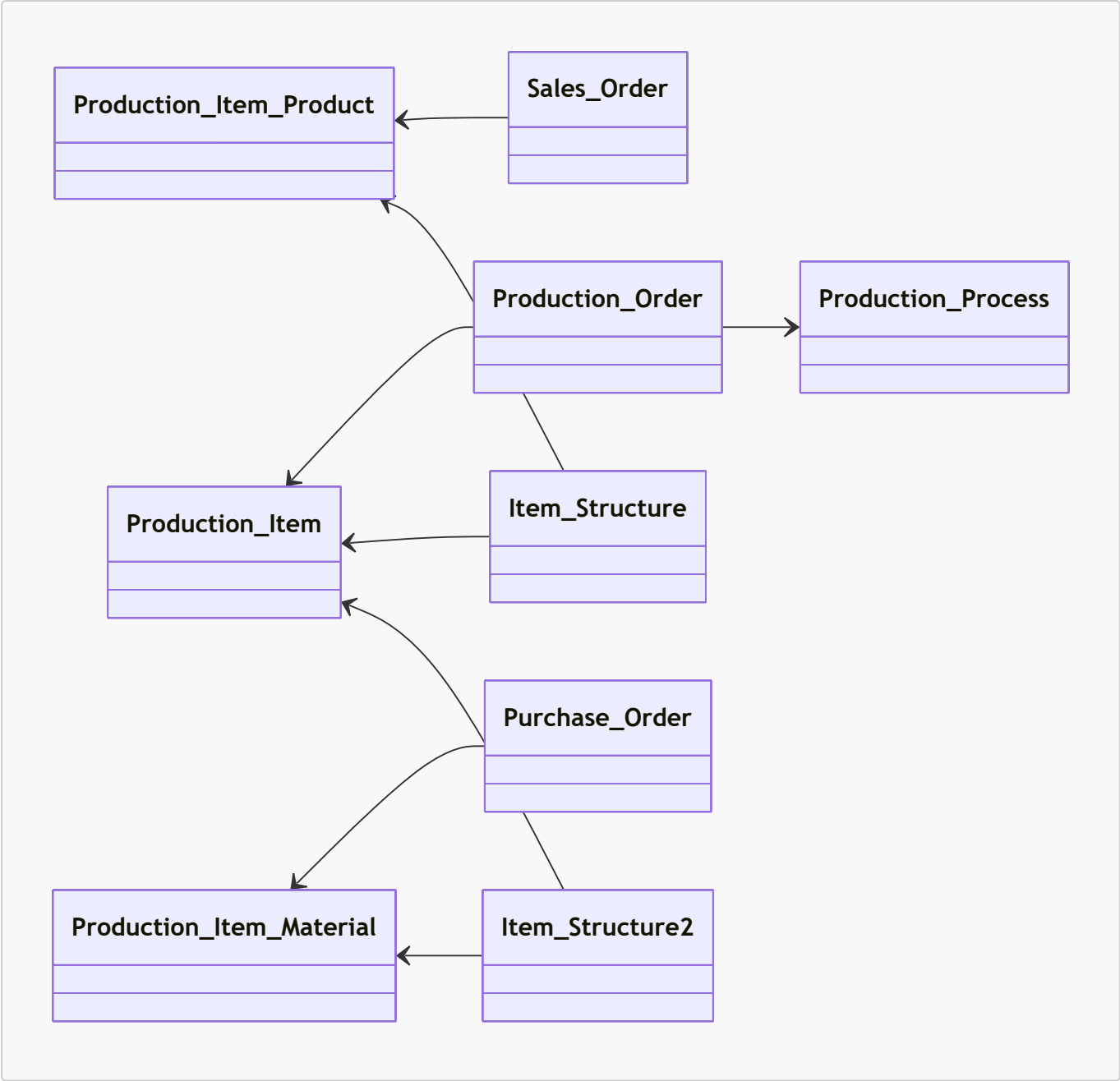


Table 55 Value Stream Map (VSM)

No.	Item Name	Entity	Attribute	Relationship
1	Production Item	Sales Order	Production Item ID	
2	Shipment Date/Time	Sales Order	Shipment Date/Time	
3	Quantity	Sales Order	Quantity	
4	Unit	Sales Order	Unit	
5	Sales Order	Production Order	Sales Order ID	*1
6	Production Item	Production Order	Production Item	*1
7	End Date/Time	Production Order	End Date/Time	*1
8	Production Process	Production Order	Production Process ID	*1

No.	Item Name	Entity	Attribute	Relationship
9	Location	Production Order	Location	*1
10	Quantity	Production Order	Quantity	*1
11	Unit	Production Order	Unit	*1
12	Production Order	Purchase Order	Production Order ID	*2
13	Production Item	Purchase Order	Production Item	*2
14	Receipt Date/Time	Purchase Order	Receipt Date/Time	*2
15	Quantity	Purchase Order	Quantity	*2
16	Unit	Purchase Order	Unit	*2
17	Item Number	Production Item	Item Number	
18	Item Name	Production Item	Production Item Name	
19	Status	Production Item	Status	
20	Quantity	Production Item	Quantity	
21	Unit	Production Item	Unit	
22	Process Number	Production Process	Process Number	*1
23	Process Name	Production Process	Production Process Name	*1
24	Status	Production Process	Status	*1
25	Quantity	Production Process	Quantity	*1
26	Unit	Production Process	Unit	*1
27	Parent Production Item	Item Structure	Parent Production Item ID	*2
28	Child Production Item	Item Structure	Child Production Item ID	*2
29	Status	Item Structure	Status	*2
30	Quantity per Unit	Item Structure	Quantity per Unit	*2
31	Unit	Item Structure	Unit	*2

Traceability Table

A table recorded so that parts, processes, and work history can be traced. Essential for tracing the cause when a quality problem occurs.

Figure 31 Traceability Table

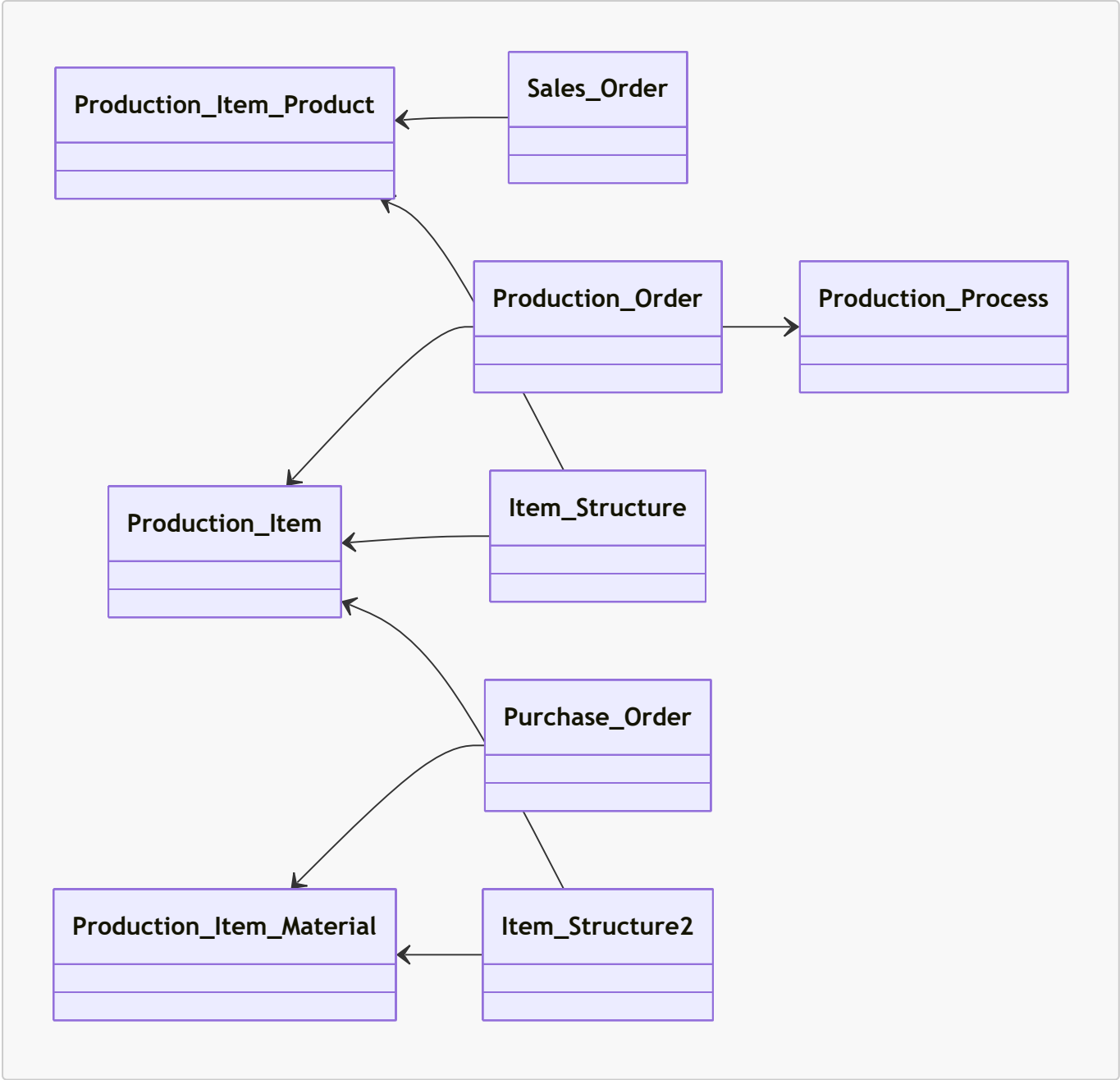


Table 56 Traceability Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Item	Sales Order	Production Item ID	
2	Shipment Date/Time	Sales Order	Shipment Date/Time	
3	Quantity	Sales Order	Quantity	
4	Unit	Sales Order	Unit	
5	Sales Order	Production Order	Sales Order ID	*1
6	Production Item	Production Order	Production Item	*1
7	End Date/Time	Production Order	End Date/Time	*1
8	Production Process	Production Order	Production Process ID	*1

No.	Item Name	Entity	Attribute	Relationship
9	Location	Production Order	Location	*1
10	Quantity	Production Order	Quantity	*1
11	Unit	Production Order	Unit	*1
12	Production Order	Purchase Order	Production Order ID	*2
13	Production Item	Purchase Order	Production Item	*2
14	Receipt Date/Time	Purchase Order	Receipt Date/Time	*2
15	Quantity	Purchase Order	Quantity	*2
16	Unit	Purchase Order	Unit	*2
17	Item Number	Production Item	Item Number	
18	Item Name	Production Item	Production Item Name	
19	Status	Production Item	Status	
20	Quantity	Production Item	Quantity	
21	Unit	Production Item	Unit	
22	Process Number	Production Process	Process Number	*1
23	Process Name	Production Process	Production Process Name	*1
24	Status	Production Process	Status	*1
25	Quantity	Production Process	Quantity	*1
26	Unit	Production Process	Unit	*1
27	Parent Production Item	Item Structure	Parent Production Item ID	*2
28	Child Production Item	Item Structure	Child Production Item ID	*2
29	Status	Item Structure	Status	*2
30	Quantity per Unit	Item Structure	Quantity per Unit	*2
31	Unit	Item Structure	Unit	*2

Quality Information (Instruction Information)

Execution-instruction-type information specifying the content to be executed on the shop floor. Operational data that concretizes plan information, such as quantity, delivery date, and equipment. Inspections and checks are also covered. Table 57 Quality Information

No.	Information Name	Product	Process	Equipment	Description
1	Product Inspection	○			A table recording the inspection items, criteria, and results for a finished product.

No.	Information Name	Product	Process	Equipment	Description
	Table				Used to manage quality assurance and shipment judgment.
2	Material Inspection Table	○			A table recording the inspection items, specifications, and results for received materials. Used to ensure quality and prevent defective materials from entering the process.
3	Equipment Inspection Table			○	A table recording the inspection items, cycle, allowable values, and results for equipment/machinery. Used to prevent failures and manage stable operation.

Product Inspection Table

A table recording the inspection items, criteria, and results for a finished product. Used to manage quality assurance and shipment judgment.

Figure 32 Product Inspection Table

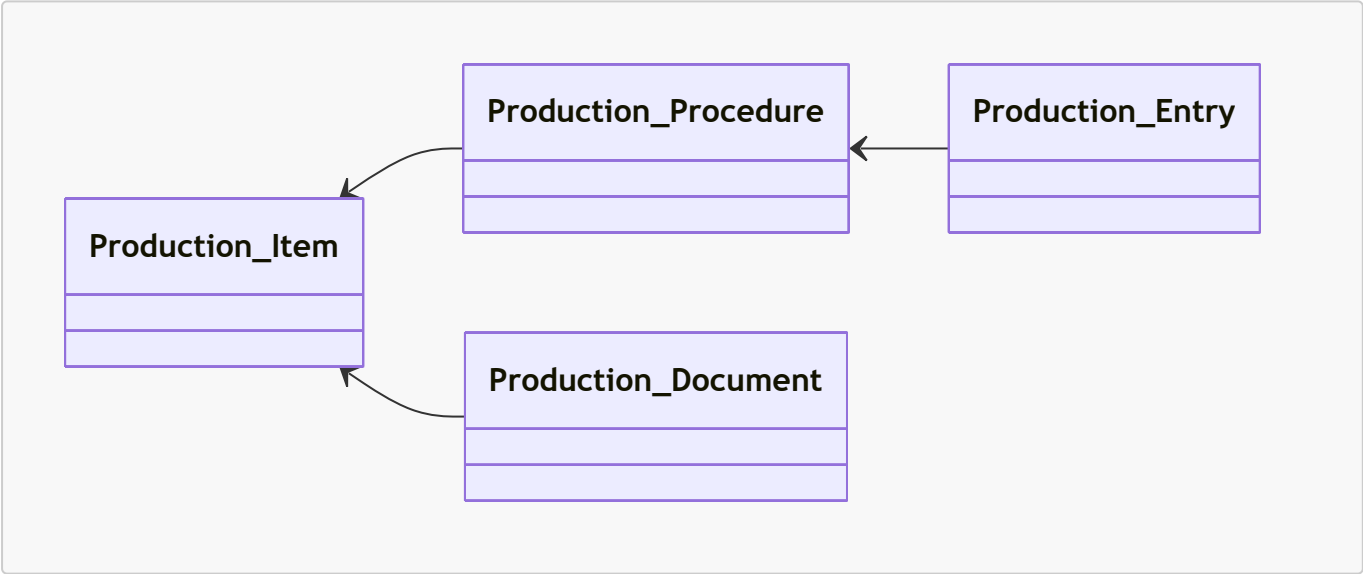


Table 59 Product Inspection Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Drawing Number	Production Item	Drawing Number	
3	Item Name	Production Item	Production Item Name	
4	Specification	Production Item	Specification	
5	Inspection Drawing	Item Document	File	

No.	Item Name	Entity	Attribute	Relationship
6	Sequence Number	Production Procedure	Sequence Number	*
7	Procedure Name	Production Procedure	Production Procedure Name	*1
8	Procedure Classification	Production Procedure	Production Procedure Classification	*1
9	Control Method	Production Procedure	Control Method	*1
10	Inspection Method	Production Procedure	Inspection Method	*1
11	Procedure Content	Production Procedure	Procedure Content	*1
12	Standard Quantity	Production Procedure	Quantity	*1
13	Unit	Production Procedure	Unit	*1
14	Sequence Number	Production Entry	Sequence Number	*2
15	Entry Name	Production Entry	Production Entry Name	*2
16	Entry Classification	Production Entry	Production Entry Classification	*2
17	Location Name	Production Entry	Production Location ID	*2
18	Jig/Tool	Production Entry	Jig/Tool ID	*2
19	Entry Content	Production Entry	Specification	*2
20	Standard Value	Production Entry	Quantity	*2
21	Measured Value	Production Entry	Measured Value	*2
22	Unit	Production Entry	Unit	*2

Material Inspection Table

A table recording the inspection items, specifications, and results for received materials. Used to ensure quality and prevent defective materials from entering the process.

Figure 33 Material Inspection Table

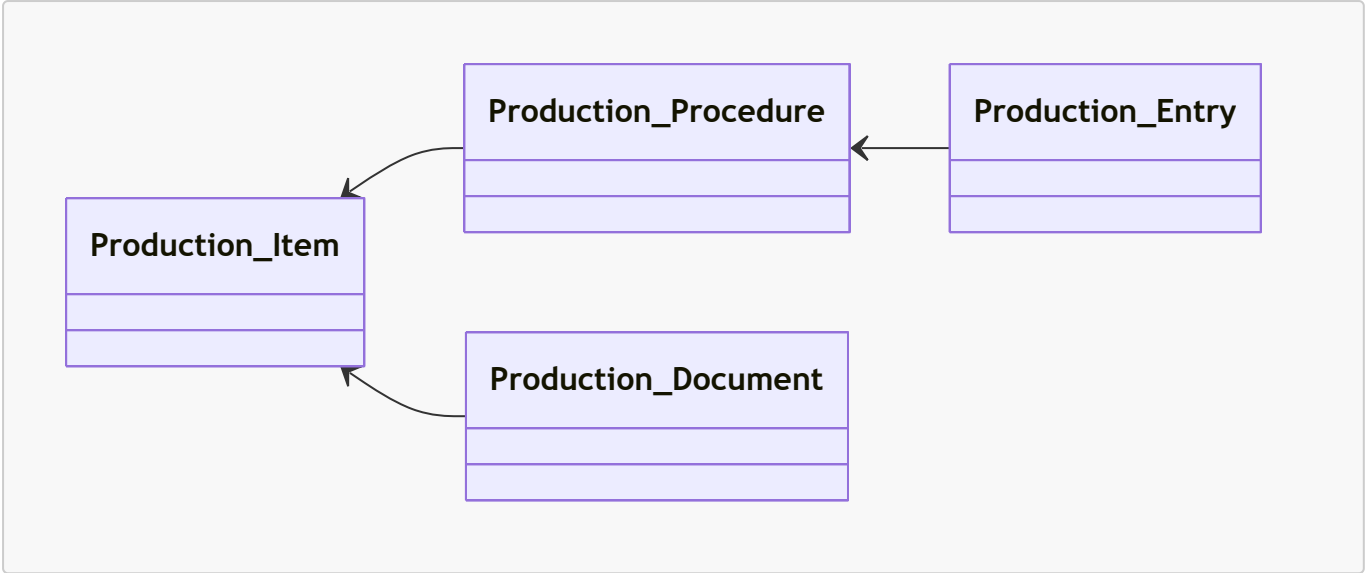


Table 60 Material Inspection Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Drawing Number	Production Item	Drawing Number	
3	Item Name	Production Item	Production Item Name	
4	Specification	Production Item	Specification	
5	Inspection Drawing	Item Document	File	
6	Sequence Number	Production Procedure	Sequence Number	*
7	Procedure Name	Production Procedure	Production Procedure Name	*1
8	Procedure Classification	Production Procedure	Production Procedure Classification	*1
9	Control Method	Production Procedure	Control Method	*1
10	Inspection Method	Production Procedure	Inspection Method	*1
11	Procedure Content	Production Procedure	Procedure Content	*1
12	Standard Quantity	Production Procedure	Quantity	*1
13	Unit	Production Procedure	Unit	*1
14	Sequence Number	Production Entry	Sequence Number	*2

No.	Item Name	Entity	Attribute	Relationship
15	Entry Name	Production Entry	Production Entry Name	*2
16	Entry Classification	Production Entry	Production Entry Classification	*2
17	Location Name	Production Entry	Production Location ID	*2
18	Jig/Tool	Production Entry	Jig/Tool ID	*2
19	Entry Content	Production Entry	Specification	*2
20	Standard Value	Production Entry	Quantity	*2
21	Measured Value	Production Entry	Measured Value	*2
22	Unit	Production Entry	Unit	*2

Equipment Inspection Table

A table recording the inspection items, cycle, allowable values, and results for equipment/machinery. Used to prevent failures and manage stable operation.

Figure 34 Equipment Inspection Table

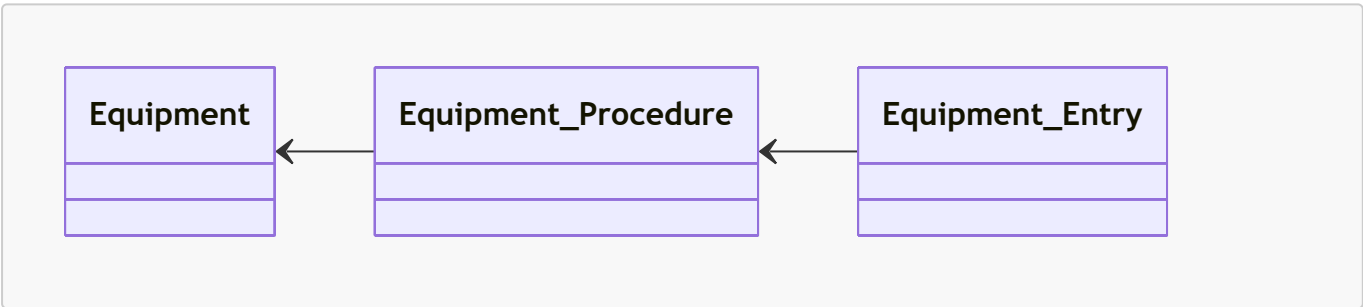


Table 58 Equipment Inspection Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Line	Production Equipment	Production Line ID	
2	Equipment Number	Production Equipment	Equipment Number	
3	Equipment Name	Production Equipment	Production Equipment Name	
4	Specification	Production Equipment	Specification	
5	Inspection Drawing	Equipment Document	File	
6	Sequence Number	Equipment Procedure	Sequence Number	*

No.	Item Name	Entity	Attribute	Relationship
7	Procedure Name	Equipment Procedure	Equipment Procedure Name	*1
8	Inspection Classification	Equipment Procedure	Equipment Procedure Classification	*1
9	Control Method	Equipment Procedure	Control Method	*1
10	Inspection Method	Equipment Procedure	Inspection Method	*1
11	Procedure Content	Equipment Procedure	Procedure Content	*1
12	Inspection Cycle	Equipment Procedure	Inspection Cycle	*1
13	Standard Quantity	Equipment Procedure	Quantity	*1
14	Unit	Equipment Procedure	Unit	*1
15	Sequence Number	Equipment Entry	Sequence Number	*2
16	Entry Name	Equipment Entry	Equipment Entry Name	*2
17	Entry Classification	Equipment Entry	Equipment Entry Classification	*2
18	Location Name	Equipment Entry	Equipment Location ID	*2
19	Jig/Tool	Equipment Entry	Jig/Tool ID	*2
20	Entry Content	Equipment Entry	Specification	*2
21	Standard Value	Equipment Entry	Quantity	*2
22	Measured Value	Equipment Entry	Measured Value	*2
23	Unit	Equipment Entry	Unit	*2

7.4. Management Information

Schedule Information

Information that organizes and manages, on a time axis, activities such as work, processes, and maintenance so that business or production activities are executed in a planned and reliable manner. By consistently visualizing everything from the overall plan down to the execution level, it facilitates grasping progress and coordinating, supporting delay prevention and effective use of resources.

Table 61 Schedule Information

No.	Information Name	Product	Process	Equipment	Description
1	Master Production Schedule (MPS)	○			A plan showing the production volume and timing of major items at the monthly-to-weekly level. Forms the basis for material and capacity planning.
2	Production Schedule (Detailed Schedule)	○	○		A plan showing the work sequence by process and by production line, on a daily or shift basis. Forms the basis of execution management for shop-floor work.
3	Maintenance Schedule Table		○	○	A table planning the timing, method, and required resources for equipment inspection, repair, and preventive maintenance. Management material for maintaining stable operation.
4	Work Breakdown Structure (WBS)		○		A table that hierarchically decomposes work and organizes process, period, and responsibility. Used for planning development and equipment introduction.

Master Production Schedule (MPS)

A plan showing the production volume and timing of major items at the monthly-to-weekly level. Forms the basis for material and capacity planning.

Figure 35 Master Production Schedule

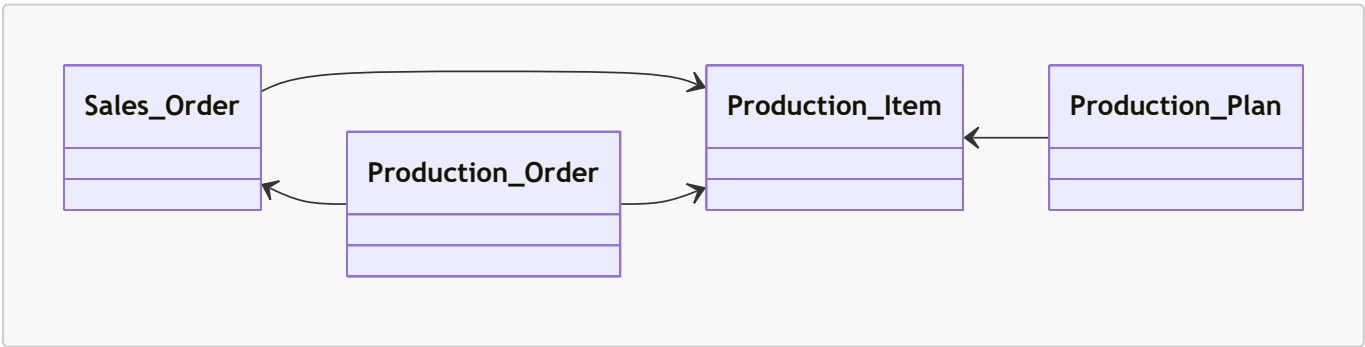


Table 62 Master Production Schedule (MPS)

No.	Item Name	Entity	Attribute	Relationship
1	Sales Order	Sales Order	Sales Order Name	
2	Content (Sales Order)	Sales Order × Date	Content (Sales Order)	**
3	Status	Production Order	Status	-1

No.	Item Name	Entity	Attribute	Relationship
4	Production Item	Production Order	Production Item ID	-1
5	Requested Date/Time	Production Order	Requested Date/Time	-1
6	Content	Production Order	Content	-1
7	Quantity	Production Order	Quantity	-1
8	Unit	Production Order	Unit	-1
9	Start Date/Time	Production Order	Start Date/Time	-1
10	End Date/Time	Production Order	End Date/Time	-1
11	Item Group	Item Group	Item Group ID	
12	Content (Item Group)	Item Group × Date	Content	**
13	Item Group	Production Plan	Item Group	-2
14	Planned Quantity	Production Plan	Planned Quantity	-2
15	Actual Quantity	Production Plan	Actual Quantity	-2
16	Capacity Quantity	Production Plan	Quantity	-2
17	Unit	Production Plan	Unit	-2

Production Schedule (Detailed Schedule)

A plan showing the work sequence by process and by production line, on a daily or shift basis. Forms the basis of execution management for shop-floor work.

Figure 36 Production Schedule (Detailed Schedule)

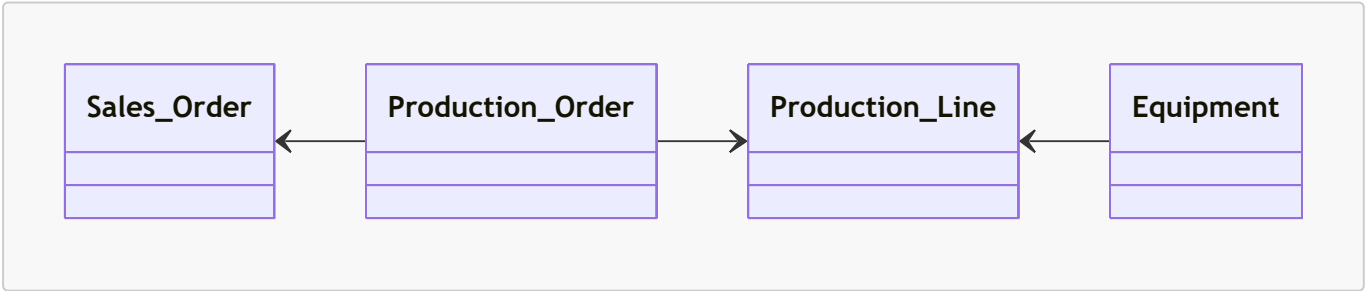


Table 63 Production Schedule (Detailed Schedule)

No.	Item Name	Entity	Attribute	Relationship
1	Status	Sales Order	Status	
2	Sales Order	Sales Order	Sales Order Name	
3	Specification	Sales Order	Specification	
4	Quantity	Sales Order	Quantity	

No.	Item Name	Entity	Attribute	Relationship
5	Unit	Sales Order	Unit	
6	Order Date/Time	Sales Order	Order Date/Time	
7	Shipment Date/Time	Sales Order	Shipment Date/Time	
8	Status	Production Order	Status	
9	Production Order	Production Order	Production Order Name	
10	Production Item	Production Order	Production Item ID	
11	Production Line	Production Order	Production Line ID	
12	Content	Production Order	Content	
13	Quantity	Production Order	Quantity	
14	Unit	Production Order	Unit	
15	Start Date/Time	Production Order	Start Date/Time	
16	End Date/Time	Production Order	End Date/Time	
17	Production Line Name	Production Line	Production Line Name	
18	Content	Production Line × Date	Content	
19	Equipment Number	Production Equipment	Equipment Number	
20	Equipment Name	Production Equipment	Production Equipment Name	
21	Specification	Production Equipment	Specification	
22	Standard Capacity	Production Equipment	Standard Capacity	
23	Unit	Production Equipment	Unit	

Maintenance Schedule Table

A table planning the timing, method, and required resources for equipment inspection, repair, and preventive maintenance. Management material for maintaining stable operation.

Figure 37 Maintenance Schedule Table

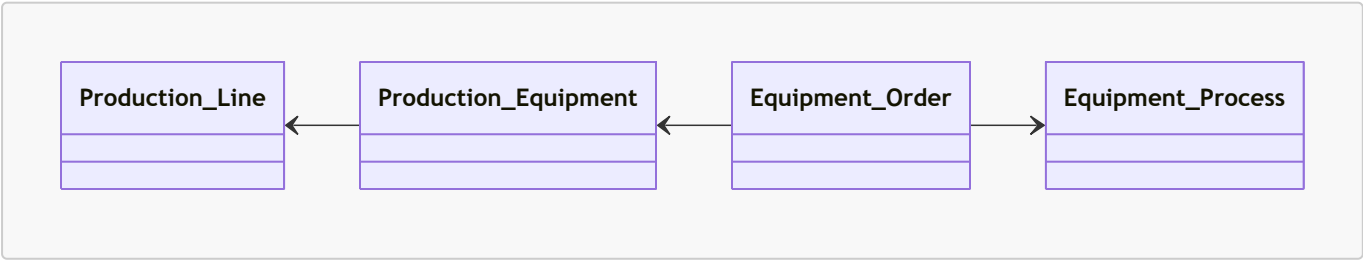


Table 64 Maintenance Schedule Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Line Name	Production Line	Production Line Name	
2	Content	Production Line × Date	Content	
3	Status	Production Equipment	Status	*1
4	Equipment Number	Production Equipment	Equipment Number	*1
5	Equipment Name	Production Equipment	Production Equipment Name	*1
6	Specification	Production Equipment	Specification	*1
7	Standard Capacity	Production Equipment	Standard Capacity	*1
8	Unit	Production Equipment	Unit	*1
9	Status	Equipment Order	Status	*2
10	Equipment Order Name	Equipment Order	Equipment Order Name	*2
11	Production Equipment	Equipment Order	Production Equipment	*2
12	Equipment Process	Equipment Order	Equipment Process	*2
13	Content	Equipment Order	Content	*2
14	Quantity	Equipment Order	Quantity	*2
15	Unit	Equipment Order	Unit	*2
16	Start Date/Time	Equipment Order	Start Date/Time	*2
17	End Date/Time	Equipment Order	End Date/Time	*2
18	Process Name	Equipment Process	Equipment Process Name	-2
19	Process Classification	Equipment Process	Equipment Process Classification	-2
20	Production Equipment	Equipment Process	Production Equipment ID	-2
21	Site	Equipment Process	Site ID	-2
22	Status	Equipment Process	Status	-2
23	Specification	Equipment Process	Specification	-2
24	Quantity	Equipment Process	Quantity	-2
25	Unit	Equipment Process	Unit	-2

Work Breakdown Structure (WBS)

A table that hierarchically decomposes work and organizes process, period, and responsibility. Used for planning development and equipment introduction.

Figure 38 Work Breakdown Structure (WBS)

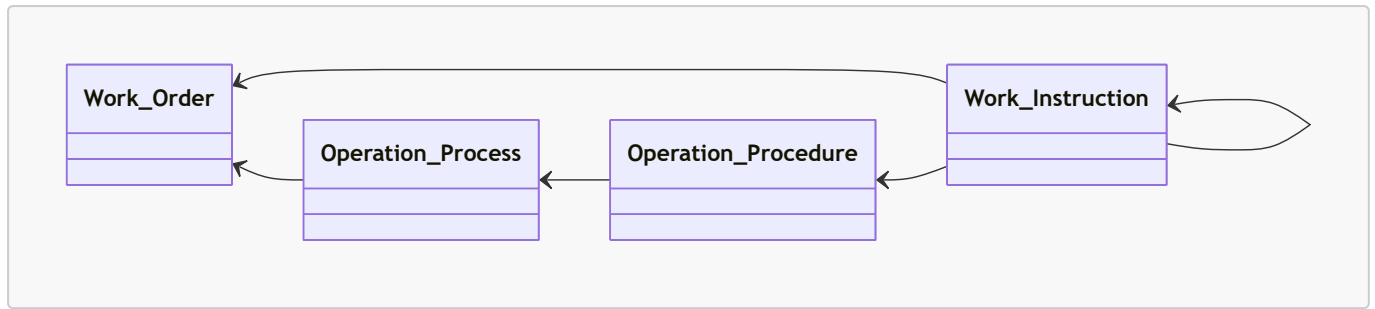


Table 65 Work Breakdown Structure (WBS)

No.	Item Name	Entity	Attribute	Relationship
1	Status	Work Order	Status	*
2	Production Order Name	Work Order	Production Order Name	*
3	Content	Work Order	Content	*
4	Start Date/Time	Work Order	Start Date/Time	*
5	End Date/Time	Work Order	End Date/Time	*
6	Work Time	Work Order	Work Time	*
7	Process Name	Operation Process	Production Process Name	-
8	Process Classification	Operation Process	Production Process Classification	-
9	Status	Operation Process	Status	-
10	Specification	Operation Process	Specification	-
11	Quantity	Operation Process	Quantity	-
12	Unit	Operation Process	Unit	-
13	Status	Work Instruction	Status	*1
14	Instruction Name	Work Instruction	Instruction Name	*1
15	Content	Work Instruction	Content	*1
16	Start Date/Time	Work Instruction	Start Date/Time	*1
17	End Date/Time	Work Instruction	End Date/Time	*1
18	Work Time	Work Instruction	Work Time	*1
19	Predecessor Event	Work Instruction	Predecessor Event	*1
20	Successor Event	Work Instruction	Successor Event	*1
21	Sequence Number	Operation Procedure	Sequence Number	*2

No.	Item Name	Entity	Attribute	Relationship
22	Procedure Name	Operation Procedure	Operation Procedure Name	*2
23	Content	Operation Procedure	Content	*2
24	Production Man-hours	Operation Procedure	Production Man-hours	*2
25	Worker	Operation Procedure	Worker Group ID	*2
26	Equipment	Operation Procedure	Equipment Group ID	*2
27	Quantity	Operation Procedure	Quantity	*2
28	Unit	Operation Procedure	Unit	*2

Instruction Information (Instruction Information)

Performance data accumulated chronologically from the results of executing production, quality, operation, and similar activities. Record-type information premised on comparison and analysis against plan information.

Table 66 Instruction Information

No.	Information Name	Product	Process	Equipment	Description
1	Production Order (Production Order)	○	○	○	A document indicating the item, quantity, delivery date, and process information to be manufactured, and instructing the start of production. A formal instruction to the manufacturing site.
2	Gantt Chart		○	○	A diagram showing work allocation and operating status per piece of equipment along a timeline. Optimizes the production schedule and equipment usage.
3	Load Chart		○	○	A table that visualizes work load on equipment/personnel along a time axis. Used for capacity planning and load leveling.

Production Order (Production Order)

A document indicating the item, quantity, delivery date, and process information to be manufactured, and instructing the start of production. A formal instruction to the manufacturing site.

Figure 39 Production Order (Production Order)

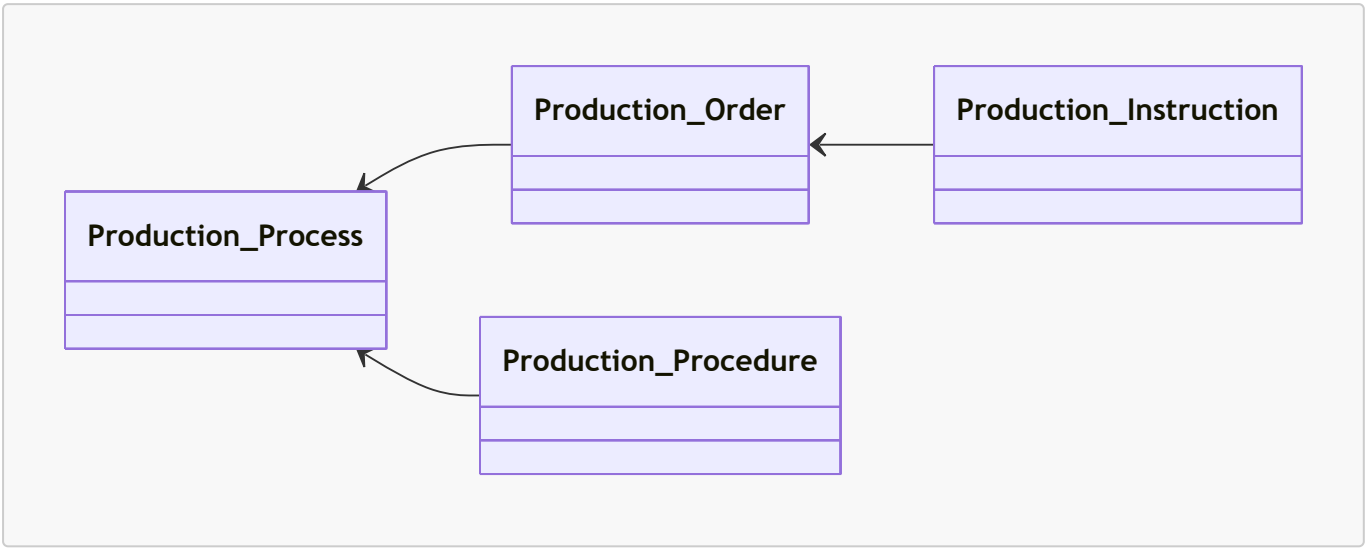


Table 69 Production Order (Production Order)

No.	Item Name	Entity	Attribute	Relationship
1	Order Name	Production Order	Production Order Name	
2	Order Classification	Production Order	Production Order Classification	
3	Unit	Production Order	Unit	
4	Quantity	Production Order	Quantity	
5	Start Date/Time	Production Order	Start Date/Time	
6	End Date/Time	Production Order	End Date/Time	
7	Content	Production Order	Content	
8	Location	Production Order	Location	
9	Issue Date/Time	Production Order	Issue Date/Time	
10	Requested Date/Time	Production Order	Requested Date/Time	
11	Process Name	Production Process	Production Process Name	-
12	Process Classification	Production Process	Production Process Classification	-
13	Specification	Production Process	Specification	-
14	Sequence Number	Production Procedure	Sequence Number	*1
15	Procedure Name	Production Procedure	Production Procedure Name	*1
16	Production Process	Production Procedure	Production Process ID	*1

No.	Item Name	Entity	Attribute	Relationship
17	Worker	Production Procedure	Worker Group ID	*1
18	Equipment	Production Procedure	Equipment Group ID	*1
19	Content	Production Procedure	Content	*1
20	Quantity	Production Procedure	Quantity	*1
21	Unit	Production Procedure	Unit	*1
22	Production Man-hours	Production Procedure	Production Man-hours	*1
23	Production Days	Production Procedure	Production Days	*1
24	Status	Production Procedure	Status	*1
25	Sequence Number	Production Instruction	Sequence Number	*2
26	Instruction Name	Production Instruction	Production Instruction Name	*2
27	Content	Production Instruction	Content	*2
28	Quantity	Production Instruction	Quantity	*2
29	Unit	Production Instruction	Unit	*2
30	Issue Date/Time	Production Instruction	Issue Date/Time	*2
31	Start Date/Time	Production Instruction	Start Date/Time	*2
32	Status	Production Instruction	Status	*2

Gantt Chart

A diagram showing work allocation and operating status per piece of equipment along a timeline. Optimizes the production schedule and equipment usage.

Figure 40 Gantt Chart

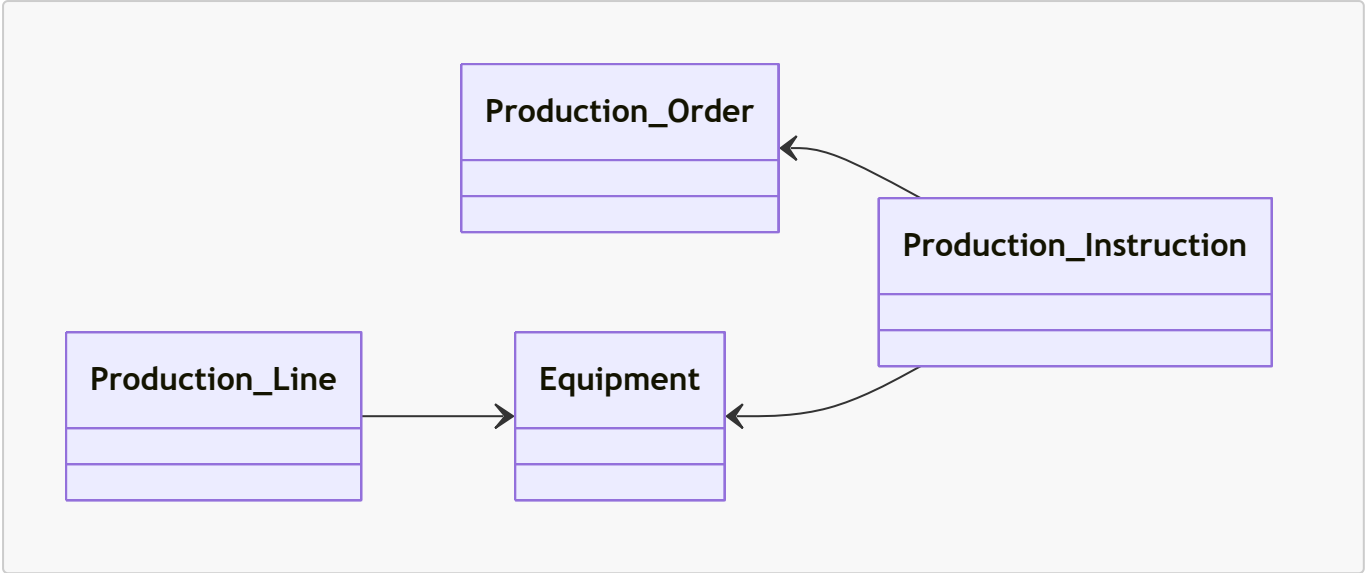


Table 67 Gantt Chart

No.	Item Name	Entity	Attribute	Relationship
1	Production Line Name	Production Line	Production Line Name	
2	Content (Production Order, etc.)	Production Line × Time	Content	**
3	Equipment Name	Production Equipment	Equipment Name	*1
4	Equipment Number	Production Equipment	Equipment Number	*1
5	Quantity	Production Equipment	Quantity	*1
6	Unit	Production Equipment	Unit	*1
7	Specification	Production Equipment	Specification	*1
8	Location	Production Equipment	Location	*1
9	Rate	Production Equipment	Rate	*1
10	Status (Production Order)	Production Order	Status	
11	Production Item	Production Order	Production Item	
12	Production Process	Production Order	Production Process	
13	Content (Production Order)	Production Order	Content	
14	Quantity (Production Order)	Production Order	Quantity	

No.	Item Name	Entity	Attribute	Relationship
15	Unit (Production Order)	Production Order	Unit	
16	Issue Date/Time	Production Order	Issue Date/Time	
17	Requested Date/Time	Production Order	Requested Date/Time	
18	Start Date/Time (Production Order)	Production Order	Start Date/Time	
19	End Date/Time (Production Order)	Production Order	End Date/Time	
20	Sequence Number	Work Instruction	Sequence Number	*2
21	Production Equipment	Work Instruction	Production Equipment	*2
22	Worker	Work Instruction	Worker	*2
23	Quantity (Work Instruction)	Work Instruction	Quantity	*2
24	Unit (Work Instruction)	Work Instruction	Unit	*2
25	Content (Work Instruction)	Work Instruction	Content	*2
26	Location	Work Instruction	Location	*2
27	Start Date/Time (Work Instruction)	Work Instruction	Start Date/Time	*2
28	End Date/Time (Work Instruction)	Work Instruction	End Date/Time	*2
29	Status	Work Instruction	Status	*2

Load Chart

A table that visualizes work load on equipment/personnel along a time axis. Used for capacity planning and load leveling.

Figure 41 Load Chart

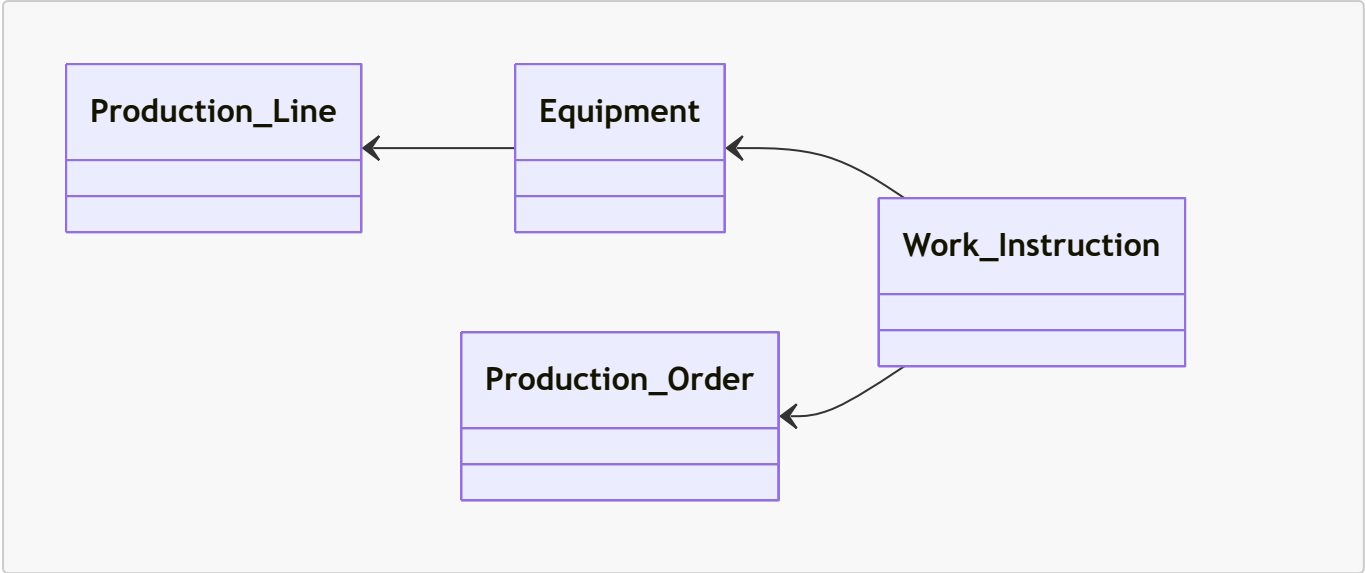


Table 68 Load Chart

No.	Item Name	Entity	Attribute	Relationship
1	Production Line Name	Production Line	Production Line Name	
2	Content (Production Order, etc.)	Production Line × Time	Content	**
3	Status	Production Order	Status	*1
4	Production Order	Production Order	Production Order Name	*1
5	Production Item	Production Order	Production Item ID	*1
6	Production Process	Production Order	Production Process ID	*1
7	Content	Production Order	Content	*1
8	Quantity	Production Order	Quantity	*1
9	Unit	Production Order	Unit	*1
10	Issue Date/Time	Production Order	Issue Date/Time	*1
11	Requested Date/Time	Production Order	Requested Date/Time	*1
12	Start Date/Time	Production Order	Start Date/Time	*1
13	End Date/Time	Production Order	End Date/Time	*1
14	Production Equipment Name	Production Equipment	Production Equipment Name	
15	Equipment Number	Production Equipment	Equipment Number	
16	Content	Production Equipment × Time	Content	**
17	Sequence Number	Production Instruction	Sequence Number	*2

No.	Item Name	Entity	Attribute	Relationship
18	Production Instruction	Production Instruction	Production Instruction Name	*2
19	Content	Production Instruction	Content	*2
20	Quantity	Production Instruction	Quantity	*2
21	Unit	Production Instruction	Unit	*2

Performance Information

Information recording the results of work or production actually carried out against a plan or instruction. By grasping facts such as quantity, time, and quality, it is used for variance analysis against the plan, business evaluation, and improving the accuracy of the next plan.

Table 70 Performance Information

No.	Information Name	Product	Process	Equipment	Description
1	Production Performance Table	○	○		A table summarizing the production quantity, plan variance, and operating status for the whole plant or per line. Evaluates the degree of achievement of the production plan.
2	Operation Performance Table		○	○	A table recording the production quantity, time, loss, and operating status per task. Basic material for daily manufacturing management and improvement.
3	Inspection Performance Table	○	○		A table recording the inspection results, measured values, and judgment for products, materials, and processes. Grasps quality status and is used for improvement.
4	Maintenance Performance Table			○	A table recording the content and results of equipment inspection, repair, and failure response. Useful for grasping equipment condition and improving maintenance.

Production Performance Table

A table summarizing the production quantity, plan variance, and operating status for the whole plant or per line. Evaluates the degree of achievement of the production plan.

Figure 42 Production Performance Table

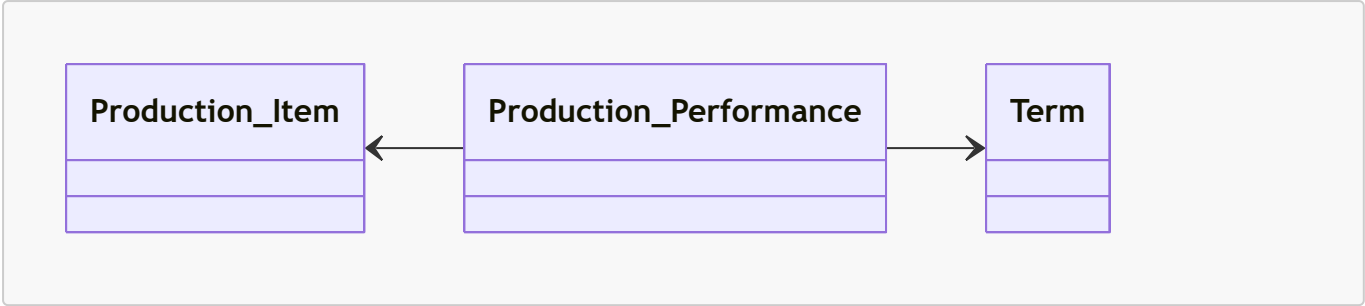


Table 71 Production Performance Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Name	Production Item	Production Item Name	
2	Item Classification	Production Item	Production Item Classification	
3	Performance	Production Item × Date	Performance	**
4	Status	Production Performance	Status	*1
5	Content	Production Performance	Content	*1
6	Result	Production Performance	Result	*1
7	Start Date/Time	Production Performance	Start Date/Time	*1
8	End Date/Time	Production Performance	End Date/Time	*1
9	Production Time	Production Performance	Production Time	*1
10	Quantity	Production Performance	Quantity	*1
11	Unit	Production Performance	Unit	*1

Operation Performance Table

A table recording the production quantity, time, loss, and operating status per task. Basic material for daily manufacturing management and improvement.

Figure 43 Operation Performance Table

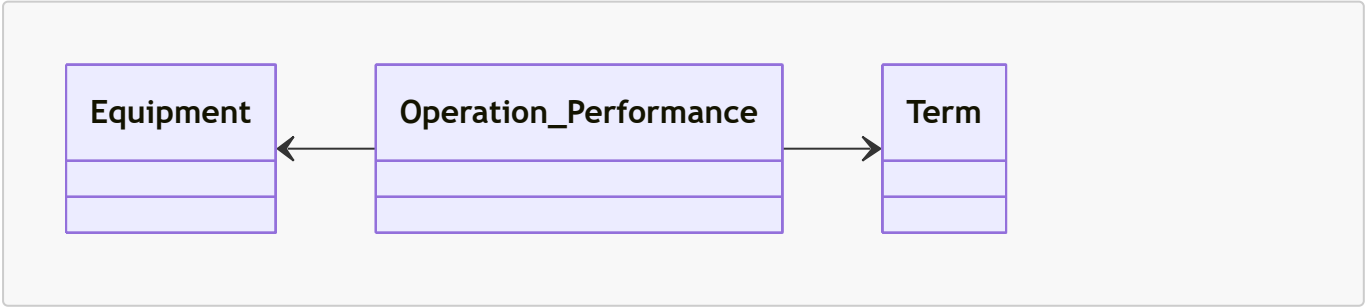


Table 72 Operation Performance Table

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Name	Production Equipment	Production Equipment Name	

No.	Item Name	Entity	Attribute	Relationship
2	Equipment Classification	Production Equipment	Production Equipment Classification	
3	Performance	Production Equipment × Date	Performance	**
4	Status	Operation Performance	Status	*1
5	Content	Operation Performance	Content	*1
6	Result	Operation Performance	Result	*1
7	Start Date/Time	Operation Performance	Start Date/Time	*1
8	End Date/Time	Operation Performance	End Date/Time	*1
9	Production Time	Operation Performance	Production Time	*1
10	Quantity	Operation Performance	Quantity	*1
11	Unit	Operation Performance	Unit	*1

Inspection Performance Table

A table recording the inspection results, measured values, and judgment for products, materials, and processes. Grasps quality status and is used for improvement.

Figure 44 Inspection Performance Table

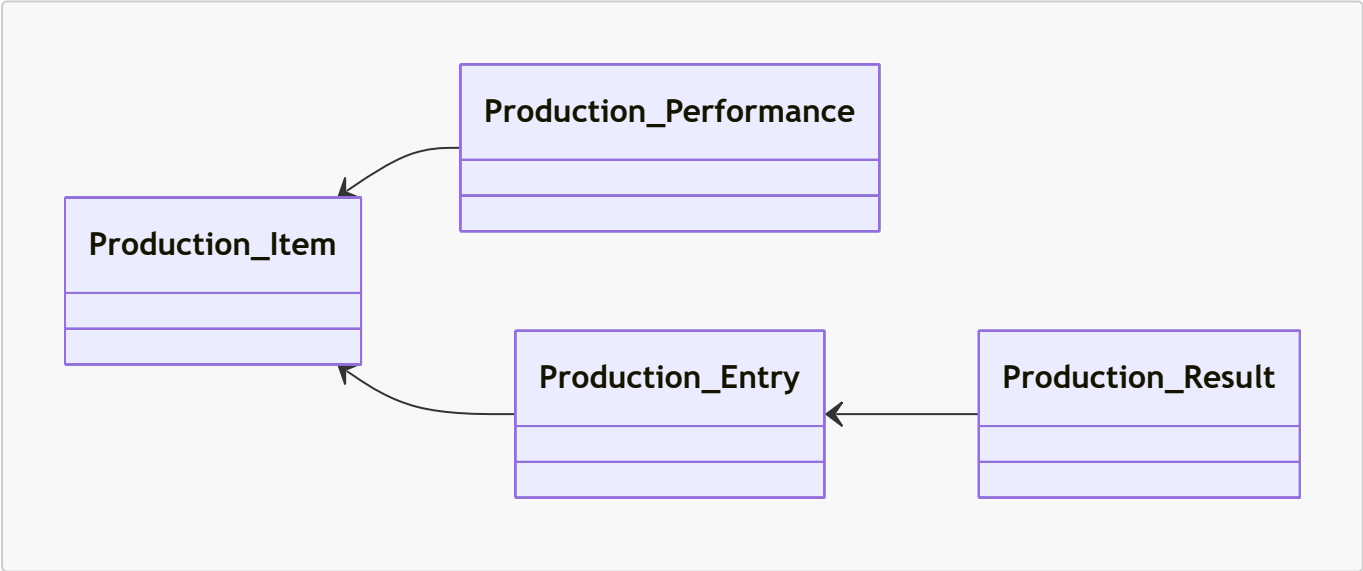


Table 73 Inspection Performance Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Drawing Number	Production Item	Drawing Number	
3	Item Name	Production Item	Production Item Name	

No.	Item Name	Entity	Attribute	Relationship
4	Specification	Production Item	Specification	
5	Sequence Number	Production Entry	Sequence Number	*
6	Entry Name	Production Entry	Production Entry Name	*
7	Location Name	Production Entry	Item Location ID	*
8	Jig/Tool	Production Entry	Jig/Tool ID	*
9	Specification	Production Entry	Specification	*
10	Standard Value	Production Entry	Standard Value	*
11	Measured Value	Production Entry	Measured Value	*
12	Unit	Production Entry	Unit	*
13	Status	Production Performance	Status	*1
14	Start Date/Time	Production Performance	Start Date/Time	*1
15	End Date/Time	Production Performance	End Date/Time	*1
16	Content	Production Performance	Content	*1
17	Result	Production Performance	Result	*1
18	Unit	Production Performance	Unit	*1
19	Judgment	Production Result	Status	*2
20	Result Name	Production Result	Production Result Name	*2
21	Result Classification	Production Result	Production Result Classification	*2
22	Specification	Production Result	Specification	*2
23	Result	Production Result	Result	*2
24	Quantity	Production Result	Quantity	*2
25	Unit	Production Result	Unit	*2
26	Worker	Production Result	Worker ID	*2
27	Work Date/Time	Production Result	Work Date/Time	*2

Maintenance Performance Table

A table recording the content and results of equipment inspection, repair, and failure response. Useful for grasping equipment condition and improving maintenance.

Figure 45 Maintenance Performance Table

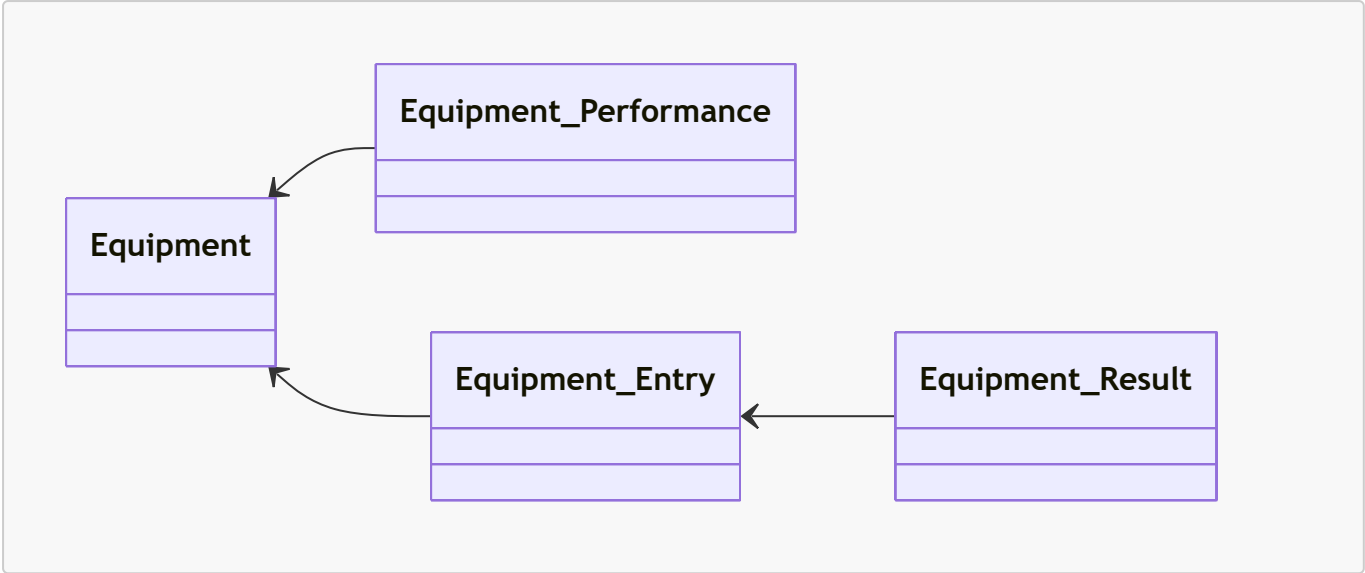


Table 74 Maintenance Performance Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Line	Production Equipment	Production Line	
2	Equipment Number	Production Equipment	Equipment Number	
3	Equipment Name	Production Equipment	Equipment Name	
4	Specification	Production Equipment	Specification	
5	Sequence Number	Equipment Entry	Sequence Number	*
6	Entry Name	Equipment Entry	Production Entry Name	*
7	Entry Classification	Equipment Entry	Production Entry Classification	*
8	Entry Content	Equipment Entry	Content	*
9	Location Name	Equipment Entry	Item Location ID	*
10	Standard Value	Equipment Entry	Standard Value	*
11	Measured Value	Equipment Entry	Measured Value	*
12	Unit	Equipment Entry	Unit	*
13	Status	Equipment Performance	Status	*1
14	Start Date/Time	Equipment Performance	Start Date/Time	*1
15	End Date/Time	Equipment Performance	End Date/Time	*1
16	Content	Equipment Performance	Content	*1
17	Result	Equipment Performance	Result	*1
18	Unit	Equipment Performance	Unit	*1
19	Judgment	Equipment Result	Status	*1
20	Result Name	Equipment Result	Production Result Name	*2

No.	Item Name	Entity	Attribute	Relationship
21	Result Classification	Equipment Result	Production Result Classification	*2
22	Specification	Equipment Result	Specification	*2
23	Result	Equipment Result	Result	*2
24	Quantity	Equipment Result	Quantity	*2
25	Unit	Equipment Result	Unit	*2
26	Worker	Equipment Result	Worker ID	*2
27	Work Date/Time	Equipment Result	Work Date/Time	*2

7.5. Shop-floor Information

4M Information (Management Information)

Management information regarding the basic perspectives from which actual production is carried out on the shop floor: Man, Machine, Material, and Method. Manages individual characteristics and management methods, current status, and past history.

Table 75 4M Information

No.	Information Name	Product	Process	Equipment	Description
1	Process Control Table		○		A table managing the work status, progress, issues, and variance from schedule for each process. Stabilizes the flow of production.
2	Quality Control Table	○	○		A table recording quality indicators, control items, measurement results, and improvement status. Used to maintain/improve process quality.
3	Equipment Control Table			○	A table managing equipment condition, maintenance history, operating status, and abnormality information. Used to improve equipment efficiency and reliability.
4	Skill Control Table		○	○	A table organizing the skill level, qualifications, and education history of workers. Useful for appropriate personnel placement and skill development.

Process Control Table

A table managing the work status, progress, issues, and variance from schedule for each process. Stabilizes the flow of production.

Figure 46 Process Control Table

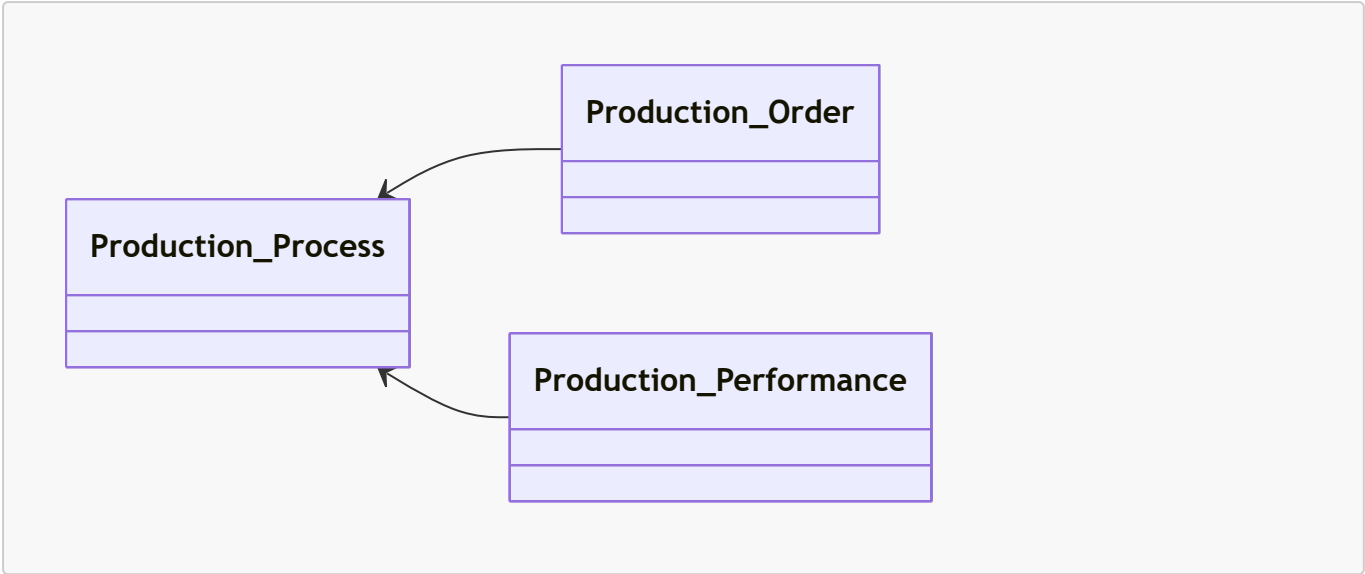


Table 76 Process Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Order	Production Order	Production Order Name	
2	Production Item	Production Order	Production Item ID	
3	Progress	Production Order × Production Process	Progress	**
4	Process Classification	Production Process	Production Process Classification	
5	Process Name	Production Process	Production Process Name	
6	Status	Production Performance	Status	*1
7	Production Order	Production Performance	Production Order ID	*1
8	Production Item	Production Performance	Production Item ID	*1
9	Content	Production Performance	Content	*1
10	Result	Production Performance	Result	*1
11	Quantity	Production Performance	Quantity	*1
12	Unit	Production Performance	Unit	*1

Quality Control Table

A table recording quality indicators, control items, measurement results, and improvement status. Used to maintain/improve process quality.

Figure 47 Quality Control Table

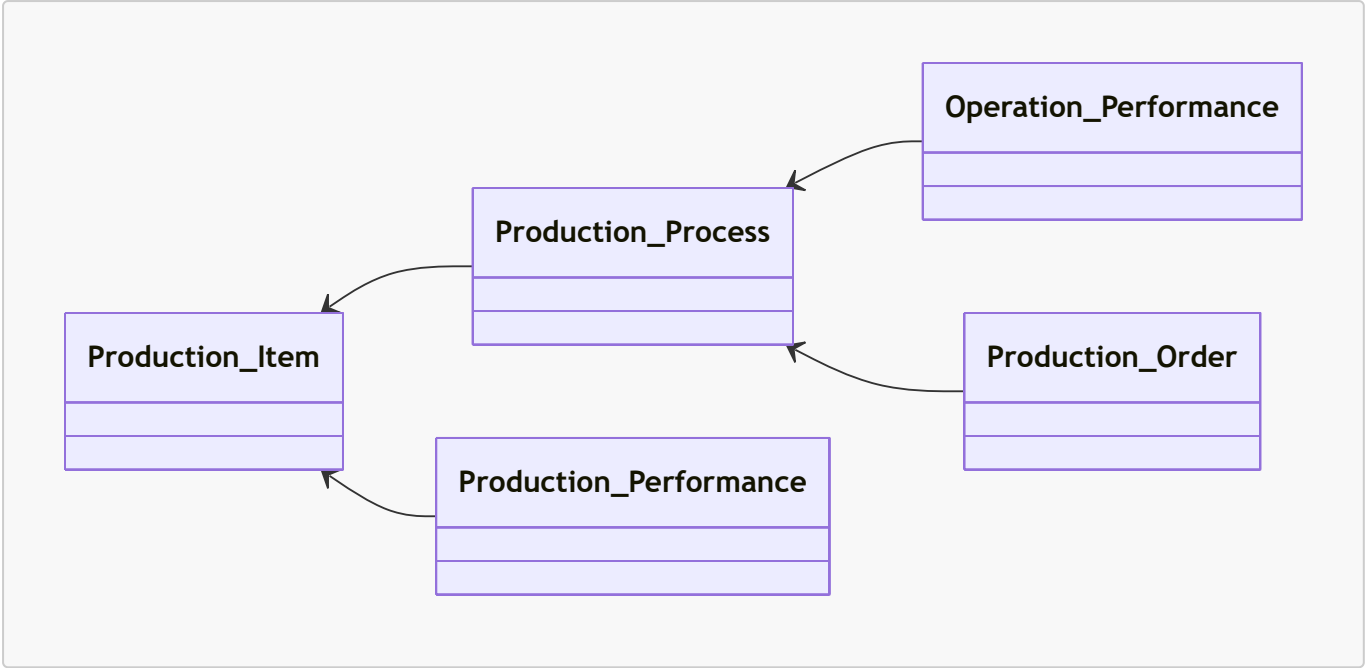


Table 77 Quality Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Drawing Number	Production Item	Drawing Number	
3	Item Name	Production Item	Production Item Name	
4	Specification	Production Item	Specification	
5	Process Name	Production Process	Process Name	1
6	Process Classification	Production Process	Process Classification	1
7	Site	Production Process	Site	1
8	Specification	Production Process	Specification	1
9	Status	Production Performance	Status	*1
10	Start Date/Time	Production Performance	Start Date/Time	*1
11	End Date/Time	Production Performance	End Date/Time	*1
12	Production Performance Name	Production Performance	Production Performance Name	*1
13	Performance Classification	Production Performance	Production Performance Classification	*1
14	Production Time	Production Performance	Production Time	*1

No.	Item Name	Entity	Attribute	Relationship
15	Content	Production Performance	Content	*1
16	Result	Production Performance	Result	*1
17	Quantity	Production Performance	Quantity	*1
18	Unit	Production Performance	Unit	*1
19	Status	Item Event	Status	*2
20	Event Name	Item Event	Item Event Name	*2
21	Event Classification	Item Event	Item Event Classification	*2
22	Occurrence Date/Time	Item Event	Occurrence Date/Time	*2
23	Event Duration	Item Event	Event Duration	*2
24	Fact	Item Event	Fact	*2
25	Cause	Item Event	Cause	*2
26	Issue	Item Event	Issue	*2
27	Location	Item Event	Location	*2
28	Quantity	Item Event	Quantity	*2
29	Unit	Item Event	Unit	*2
30	Sequence Number	Production Entry	Sequence Number	*3
31	Entry Name	Production Entry	Production Entry Name	*3
32	Specification	Production Entry	Specification	*3
33	Standard Quantity	Production Entry	Standard Quantity	*3
34	Measured Value	Production Entry	Measured Value	*3
35	Unit	Production Entry	Unit	*3
36	Status	Production Entry	Status	*3

Equipment Control Table

A table managing equipment condition, maintenance history, operating status, and abnormality information. Used to improve equipment efficiency and reliability.

Figure 48 Equipment Control Table

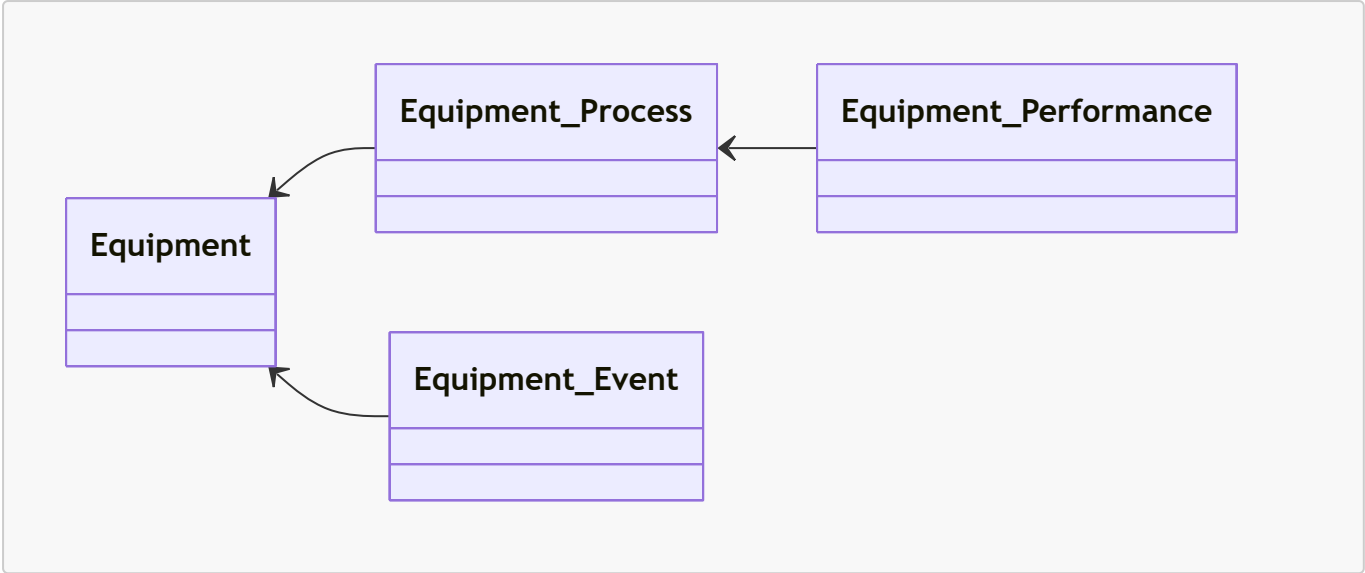


Table 78 Equipment Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Line	Production Equipment	Production Line	
2	Equipment Number	Production Equipment	Equipment Number	
3	Equipment Name	Production Equipment	Production Equipment Name	
4	Specification	Production Equipment	Specification	
5	Process Name	Equipment Process	Process Name	1
6	Process Classification	Equipment Process	Process Classification	1
7	Site	Equipment Process	Site	1
8	Specification	Equipment Process	Specification	1
9	Status	Maintenance Performance	Status	*1
10	Start Date/Time	Maintenance Performance	Start Date/Time	*1
11	End Date/Time	Maintenance Performance	End Date/Time	*1
12	Production Performance Name	Maintenance Performance	Production Performance Name	*1
13	Performance Classification	Maintenance Performance	Production Performance Classification	*1

No.	Item Name	Entity	Attribute	Relationship
14	Production Time	Maintenance Performance	Production Time	*1
15	Content	Maintenance Performance	Content	*1
16	Result	Maintenance Performance	Result	*1
17	Quantity	Maintenance Performance	Quantity	*1
18	Unit	Maintenance Performance	Unit	*1
19	Status	Maintenance Event	Status	*2
20	Event Name	Maintenance Event	Item Event Name	*2
21	Event Classification	Maintenance Event	Item Event Classification	*2
22	Occurrence Date/Time	Maintenance Event	Occurrence Date/Time	*2
23	Event Duration	Maintenance Event	Event Duration	*2
24	Fact	Maintenance Event	Fact	*2
25	Cause	Maintenance Event	Cause	*2
26	Issue	Maintenance Event	Issue	*2
27	Location	Maintenance Event	Location	*2
28	Quantity	Maintenance Event	Quantity	*2
29	Unit	Maintenance Event	Unit	*2
30	Status	Operation Performance	Status	*3
31	Start Date/Time	Operation Performance	Start Date/Time	*3
32	End Date/Time	Operation Performance	End Date/Time	*3
33	Performance Name	Operation Performance	Operation Performance Name	*3
34	Performance Classification	Operation Performance	Operation Performance Classification	*3
35	Production Equipment	Operation Performance	Production Equipment	*3

No.	Item Name	Entity	Attribute	Relationship
36	Operating Time	Operation Performance	Operating Time	*3
37	Content	Operation Performance	Content	*3
38	Result	Operation Performance	Result	*3
39	Quantity	Operation Performance	Quantity	*3
40	Unit	Operation Performance	Unit	*3

Skill Control Table

A table organizing the skill level, qualifications, and education history of workers. Useful for appropriate personnel placement and skill development.

Figure 49 Skill Control Table

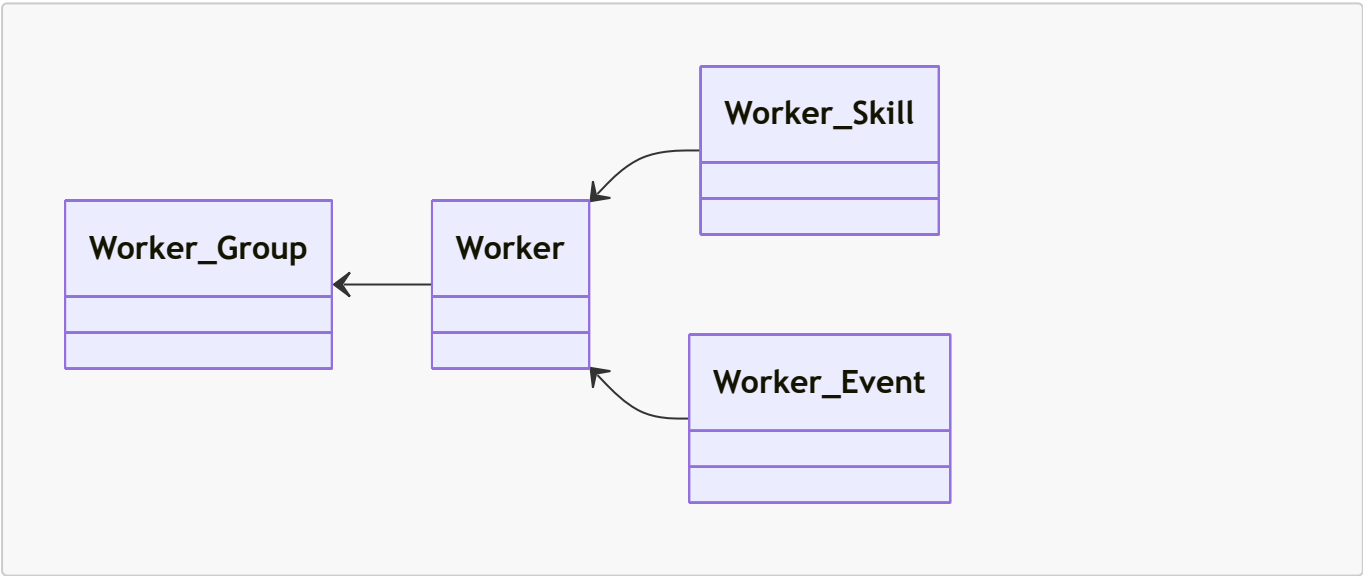


Table 79 Skill Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Group Classification	Worker Group	Worker Group Name	
2	Group Name	Worker Group	Worker Group Classification	
3	Worker Number	Worker	Worker Number	
4	Worker Name	Worker	Worker Name	
5	Worker Rank	Worker	Worker Classification	
6	Rate	Worker	Rate	

No.	Item Name	Entity	Attribute	Relationship
7	Standard Capacity	Worker	Standard Capacity	
8	Standard Quantity	Worker	Quantity	
9	Unit	Worker	Unit	
10	Specification	Worker	Specification	
11	Skill Classification	Worker Skill	Worker Skill Classification	
12	Skill Name	Worker Skill	Worker Skill Name	
13	Level	Worker Skill	Status	
14	Content	Worker Skill	Specification	
15	Quantity	Worker Skill	Quantity	
16	Unit (Worker Skill)	Worker Skill	Unit	
17	Event	Worker Event	Worker Event Name	
18	Issue	Worker Event	Issue	
19	Start Date	Worker Event	Start Date/Time	
20	Completion Date	Worker Event	End Date/Time	
21	Status	Worker Event	Status	

Status Information

Information used to grasp and manage the execution status of the shop floor or business operations, aiming to continuously visualize operating status, inventory status, work progress, etc. Forms the basis for catching business stagnation or abnormalities early and leading to swift judgment and improvement activities. Table 80 Status Information

No.	Information Name	Product	Process	Equipment	Description
1	Production Progress Table (Andon)	○	○		A control table that displays the progress, abnormality, and stoppage information of each process in real time. Enables shop-floor visualization and swift response.
2	Operation Monitoring Table			○	A table that records equipment operating time, downtime, and downtime reasons and calculates the operation rate. Used to improve equipment efficiency.
3	Inventory Control Table	○			A table that grasps the inventory quantity, receipts/issues, and appropriate inventory levels of materials, parts, and products.

No.	Information Name	Product	Process	Equipment	Description
					Used to prevent shortages and optimize inventory.

Production Progress Table (Andon)

A control table that displays the progress, abnormality, and stoppage information of each process in real time. Enables shop-floor visualization and swift response.

Figure 50 Production Progress Table (Andon)

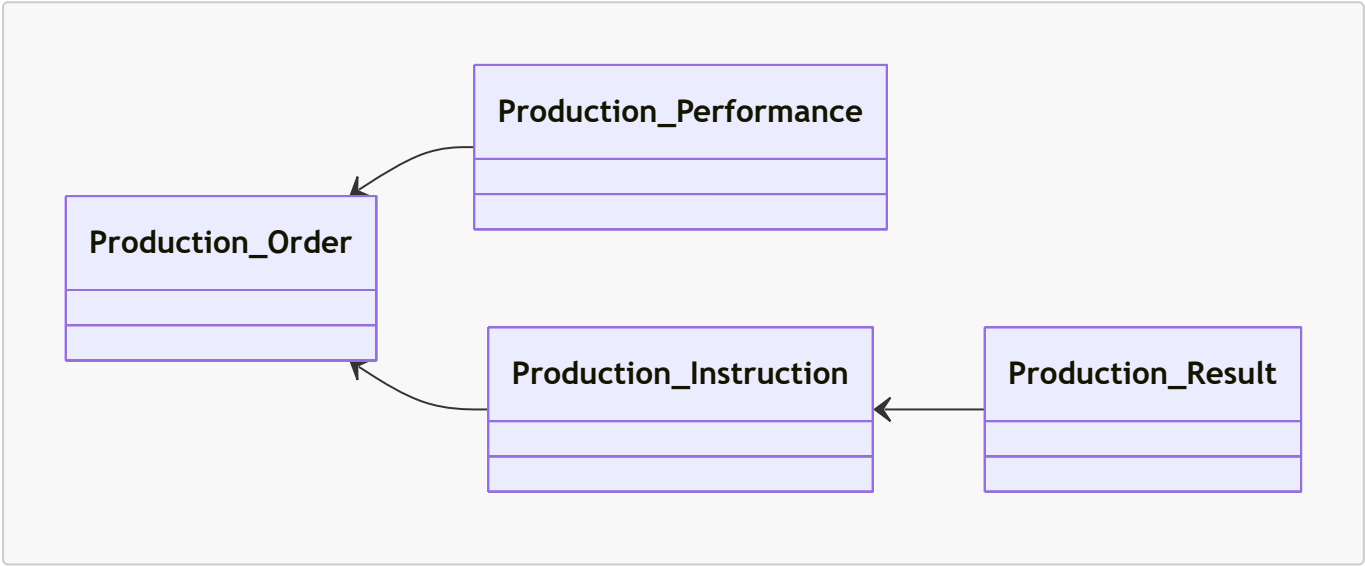


Table 83 Production Progress Table (Andon)

No.	Item Name	Entity	Attribute	Relationship
1	Order Name	Production Order	Production Order Name	
2	Production Line	Production Order	Production Line ID	
3	Production Item	Production Order	Production Item ID	
4	Content	Production Order	Content	
5	Location	Production Order	Location	
6	Manufacturing Delivery Date	Production Order	Requested Date/Time	
7	Start Date/Time	Production Order	Start Date/Time	
8	End Date/Time	Production Order	End Date/Time	
9	Quantity	Production Order	Quantity	
10	Unit	Production Order	Unit	
11	Sequence Number	Production Instruction	Sequence Number	*1
12	Status	Production Instruction	Status	*1

No.	Item Name	Entity	Attribute	Relationship
13	Content	Production Instruction	Content	*1
14	Start Date/Time	Production Instruction	Start Date/Time	*1
15	Quantity	Production Instruction	Quantity	*1
16	Unit	Production Instruction	Unit	*1
17	Content	Production Performance	Content	
18	Quantity	Production Performance	Quantity	
19	Unit	Production Performance	Unit	
20	Status	Production Performance	Status	
21	Start Date/Time	Production Performance	Start Date/Time	
22	End Date/Time	Production Performance	End Date/Time	
23	Production Time	Production Performance	Production Time	
24	Result	Production Performance	Result	
25	Result	Production Result	Result	*2
26	Status	Production Result	Status	*2
27	Specification	Production Result	Specification	*2
28	Location	Production Result	Location	*2
29	Quantity	Production Result	Quantity	*2
30	Unit	Production Result	Unit	*2

Operation Monitoring Table

A table that records equipment operating time, downtime, and downtime reasons and calculates the operation rate. Used to improve equipment efficiency.

Figure 51 Operation Monitoring Table

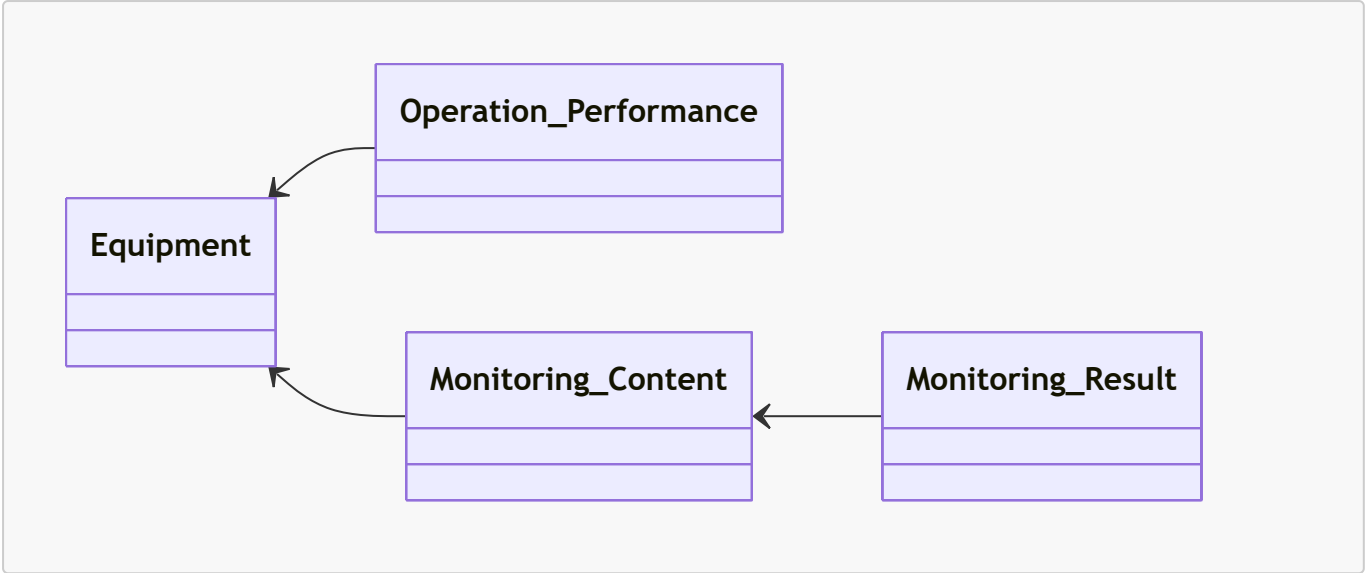


Table 81 Operation Monitoring Table

No.	Item Name	Entity	Attribute	Relationship
1	Equipment Name	Production Equipment	Production Equipment Name	
2	Equipment Number	Production Equipment	Equipment Number	
3	Status	Operation Performance	Status	*
4	Start Date/Time	Operation Performance	Start Date/Time	*
5	End Date/Time	Operation Performance	End Date/Time	*
6	Content	Operation Performance	Content	*
7	Result	Operation Performance	Result	*
8	Quantity	Operation Performance	Quantity	*
9	Unit	Operation Performance	Unit	*
10	Status	Monitoring Content	Status	*1
11	Equipment Location	Monitoring Content	Equipment Location ID	*1
12	Monitoring Item	Monitoring Content	Monitoring Item Name	*1
13	Monitoring Content	Monitoring Content	Specification	*1
14	Standard Value	Monitoring Content	Quantity	*1
15	Unit	Monitoring Content	Unit	*1
16	Monitoring Name	Monitoring Result	Monitoring Result Name	*2
17	Monitoring Content	Monitoring Result	Specification	*2
18	Result	Monitoring Result	Result	*2
19	Quantity	Monitoring Result	Quantity	*2
20	Unit	Monitoring Result	Unit	*2

No.	Item Name	Entity	Attribute	Relationship
21	Judgment	Monitoring Result	Status	*2

Inventory Control Table

A table that grasps the inventory quantity, receipts/issues, and appropriate inventory levels of materials, parts, and products. Used to prevent shortages and optimize inventory.

Figure 52 Inventory Control Table

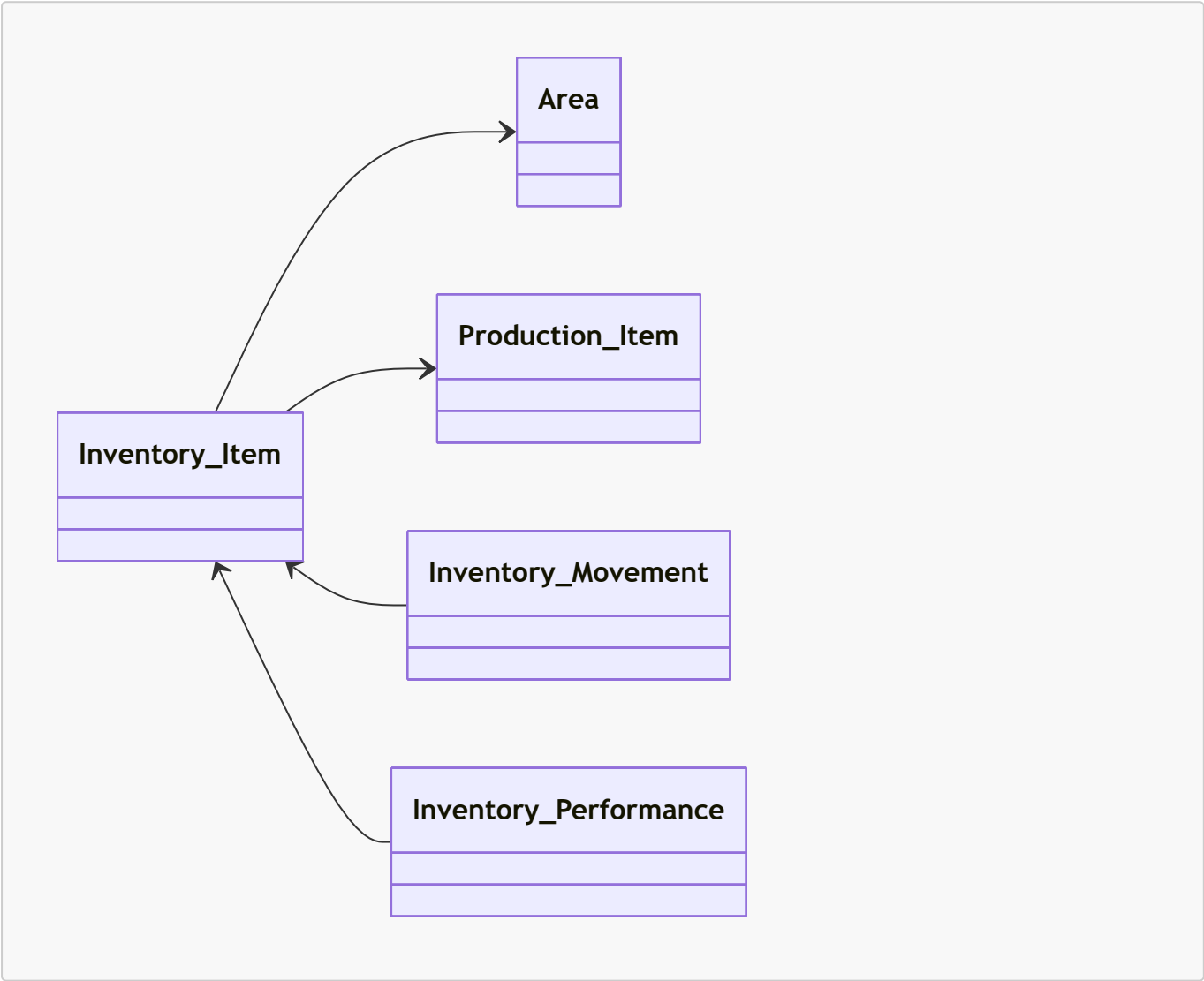


Table 82 Inventory Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Area Name	Area	Area Name	
2	Inventory Item Name	Inventory Item	Inventory Item Name	
3	Standard Value	Inventory Item	Standard Value	
4	Inventory Quantity	Inventory Item × Date	Inventory Quantity	**
5	Status	Inventory Movement	Status	*1

No.	Item Name	Entity	Attribute	Relationship
6	Movement Date/Time	Inventory Movement	Movement Date/Time	*1
7	Destination	Inventory Movement	Destination	*1
8	Inventory Item	Inventory Movement	Inventory Item ID	*1
9	Specification	Inventory Movement	Specification	*1
10	Worker	Inventory Movement	Worker ID	*1
11	Quantity	Inventory Movement	Quantity	*1
12	Unit	Inventory Movement	Unit	*1
13	Status	Inventory Performance	Status	*2
14	Inventory Date/Time	Inventory Performance	Inventory Date/Time	*2
15	Inventory Item	Inventory Performance	Inventory Item ID	*2
16	Location	Inventory Performance	Location	*2
17	Specification	Inventory Performance	Specification	*2
18	Quantity	Inventory Performance	Quantity	*2
19	Unit	Inventory Performance	Unit	*2
20	Previous Quantity	Inventory Performance	Previous Quantity	*2
21	Theoretical Quantity	Inventory Performance	Theoretical Quantity	*2

Improvement Information

Information that organizes the cause, countermeasure, and effect for issues or problems that become apparent during business or production activities. Used for the purpose of continuous improvement, leading to recurrence prevention, standardization, and improved business quality.

Table 84 Improvement Information

No.	Information Name	Product	Process	Equipment	Description
1	Issue Control Table		○	○	A table organizing the content, cause, person in charge, deadline, and response status of an issue that has occurred. Useful for progress management of improvement activities.
2	Defect Control Table		○	○	A table managing the content, source, cause, countermeasure, and recurrence-prevention status of a defect that has occurred. Essential for quality improvement.

No.	Information Name	Product	Process	Equipment	Description
3	Defective Product Control Table	○	○		A table recording the quantity, type, occurrence process, and loss of defective products. Used to grasp defect trends and for improvement measures.

Issue Control Table

A table organizing the content, cause, person in charge, deadline, and response status of an issue that has occurred. Useful for progress management of improvement activities.

Figure 53 Issue Control Table

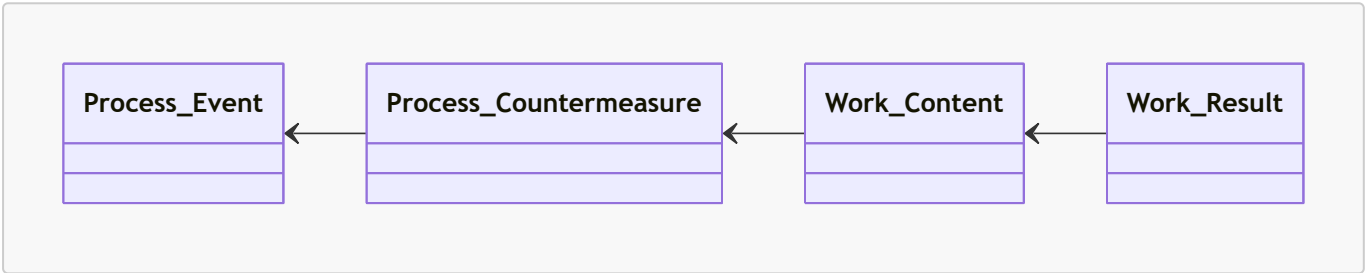


Table 85 Issue Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Status	Process Event	Status	
2	Event Name	Process Event	Process Event Name	
3	Event Classification	Process Event	Process Event Classification	
4	Start Date/Time	Process Event	Start Date/Time	
5	End Date/Time	Process Event	End Date/Time	
6	Sequence Number	Process Countermeasure	Sequence Number	*1
7	Status	Process Countermeasure	Status	*1
8	Countermeasure Name	Process Countermeasure	Process Countermeasure Name	*1
9	Countermeasure Classification	Process Countermeasure	Process Countermeasure Classification	*1
10	Start Date/Time	Process Countermeasure	Start Date/Time	*1
11	Completion Date/Time	Process Countermeasure	Completion Date/Time	*1

No.	Item Name	Entity	Attribute	Relationship
12	Countermeasure	Process Countermeasure	Countermeasure	*1
13	Result	Process Countermeasure	Result	*1
14	Location	Process Countermeasure	Location	*1
15	Worker	Process Countermeasure	Worker	*1
16	Sequence Number	Work Item	Sequence Number	
17	Status	Work Item	Status	
18	Item Name	Work Item	Work Item Name	
19	Item Classification	Work Item	Work Item Classification	
20	Specification	Work Item	Specification	
21	Quantity	Work Item	Quantity	
22	Unit	Work Item	Unit	
23	Status	Work Result	Status	*2
24	Result Name	Work Result	Work Result Name	*2
25	Result Classification	Work Result	Work Result Classification	*2
26	Specification	Work Result	Specification	*2
27	Result	Work Result	Result	*2
28	Work Date/Time	Work Result	Work Date/Time	*2
29	Worker	Work Result	Worker	*2
30	Location	Work Result	Location	*2
31	Quantity	Work Result	Quantity	*2
32	Unit	Work Result	Unit	*2

Defect Control Table

A table managing the content, source, cause, countermeasure, and recurrence-prevention status of a defect that has occurred.

Figure 54 Defect Control Table

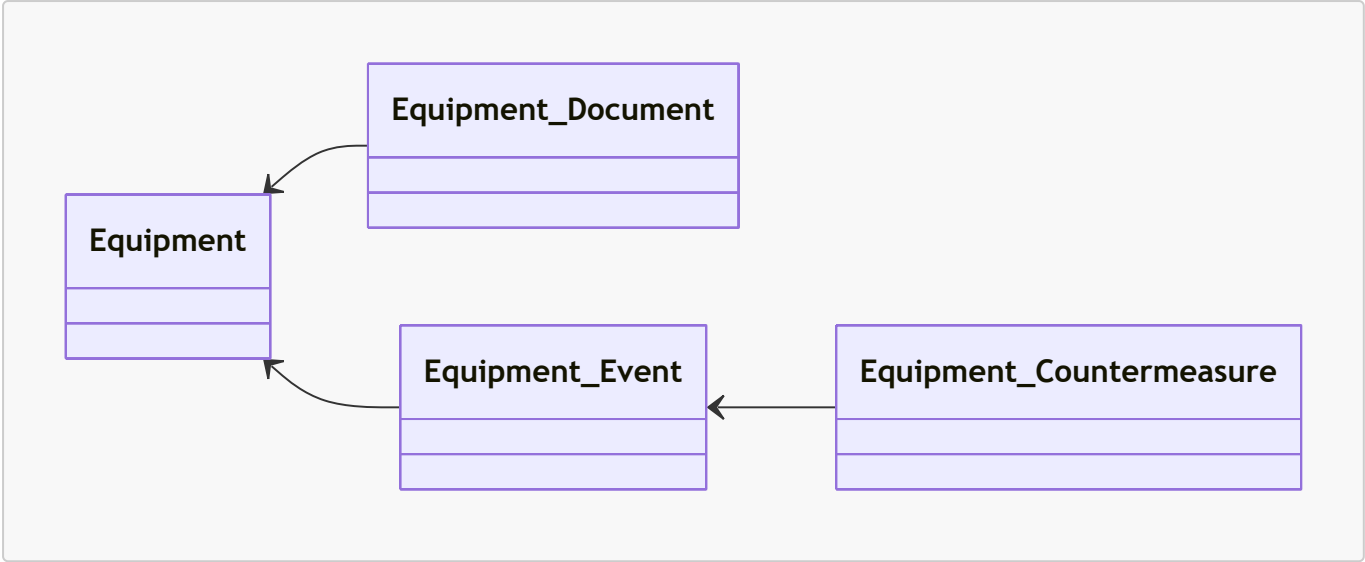


Table 86 Defect Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Production Line	Production Equipment	Production Line ID	
2	Equipment Number	Production Equipment	Production Equipment Number	
3	Equipment Name	Production Equipment	Production Equipment Name	
4	Specification	Production Equipment	Specification	
5	Report Name	Equipment Document	Equipment Document Name	*1
6	Classification	Equipment Document	Equipment Document Classification	*1
7	File	Equipment Document	File	*1
8	Summary	Equipment Document	Specification	*1
9	Registration Date/Time	Equipment Document	Registration Date/Time	*1
10	Author	Equipment Document	Author	*1
11	Status	Equipment Event	Status	
12	Event Name	Equipment Event	Equipment Event Name	
13	Event Classification	Equipment Event	Equipment Event Classification	
14	Start Date/Time	Equipment Event	Start Date/Time	
15	Event Duration	Equipment Event	Event Duration	
16	Fact	Equipment Event	Fact	
17	Cause	Equipment Event	Cause	
18	Issue	Equipment Event	Issue	

No.	Item Name	Entity	Attribute	Relationship
19	Location	Equipment Event	Location	
20	Quantity	Equipment Event	Quantity	
21	Unit	Equipment Event	Unit	
22	Sequence Number	Equipment Countermeasure	Sequence Number	*2
23	Countermeasure Name	Equipment Countermeasure	Equipment Countermeasure Name	*2
24	Countermeasure Classification	Equipment Countermeasure	Equipment Countermeasure Classification	*2
25	Start Date/Time	Equipment Countermeasure	Start Date/Time	*2
26	Completion Date/Time	Equipment Countermeasure	Completion Date/Time	*2
27	Countermeasure	Equipment Countermeasure	Countermeasure	*2
28	Result	Equipment Countermeasure	Result	*2
29	Location	Equipment Countermeasure	Location	*2
30	Worker	Equipment Countermeasure	Worker ID	*2
31	Quantity	Equipment Countermeasure	Quantity	*2
32	Unit	Equipment Countermeasure	Unit	*2
33	Status	Equipment Countermeasure	Status	*2

Defective Product Control Table

A table recording the quantity, type, occurrence process, and loss of defective products. Used to grasp defect trends and for improvement measures.

Figure 55 Defective Product Control Table

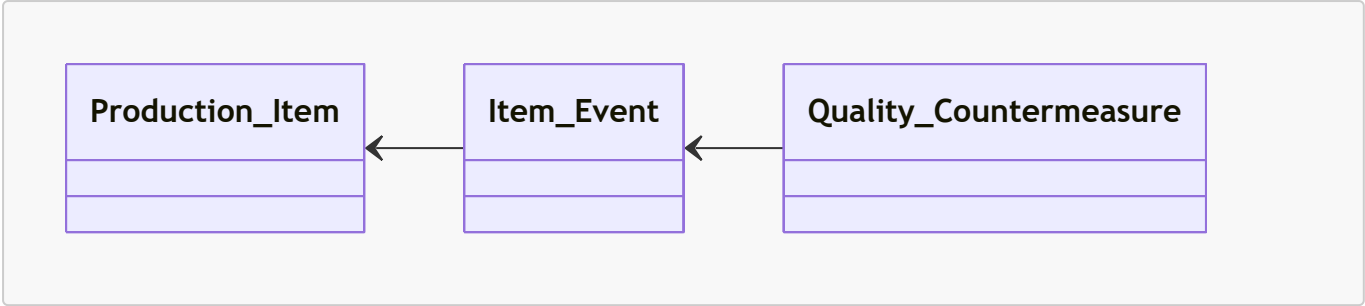


Table 87 Defective Product Control Table

No.	Item Name	Entity	Attribute	Relationship
1	Item Number	Production Item	Item Number	
2	Drawing Number	Production Item	Drawing Number	
3	Item Name	Production Item	Item Name	
4	Specification	Production Item	Specification	
5	Report Name	Item Document	Item Document Name	*1
6	Classification	Item Document	Item Document Classification	*1
7	File	Item Document	File	*1
8	Summary	Item Document	Specification	*1
9	Registration Date/Time	Item Document	Registration Date/Time	*1
10	Author	Item Document	Author	*1
11	Status	Item Event	Status	
12	Event Name	Item Event	Item Event Name	
13	Event Classification	Item Event	Item Event Classification	
14	Start Date/Time	Item Event	Start Date/Time	
15	Event Duration	Item Event	Event Duration	
16	Fact	Item Event	Fact	
17	Cause	Item Event	Cause	
18	Issue	Item Event	Issue	
19	Location	Item Event	Location	
20	Quantity	Item Event	Quantity	
21	Unit	Item Event	Unit	
22	Sequence Number	Item Countermeasure	Sequence Number	*2

No.	Item Name	Entity	Attribute	Relationship
23	Countermeasure Name	Item Countermeasure	Item Countermeasure Name	*2
24	Countermeasure Classification	Item Countermeasure	Item Countermeasure Classification	*2
25	Start Date/Time	Item Countermeasure	Start Date/Time	*2
26	Completion Date/Time	Item Countermeasure	Completion Date/Time	*2
27	Countermeasure	Item Countermeasure	Countermeasure	*2
28	Result	Item Countermeasure	Result	*2
29	Location	Item Countermeasure	Location	*2
30	Worker	Item Countermeasure	Worker ID	*2
31	Quantity	Item Countermeasure	Quantity	*2
32	Unit	Item Countermeasure	Unit	*2
33	Status	Item Countermeasure	Status	*2

8. Data Model (normative)

The data model shows the structure needed to process, on hardware such as a computer, the data required to implement the information model. The data model is defined by entities and attributes. For the content of the data model, refer separately to the content prescribed in "PSLX4.2 Technical Standard IVI-TS-PSLX-2026-01."

8.1. List of Data Models

The list of 72 data models is shown below.

Figure 56 List of Data Models

No.	Table Name	Japanese Name	Description
1	enterprise	Business Entity	Represents one's own company and the companies that are business partners.

No.	Table Name	Japanese Name	Description
2	site	Site	Represents a plant or base where production activity is carried out.
3	area	Area	Represents a section, floor, or management unit within a site.
4	production_line	Production Line	Represents the line or group of equipment that executes a production process.
5	calendar	Calendar	Represents operating conditions such as working days, holidays, and shifts.
6	term	Term	Represents the time interval that makes up the calendar.
7	asset	Asset	Represents common information about an asset involved in production activity.
8	asset_group	Asset Group	Represents a classification unit for assets sharing common attributes.
9	asset_specification	Asset Specification	Represents the specification or performance conditions required of an asset.
10	asset_structure	Asset Structure	Represents the structural relationship or hierarchy between assets.
11	production_item	Production Item	Represents the product, part, or material to be produced.
12	production_item_group	Item Group	Represents a classification unit for production items sharing common characteristics.
13	production_item_structure	Item Structure	Represents the bill of materials or structural relationship of a production item.
14	production_item_specification	Item Specification	Represents the specification, quality conditions, and performance conditions of a production item.
15	production_item_portion	Item Location	Represents an identifiable location/part making up a production item.
16	production_item_function	Item Function	Represents the function or role provided by a production item.
17	production_item_document	Item Document	Represents a drawing or specification document related to a production item.

No.	Table Name	Japanese Name	Description
18	production_item_issue	Item Event	Represents a defect or issue related to a production item.
19	production_item_countermeasure	Item Countermeasure	Represents the content of a countermeasure for an item event.
20	production_item_result	Item Result	Represents the actual outcome obtained from producing a production item.
21	equipment	Equipment	Represents the machine or device that executes production work.
22	equipment_group	Equipment Group	Represents a classification unit for equipment sharing a common function.
23	equipment_specification	Equipment Specification	Represents the performance or specification conditions required of equipment.
24	equipment_structure	Equipment Structure	Represents the parent-child relationship or structural information of equipment.
25	equipment_portion	Equipment Location	Represents an identifiable location/part making up equipment.
26	equipment_function	Equipment Function	Represents the function or role possessed by equipment.
27	equipment_document	Equipment Document	Represents a drawing or manual related to equipment.
28	equipment_issue	Equipment Event	Represents a failure or abnormality that occurred in equipment.
29	equipment_countermeasure	Equipment Countermeasure	Represents the content of a countermeasure for an equipment event.
30	personnel	Worker	Represents a worker engaged in production activity.
31	personnel_group	Worker Group	Represents a group of workers sharing a common role.
32	personnel_specification	Worker Specification	Represents the qualifications or conditions required of a worker.
33	personnel_structure	Worker Structure	Represents the hierarchy or substitution relationship among workers.

No.	Table Name	Japanese Name	Description
34	personnel_portion	Worker Location	Represents a physical or functional location/part of a worker.
35	personnel_skill	Worker Skill	Represents a skill or qualification held by a worker.
36	personnel_document	Worker Document	Represents education material or records related to a worker.
37	personnel_issue	Worker Event	Represents a problem or accident related to a worker.
38	personnel_countermeasure	Worker Countermeasure	Represents an improvement or countermeasure for a worker event.
39	production_process	Production Process	Represents a technically defined production process.
40	production_operation	Production Procedure	Represents the work procedure that concretizes a production process.
41	production_content	Production Entry	Represents the detailed work item making up a production procedure.
42	production_assignment	Production Assignment	Represents the resource assignment against a production instruction.
43	production_order	Production Order	Represents the instruction unit that requests production.
44	production_schedule	Production Instruction	Represents the execution instruction based on a production order.
45	production_result	Production Result	Represents the execution result obtained as the result of a production instruction.
46	production_capacity	Production Capacity	Represents capacity information for a production line or process.
47	production_performance	Production Performance	Represents actual production volume or operation performance.
48	equipment_process	Equipment Process	Represents a maintenance or operation process related to equipment.
49	equipment_operation	Equipment Procedure	Represents the concrete procedure that executes an equipment process.
50	equipment_content	Equipment Entry	Represents the detailed work content making up an equipment procedure.

No.	Table Name	Japanese Name	Description
51	equipment_assignment	Equipment Assignment	Represents the equipment assignment against a production procedure.
52	equipment_order	Equipment Order	Represents a work request against equipment.
53	equipment_schedule	Equipment Instruction	Represents the execution instruction based on an equipment order.
54	equipment_result	Equipment Result	Represents the result of equipment work.
55	equipment_capacity	Equipment Capacity	Represents the processing capacity or performance possessed by equipment.
56	equipment_performance	Operation Performance	Represents the operating performance or operation result of equipment.
57	work_process	Operation Process	Represents a process defined as manual/human work.
58	work_operation	Operation Procedure	Represents the concrete procedure that executes an operation process.
59	work_content	Work Content	Represents the detailed work content making up an operation procedure.
60	work_assignment	Work Assignment	Represents the resource assignment against a work instruction.
61	work_order	Work Order	Represents a work request against a worker.
62	work_schedule	Work Instruction	Represents the execution instruction based on a work order.
63	work_result	Work Result	Represents the execution result of work.
64	work_capacity	Work Capacity	Represents the capacity of a worker or work team.
65	work_performance	Work Performance	Represents the performance of a worker or work team.
66	production_document	Production Document	Represents a technical document related to a production process.
67	production_issue	Production Event	Represents a problem or abnormality that occurred in production activity.
68	production_countermeasure	Production Countermeasure	Represents an improvement or countermeasure for a production event.

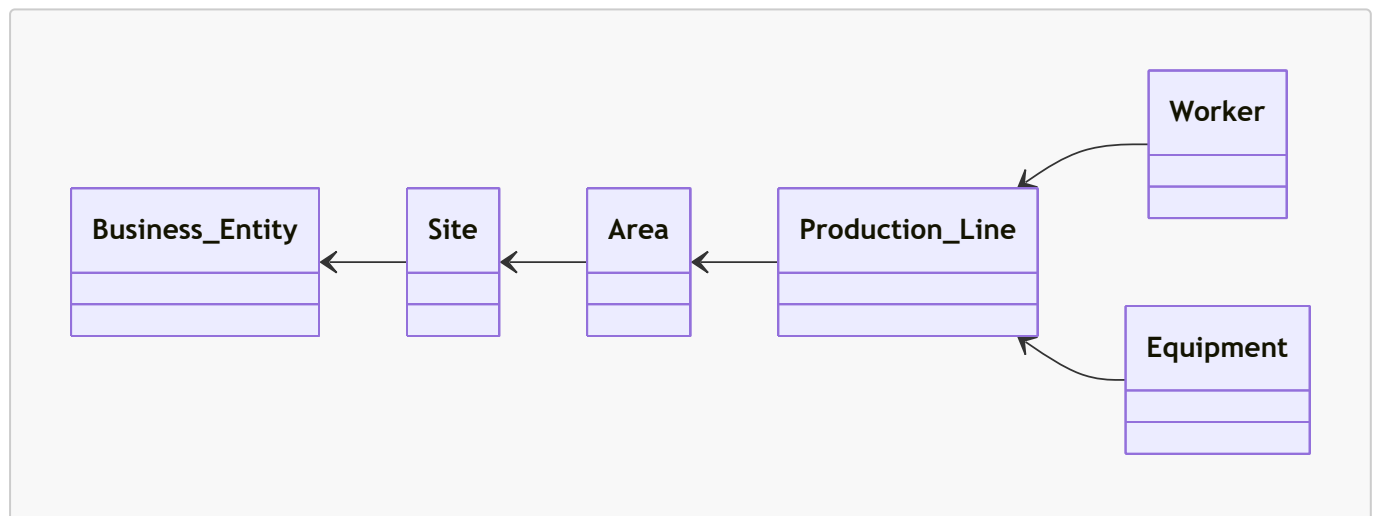
No.	Table Name	Japanese Name	Description
69	production_plan	Production Plan	Represents a plan regarding production quantity or period.
70	capacity_plan	Capacity Plan	Represents a plan for securing or allocating production capacity.
71	monitoring_content	Monitoring Content	Represents the target item monitored at equipment or a process.
72	monitoring_result	Monitoring Result	Represents the measured value or status result against a monitoring item.

8.2. Business Entity

When manufacturing is viewed as a single business entity across the whole company, it is composed, as a hierarchical structure, of the entities Site, Area, and Production Line.

The structure is as shown in Figure 58. This definition is prescribed in IEC 62264 (the international standard for MES-ERP integration) and is widely used. Calendar and Term are entities for showing the passage of discrete time. These can be set not only for the business entity as a whole but also per site or per area.

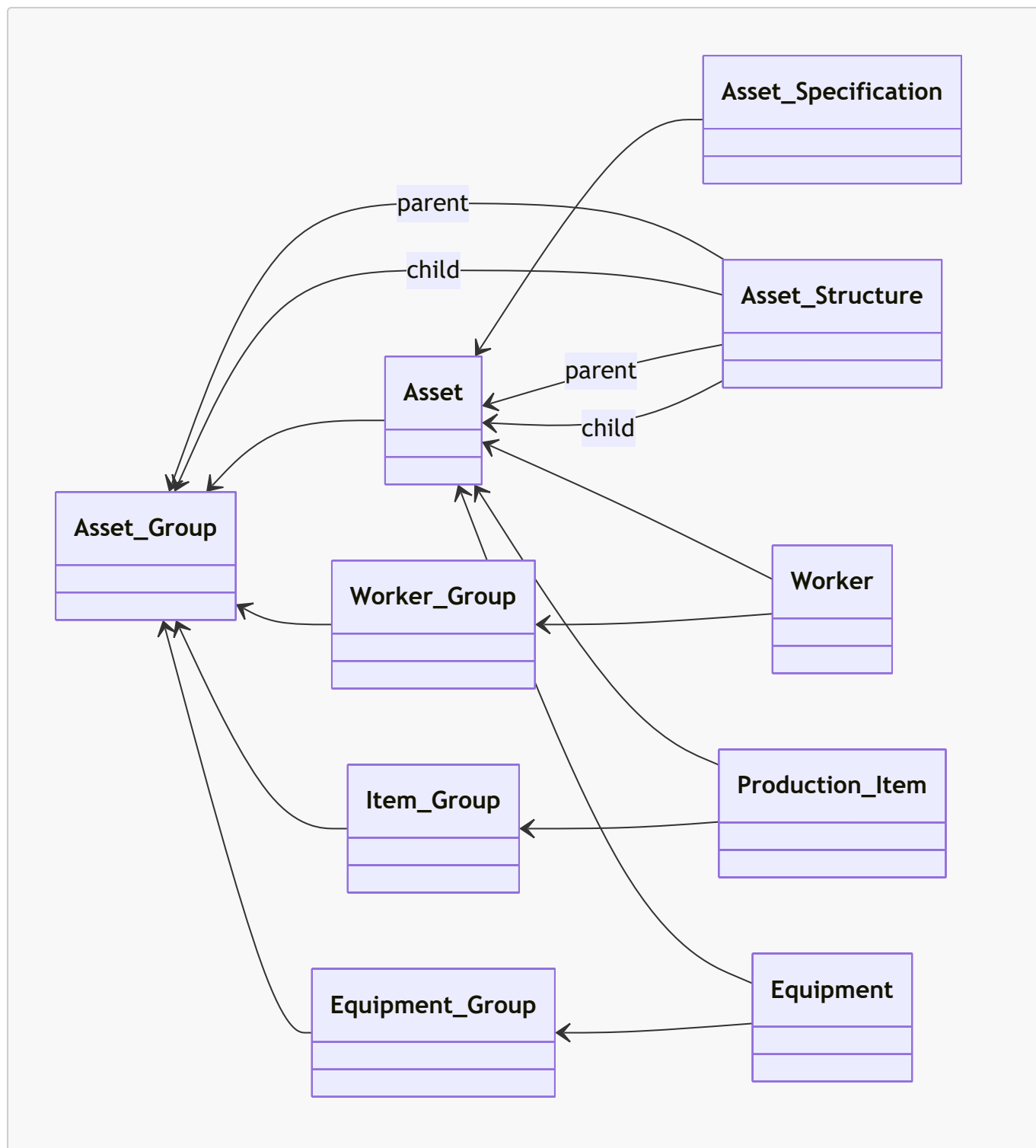
Figure 57 Hierarchical Structure of the Business Entity



8.3. Assets

An asset is an abstract concept referring to production-related assets in general. Production items, equipment, workers, and their groups all correspond to it.

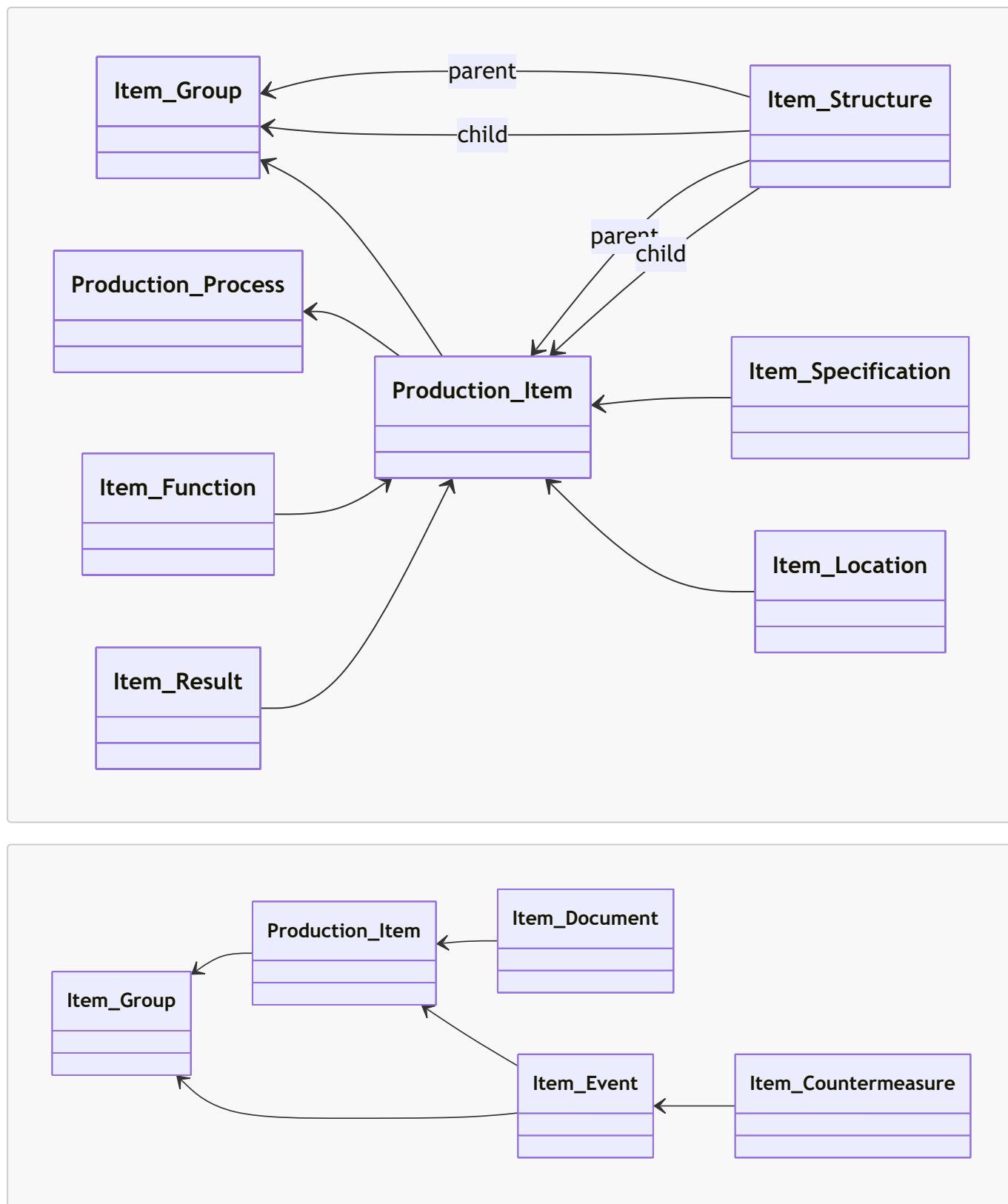
Figure 58 Entities Related to Assets



8.4. Production Items

The entities related to a production item are: Item Group, Item Structure, Item Location, and Item Function. The figure shows their relationship. The following sections explain the structure of each.

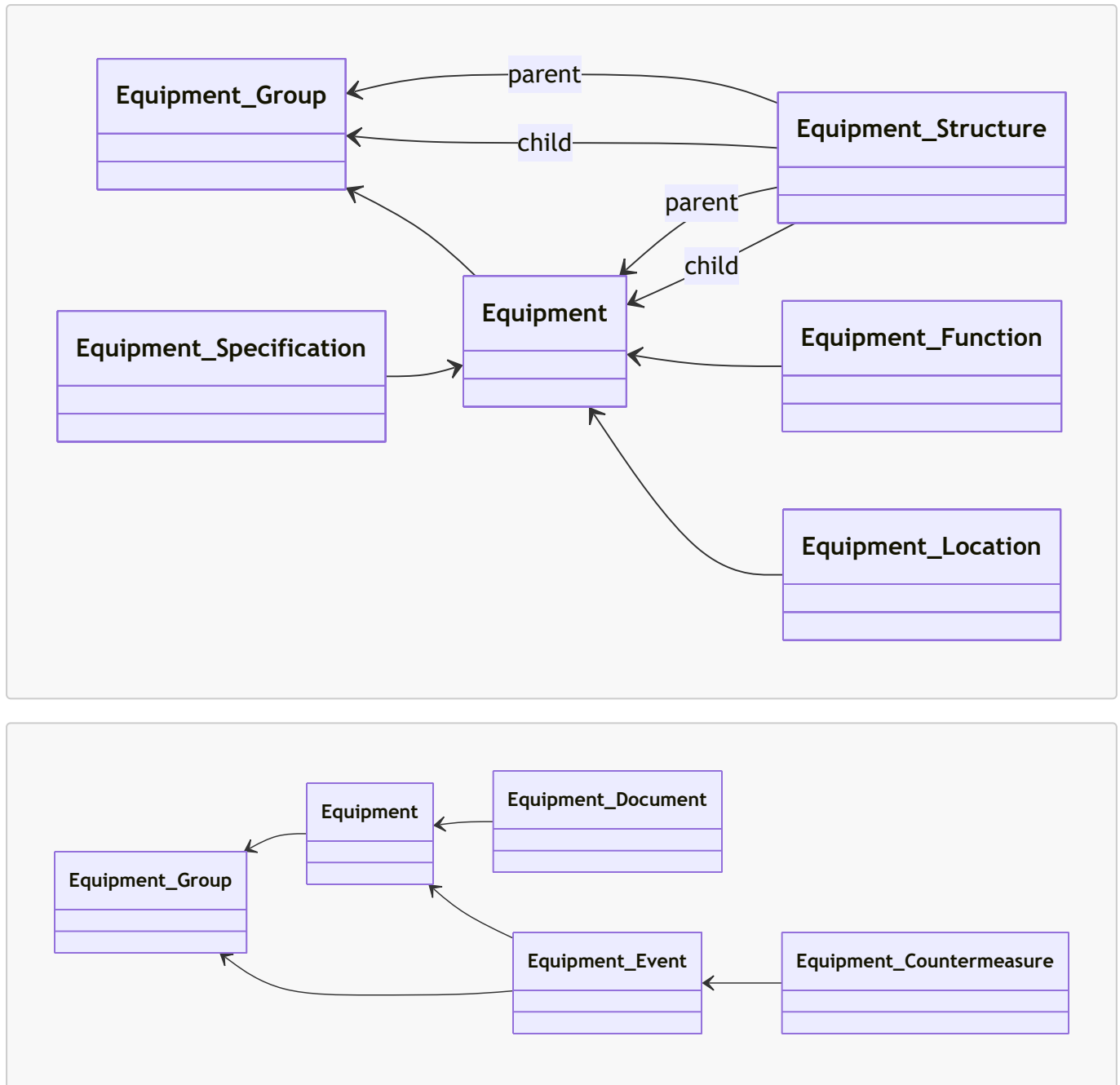
Figure 59 Entities Related to Production Items



8.5. Equipment

The entities related to equipment are: Equipment Group, Equipment Structure, Equipment Location, and Equipment Function. The figure shows their relationship. The following sections explain the structure of each.

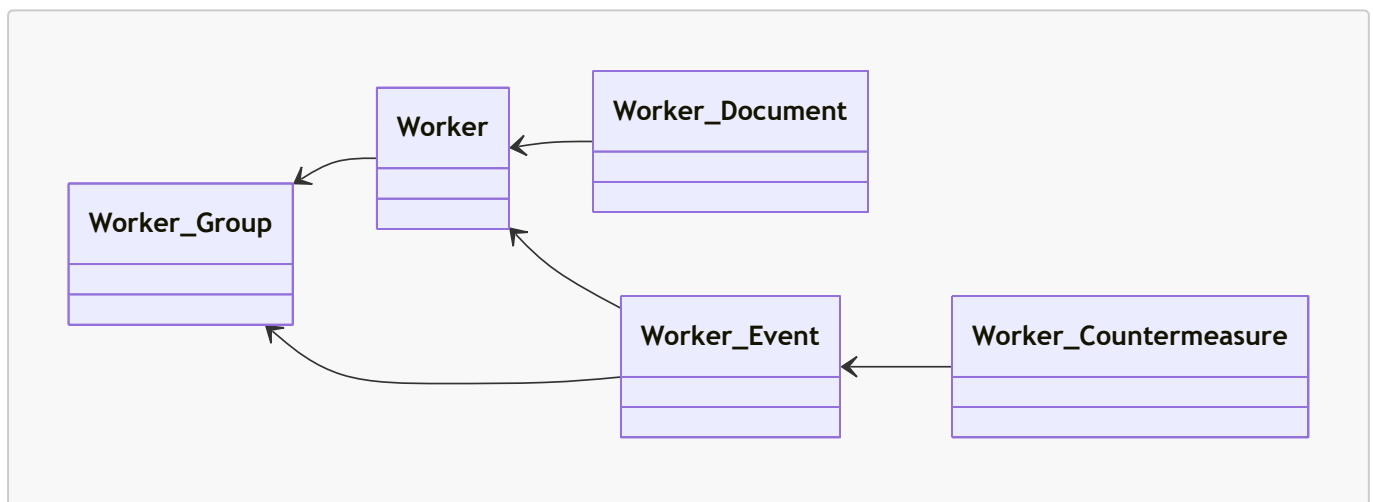
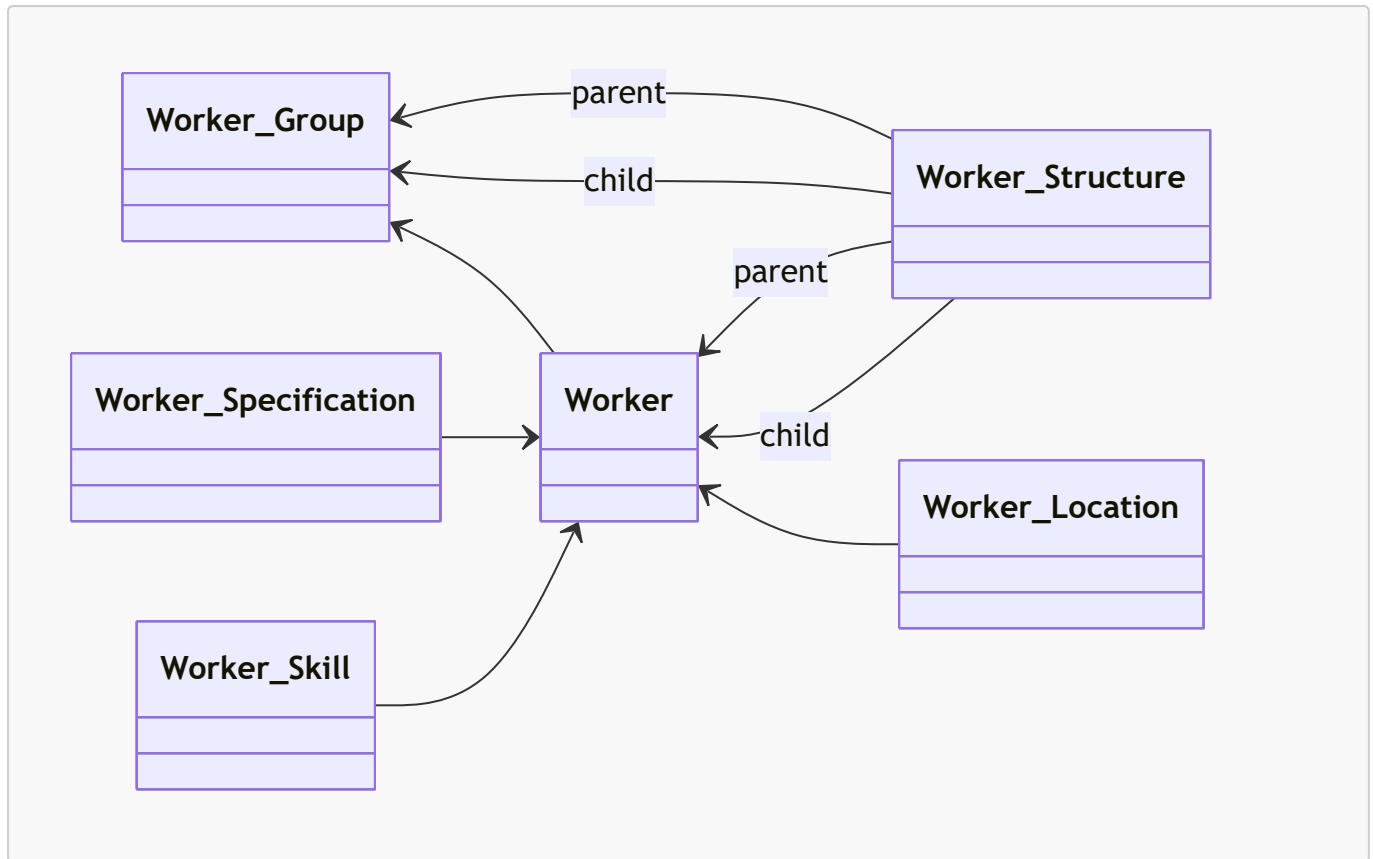
Figure 60 Entities Related to Equipment



8.6. Workers

The entities related to a worker are: Worker Group, Worker Structure, Worker Location, and Worker Skill. The figure shows their relationship. The following sections explain the structure of each. Worker Structure actually defines the composition between worker groups, but for notational purposes in the figure it is aligned with the structure of items and equipment.

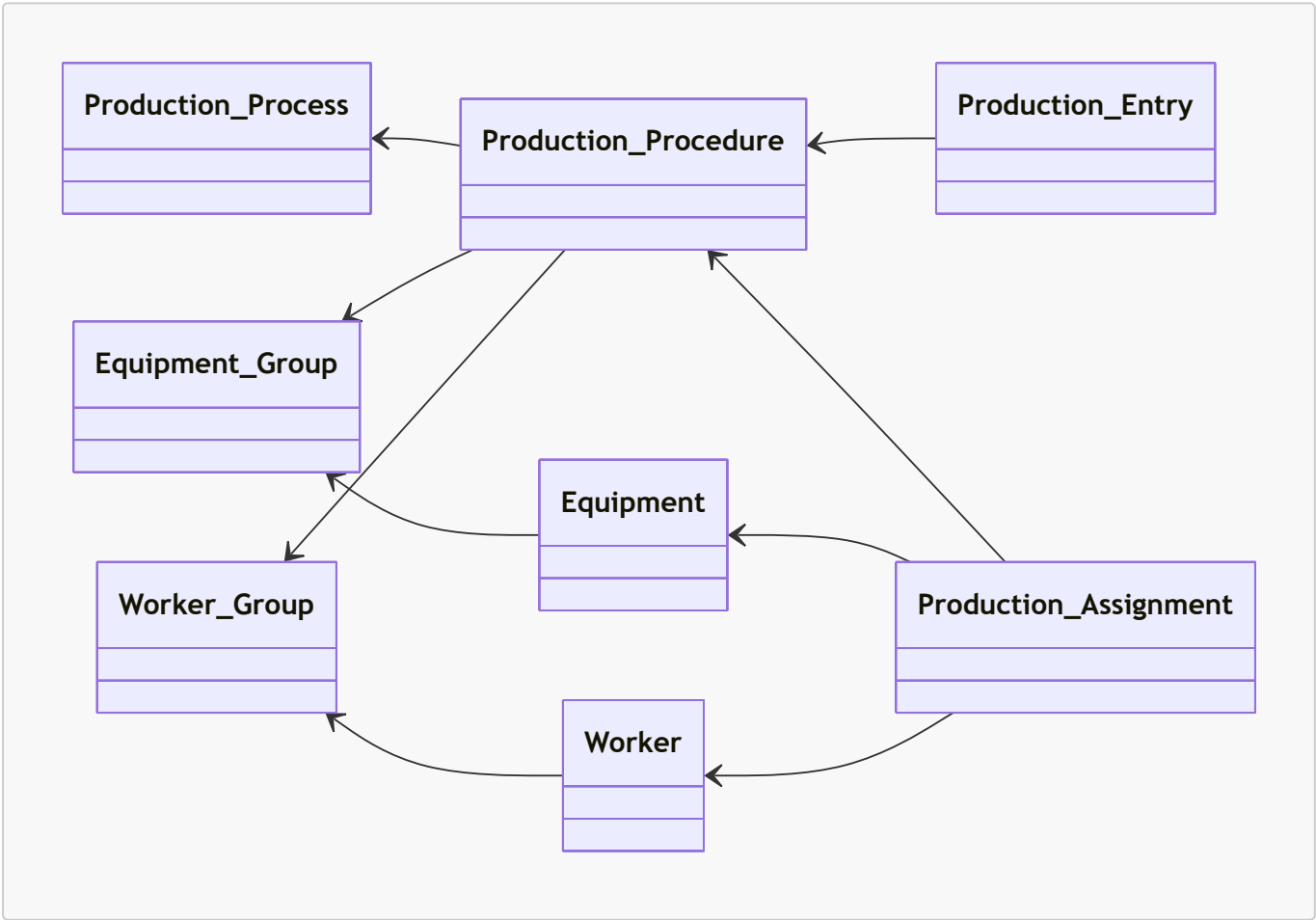
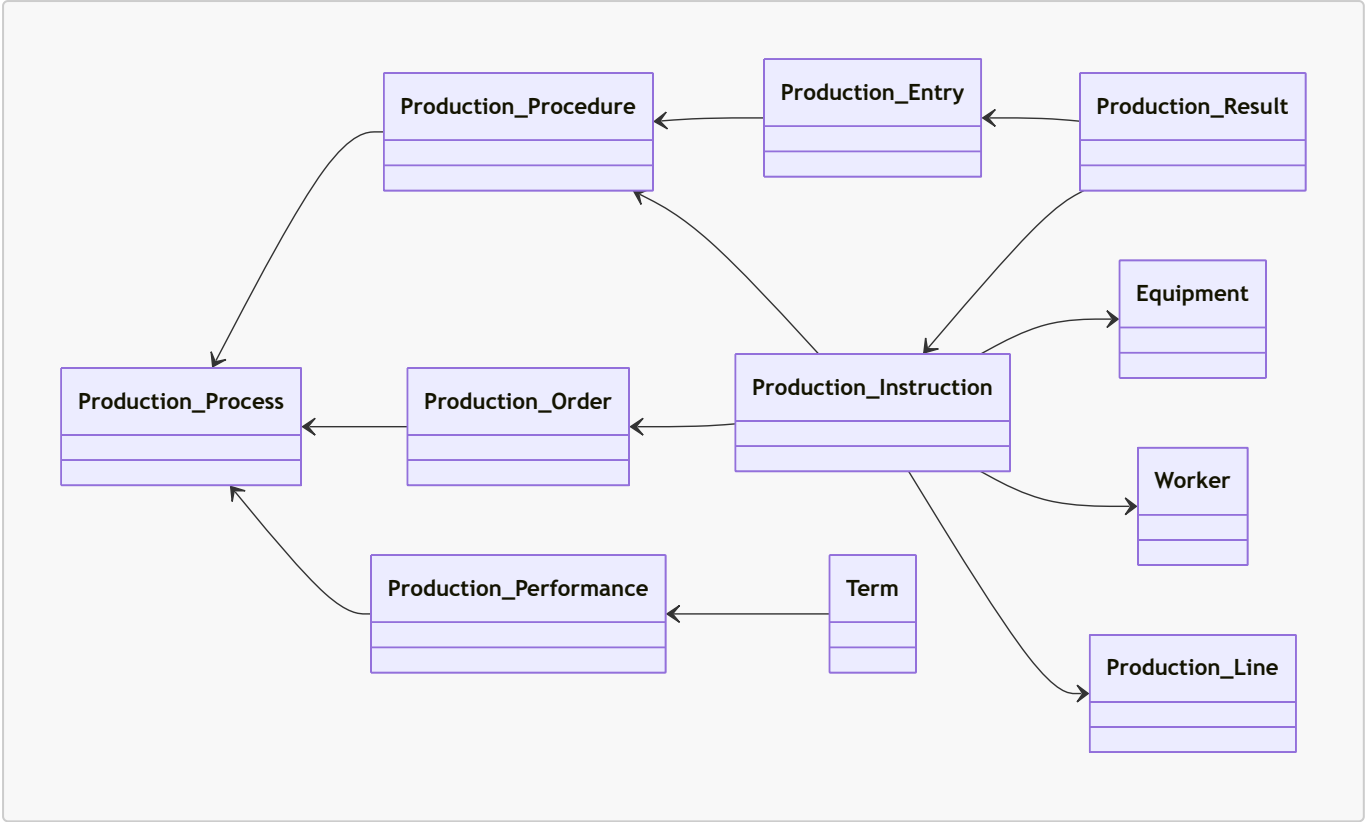
Figure 61 Entities Related to Workers

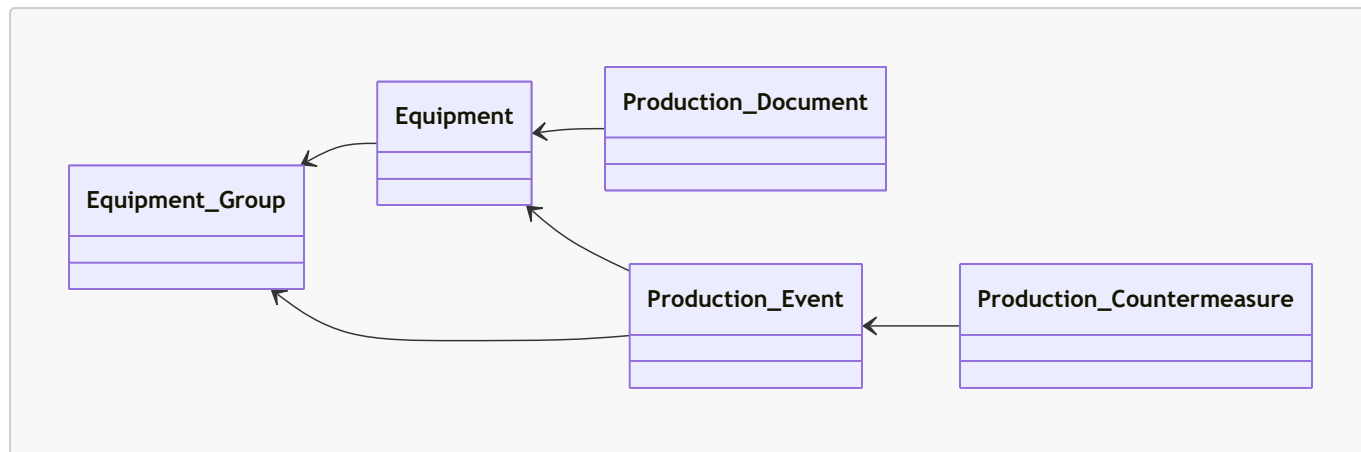


8.7. Production Process

The entities related to a production process include, as master data, Process Group, Production Procedure, and Production Entry, in addition to Production Process itself. As transaction data there are Production Order, Production Instruction, and Production Result, and Production Performance is defined corresponding to a Production Order. The figure below shows their relationship.

Figure 62 Entities Related to a Production Process

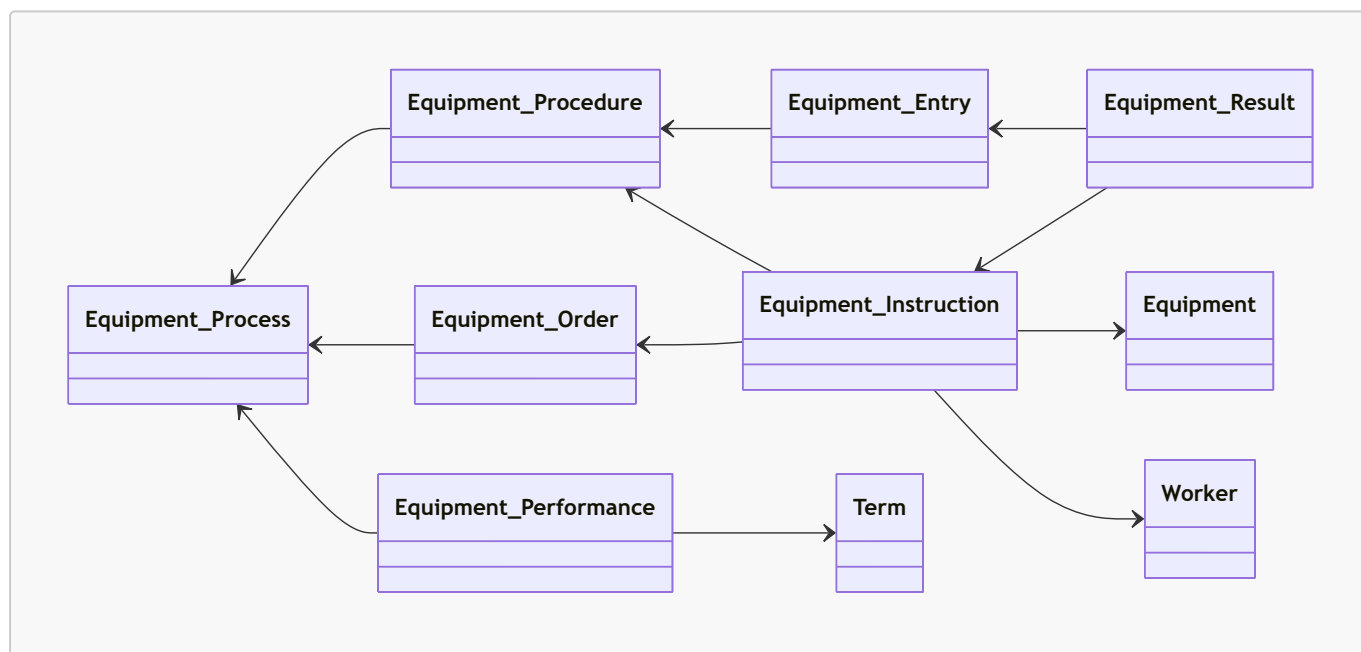


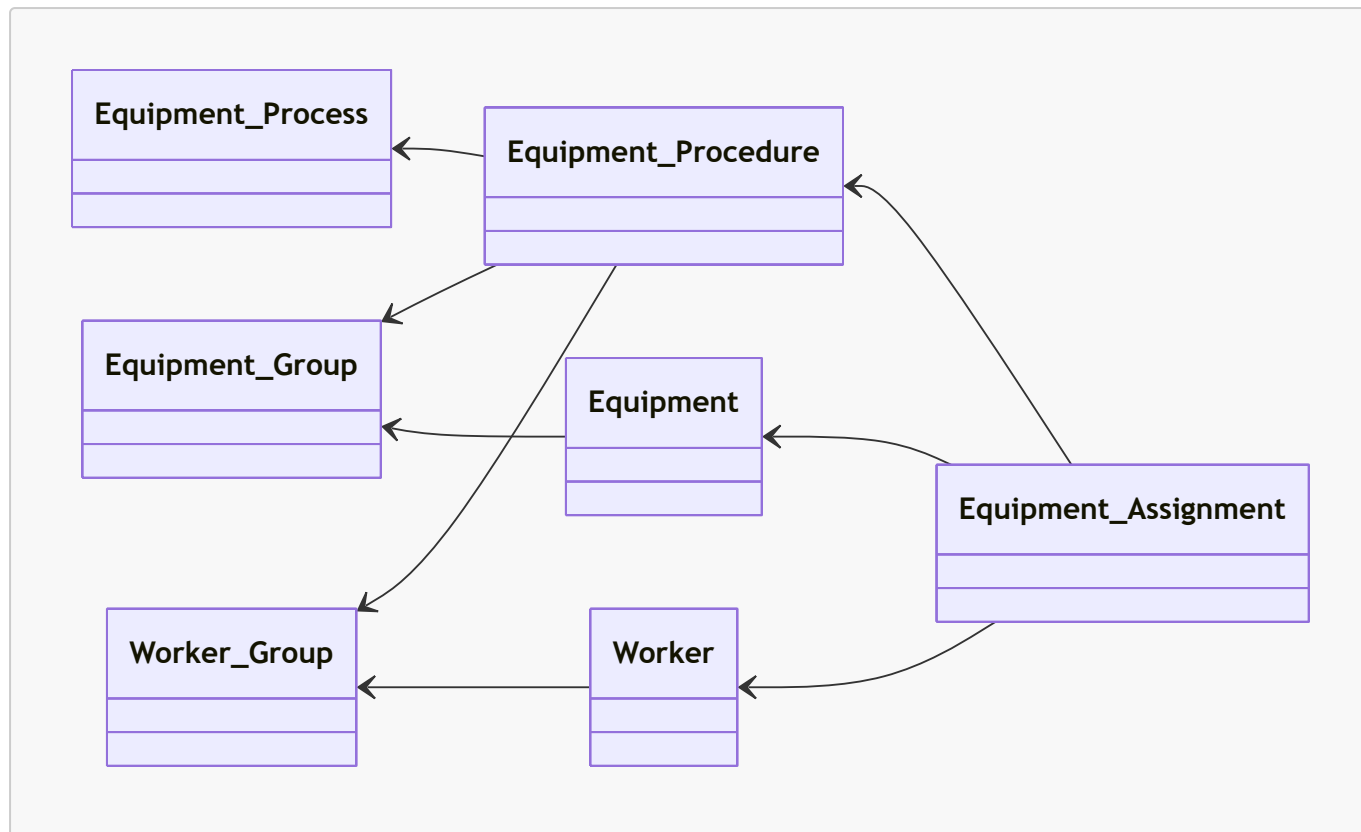


8.8. Equipment Process

The entities related to an equipment process have the same structure as those corresponding to a production process, but correspond to a process targeting equipment rather than the production of a production item. The structure is shown below.

Figure 63 Entities Related to an Equipment Process

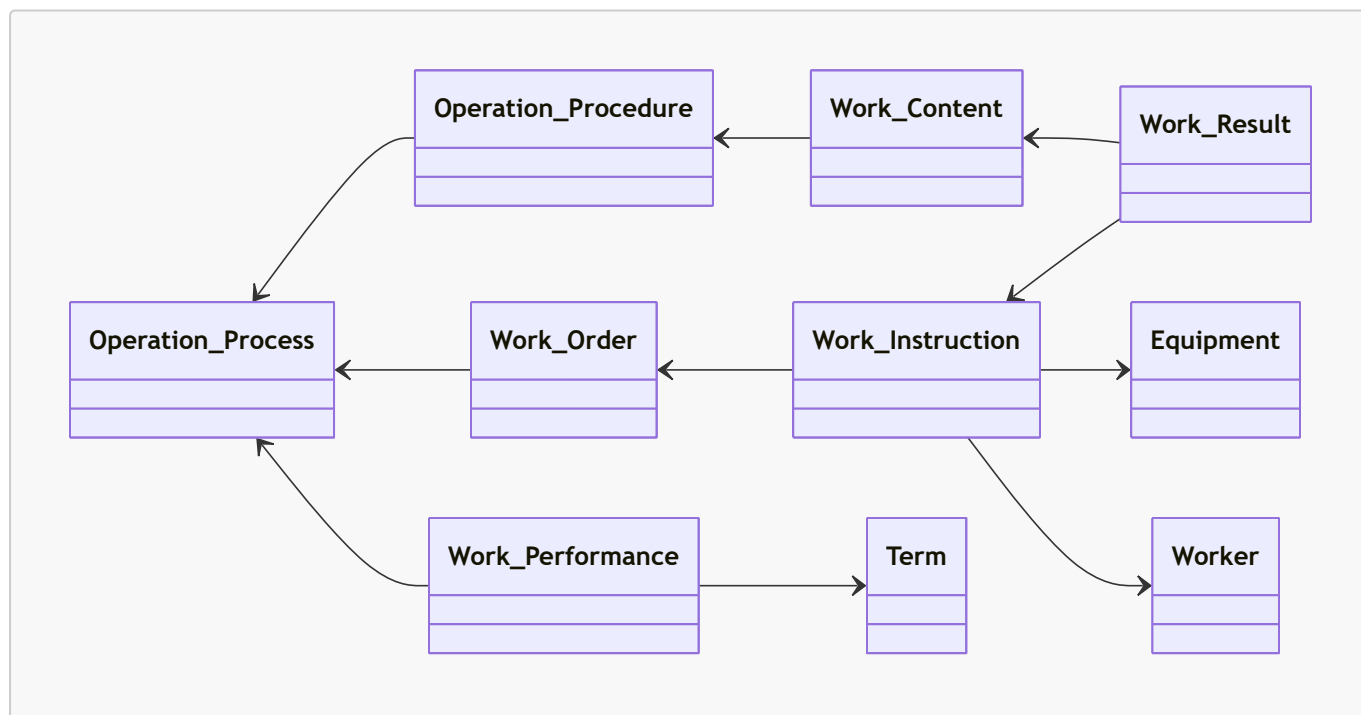


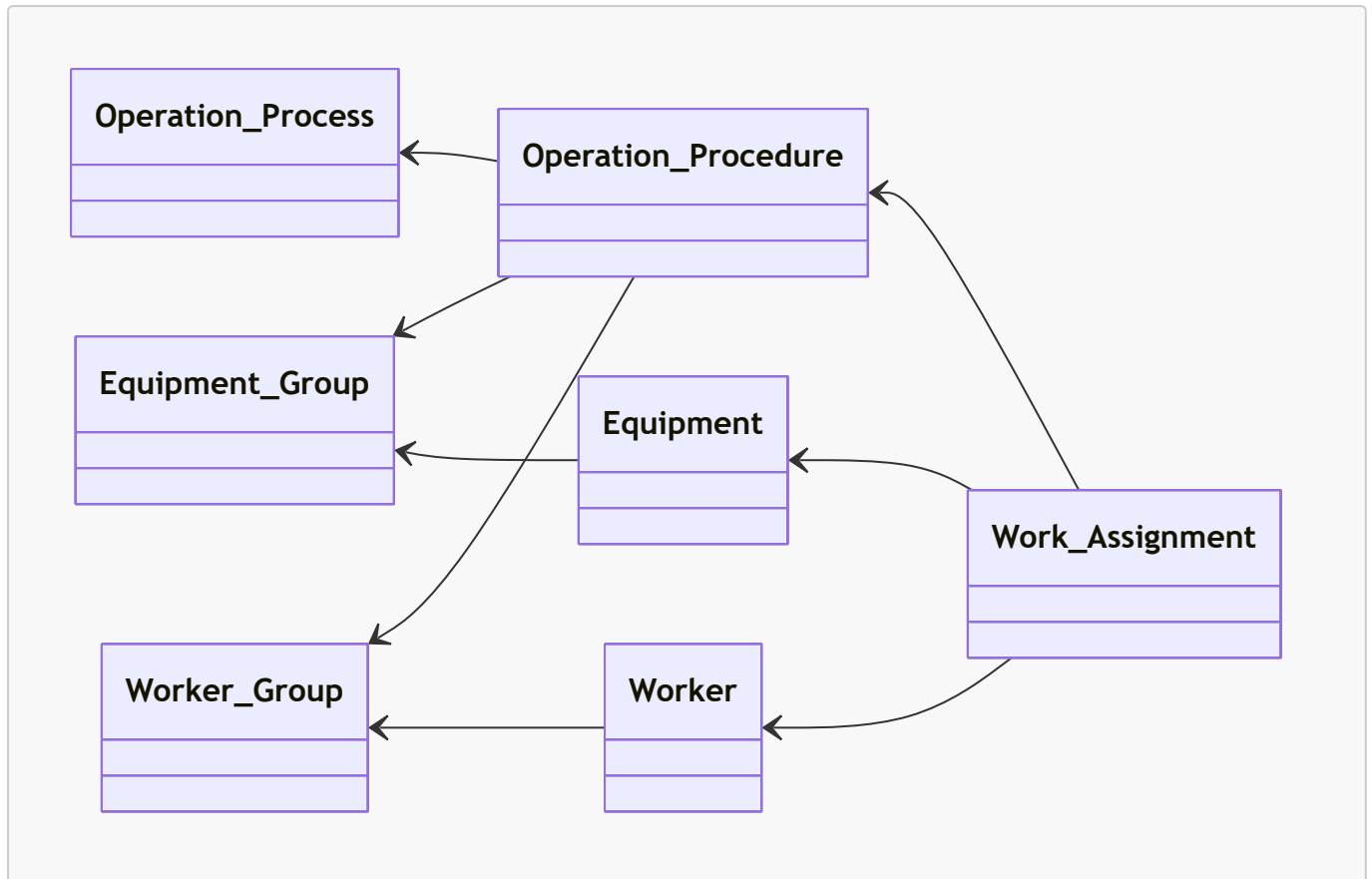


8.9. Operation Process

The operation process is a process whose executing subject is a worker, targeting cases that do not correspond to a production process or equipment process. The entities that make it up take the same form.

Figure 64 Entities Related to an Operation Process

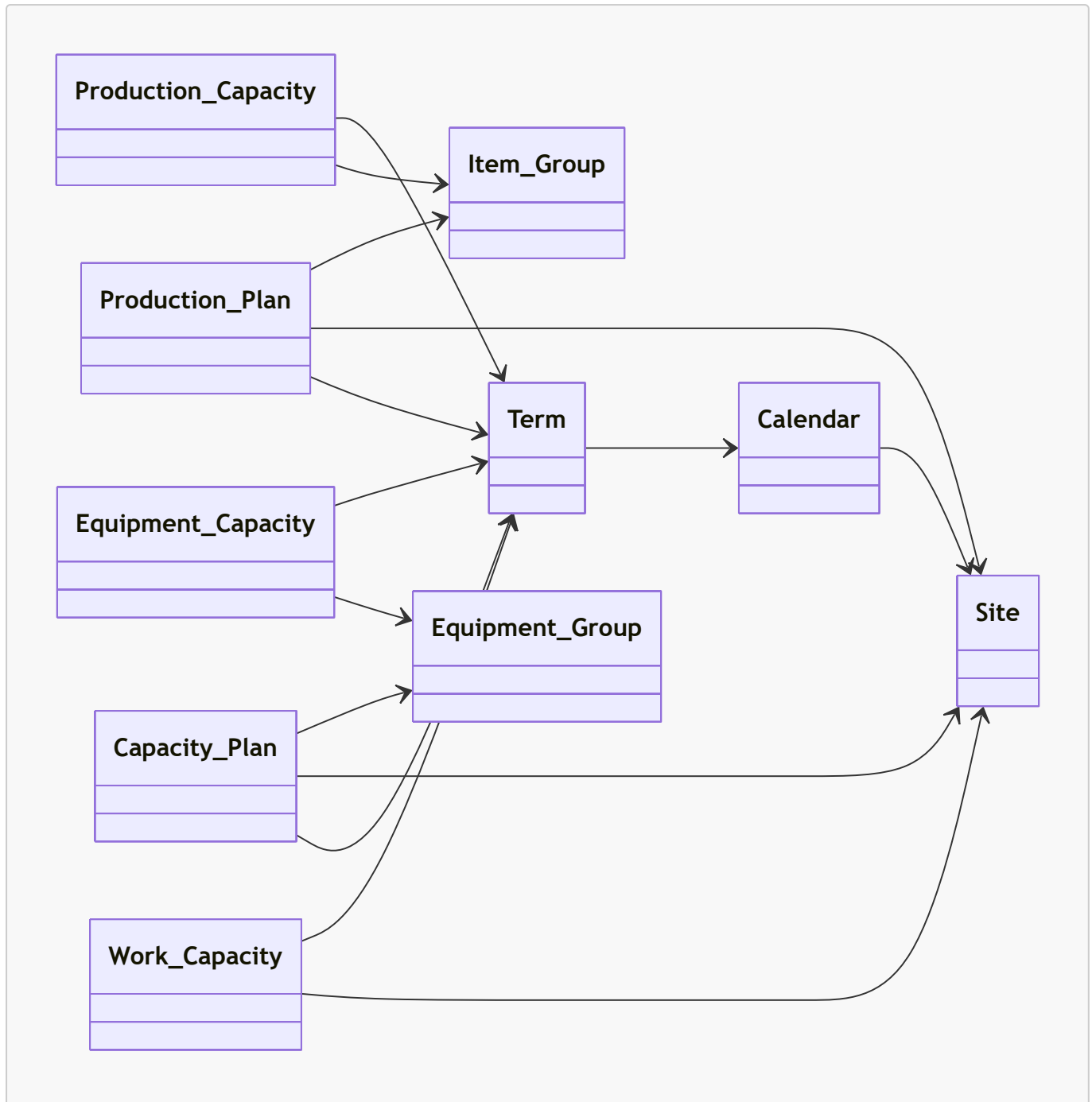




8.10. Plan Management

This is data corresponding to business that is carried out against figures aggregated over a period such as a month or week, such as the production plan or production capacity. For production equipment, Capacity Plan and Equipment Capacity correspond; for workers, Personnel Capacity and Personnel Plan correspond. The items for Production Capacity and Production Plan are shown. The attributes related to production equipment and workers follow the items of Production Plan and Production Capacity.

Figure 66 Entities Related to Planning



9. System Integration Procedure (normative)

9.1. Basic Procedure for Defining the Integration Model

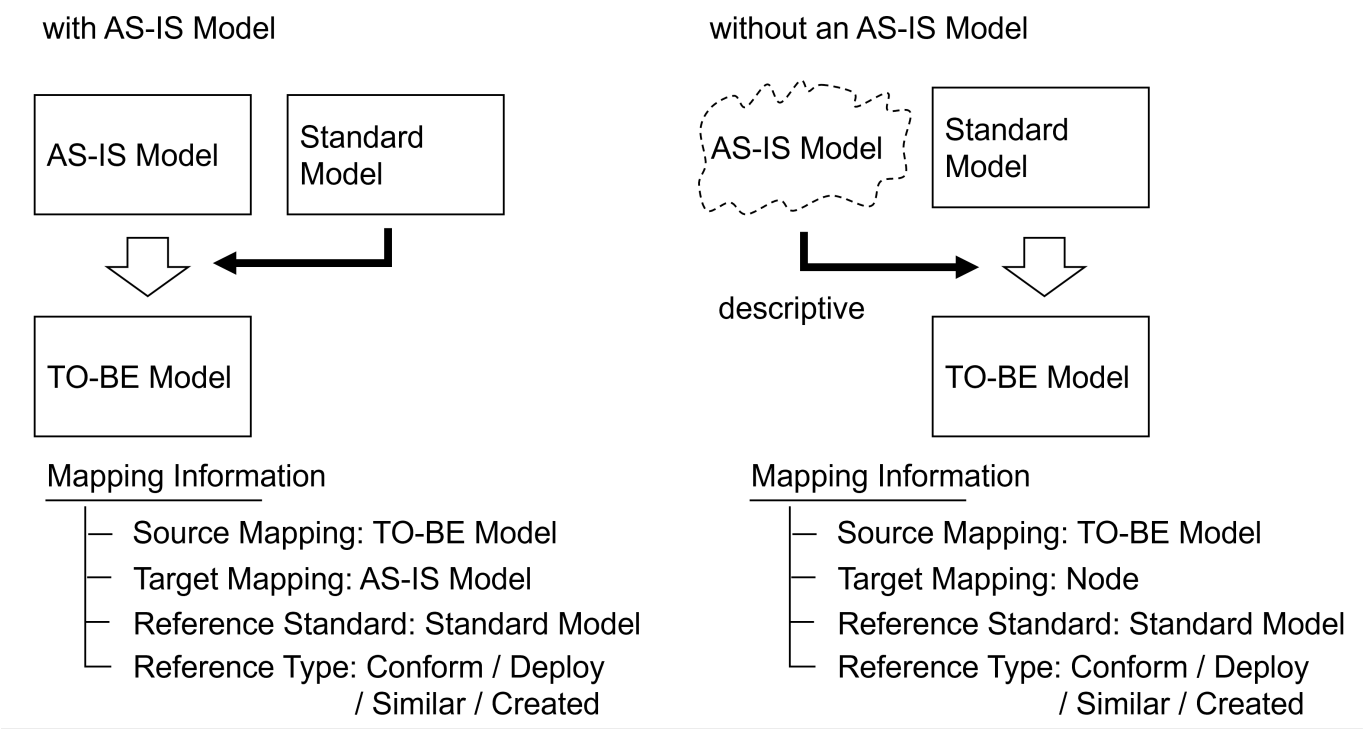
This section shows, as a procedure for ensuring interoperability, the method for turning a stakeholder's function model, information model, and data model into an integration model mapped to the standard model defined in this standard specification. For the data model, it also shows the procedure for clarifying, after defining an integration model based on the framework of the standard model, the method for converting data content between the actual model and the integration model.

In this technical specification, a model showing the current state without considering business collaboration is called the AS-IS model, and an integration model that takes business collaboration into account is called the TO-BE model. The current state of the target stakeholder is investigated, and based on those AS-IS models, a TO-BE model is defined as the desired future state.

However, there are cases where the AS-IS model showing the current state is explicitly described and presented in a form understandable by a third party, and cases where it functions in practice but no systematically described information exists for it.

As shown in Figure 66, the method of setting the TO-BE model, which is the integration model, differs depending on whether an AS-IS model exists or not. Where an AS-IS model exists, the TO-BE model is created based on it; where no AS-IS model exists, the TO-BE model is created based on the standard model.

Figure 67 Overview of the TO-BE Model Definition Procedure

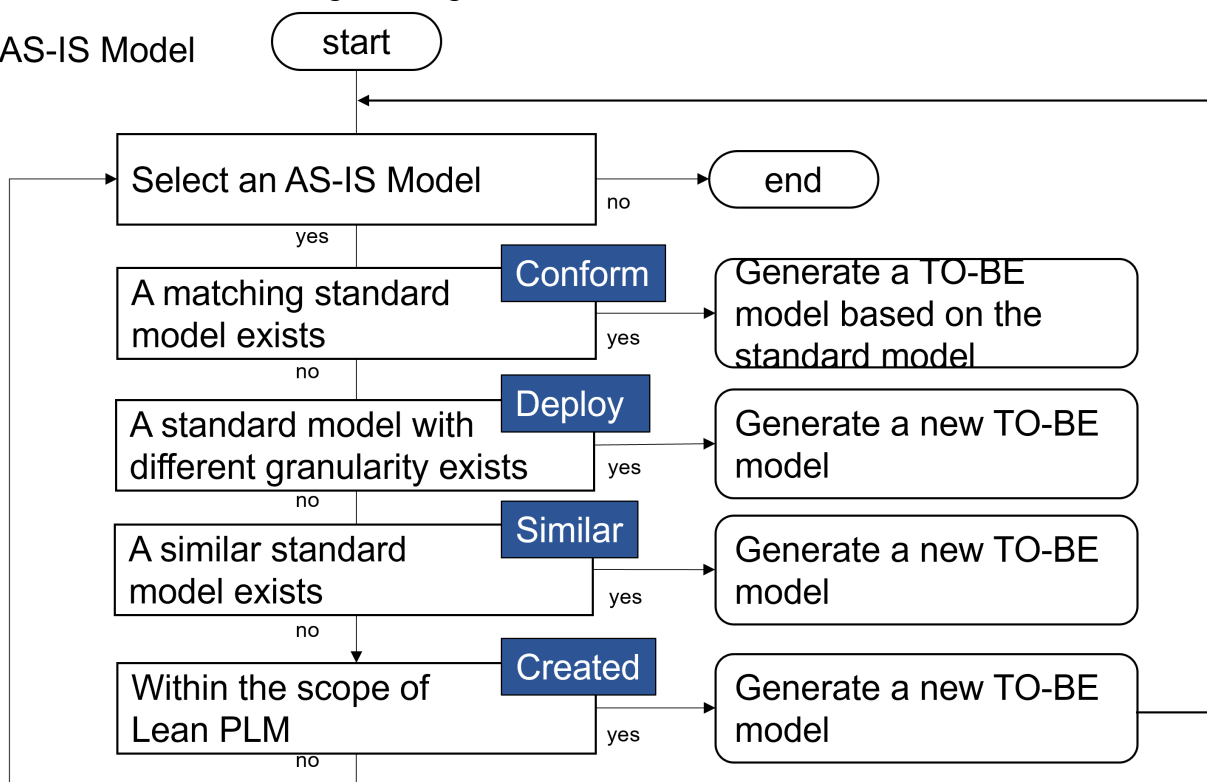


Where an AS-IS Model Is Defined

Where the current state of the target stakeholder's function model and information model has been compiled as an AS-IS model, a new TO-BE model is generated based on these AS-IS models. The method of setting the TO-BE model, which becomes the integration model, for the function model and information model is explained step by step using Figure 68.

Figure 68 Procedure for Generating the Integration Model (with an AS-IS Model)

with AS-IS Model



Elements already defined as the AS-IS model are each compared with the content defined as the standard model.

1. **Standard Match:** Where a standard model exists corresponding to the content defined in the AS-IS model, it is defined as the TO-BE model, and the corresponding standard model is set as mapping information. The mapping type is "Match."
2. **Standard Deployment:** Where the content defined in the AS-IS model is within scope but differs in granularity from the referenced standard model, it is newly added to the TO-BE model, and the newly added model is set as mapping information. The referenced standard model is set as "Deployment" in the mapping information.
3. **Standard Similarity:** Where the content defined in the AS-IS model is within scope but differs in meaning from the referenced standard model, it is newly added to the TO-BE model, and the newly added data model is set as mapping information. The referenced standard model, as the standard model with the closest meaning, is set as "Similarity" in the mapping information.
4. **New Addition:** Where the content defined in the AS-IS model is within scope but does not exist in the standard model, it is newly added to the TO-BE model, and the newly added data model is set as mapping information. The type is "New."
5. **No Target:** Where the content defined in the AS-IS model is outside the scope of this standard specification, nothing is done and it is not set in the TO-BE model.

For an information model, information items are set for each model. Accordingly, for each information model, the above procedure is further used to compare the information items in the AS-IS model with the information items in the standard model, and they are set in the TO-BE model.

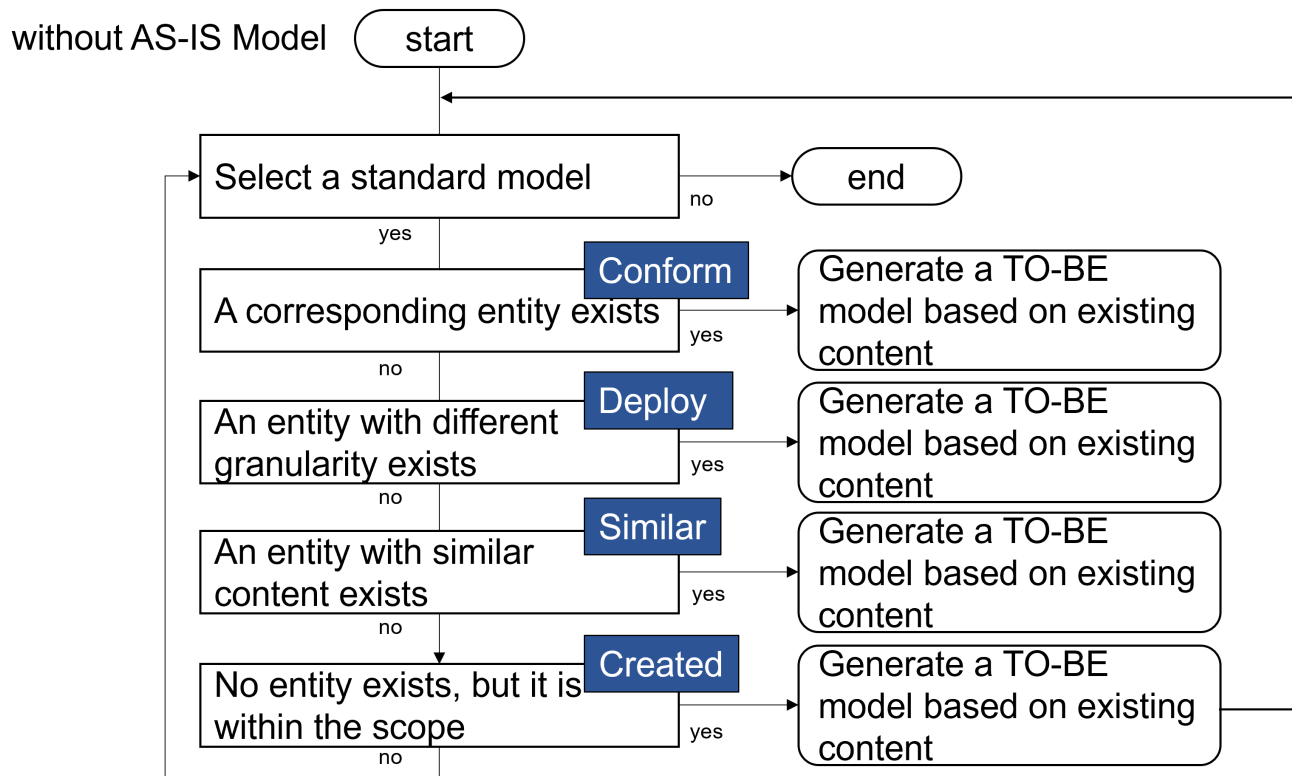
Here, "differs in granularity" refers to a case where the target range differs, such as the difference between "managing the design BOM" and "managing the BOM."

As a result, a TO-BE model is generated, and the mapping source, mapping destination, referenced standard, and reference type are set as mapping information.

Where an AS-IS Model Is Not Defined

Where the stakeholder's current state has not been compiled as an AS-IS model, a TO-BE model is newly added, following the procedure in Figure 69, based on the content defined as the standard model. For the data model, the presence of an AS-IS model is a precondition, so the following is the procedure for the function model and information model.

Figure 69 Procedure for Generating the Integration Model (without an AS-IS Model)



1. Standard Match: Where information corresponding to the content defined in the standard model actually exists, the content of the standard model is defined as the TO-BE model. The added TO-BE model is set as mapping information against the standard model, and the relationship with the standard model is "Match."
2. Standard Deployment: Where the content defined in the standard model actually exists but differs in granularity, it is newly added as the TO-BE model. The newly added TO-BE model is set as mapping information against the target standard model, and the relationship with the standard model is "Deployment."
3. Standard Similarity: Where the content defined in the standard model actually exists but differs in meaning, it is newly added as the TO-BE model. The newly added TO-BE model is set as mapping information against the target standard model, and the relationship with the standard model is "Similarity."
4. New Addition: Where there is no explicit information, but content that should be defined as an AS-IS model does not exist in the standard model, it is newly added as the TO-BE model. The newly added TO-BE model is set as mapping information, and the standard model with the closest meaning to the added model is defined as "New."

5. No Target: Where nothing corresponding to the content defined in the standard model exists in the function model that should be defined as AS-IS, no TO-BE model is generated. The mapping information against the target standard model is set as "None."

For an information model, it is necessary to set information items for each information model; where a standard model is referenced, they are set following it, and as needed, a provisional AS-IS information model is investigated based on the actual current state and its information items are set.

9.2. Defining the Data Model for Integration

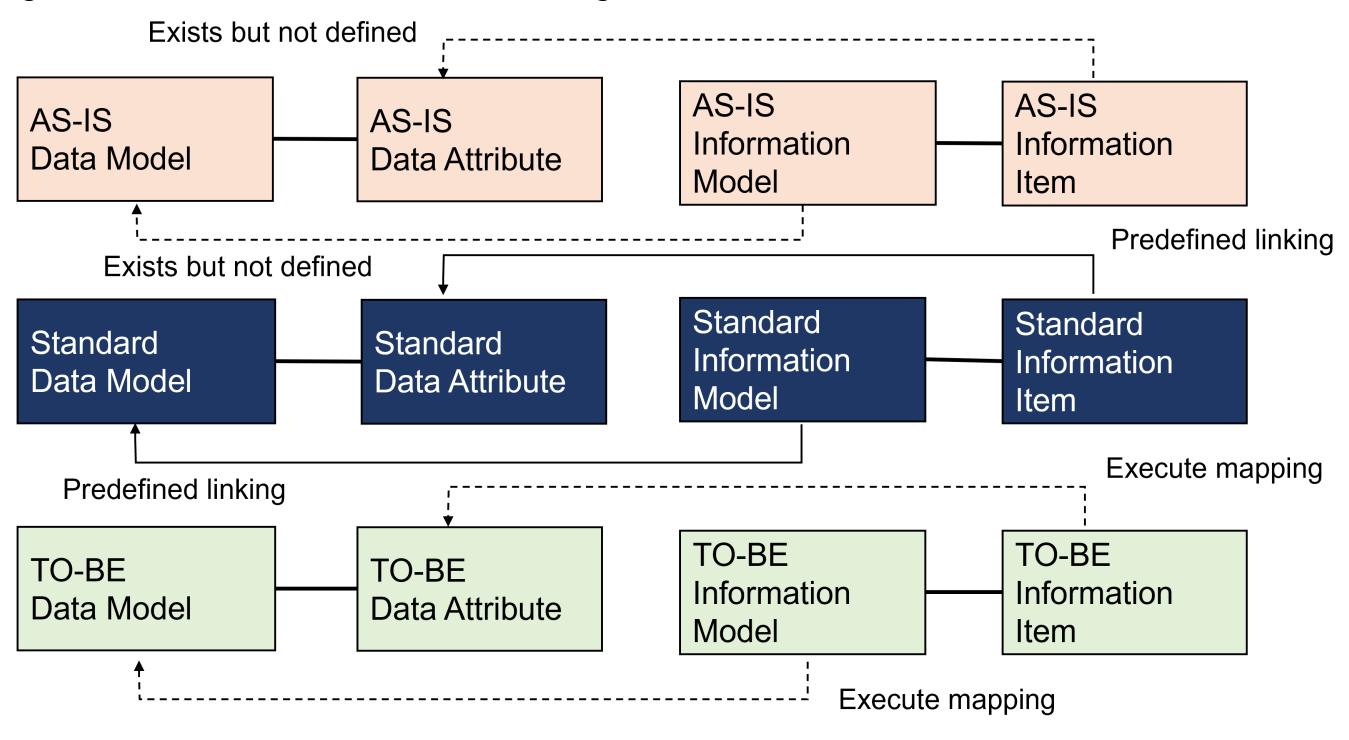
For the data model, it is assumed that all stakeholders have an AS-IS model. However, the TO-BE model, which is the integration data model, must basically be set based on the standard model.

Also, in defining the TO-BE data model, the previously defined TO-BE information model and its corresponding standard information model are used. In defining the data model for integration, the correspondence between the data model and information model, as well as the correspondence between data attributes and information items, is extremely important.

Figure 70 shows the overall structure for setting the data model for integration. The left side of the figure is the data model, and the right side is the information model. The ultimate goal is to show the correspondence between the AS-IS data model and TO-BE data model, and between the AS-IS data attributes and TO-BE data attributes, and in that process the correspondence with the information model is used as a reference.

Note that whereas for the AS-IS model, the correspondence between the data model and information model may or may not be defined, for the standard model, this technical specification explicitly specifies that relationship. For the TO-BE model, the relationship between the data model and information model is defined according to the following procedure.

Figure 70 Definition of the Data Model for Integration



Generating the Integration Data Model

First, the following procedure is applied to the TO-BE information model.

1. Search whether a standard information model exists corresponding to the target TO-BE information model.
2. Where a standard information model exists, further obtain the standard data model that is pre-linked to that standard information model within the standard specification.
3. Newly set a TO-BE data model corresponding to the standard data model, and also set TO-BE data attributes corresponding to the standard data attributes.
4. Map the newly set TO-BE data model to the TO-BE information model.

In 2. above, where no standard information model exists, the standard data model is referenced directly, and the one closest to the content of the TO-BE information model is searched for. If found there, the TO-BE data model is set following the procedure in 3. and 4. above.

Note that where no corresponding standard data model is found, a new TO-BE data model is set, and in this case it becomes a data model that does not correspond to a standard data model.

Setting Data Attributes

Data attributes and information items are each defined for the data model and information model. Here the procedure for setting a data attribute corresponding to an information item is shown.

1. For all TO-BE information items belonging to all TO-BE information models, set the corresponding standard information item. However, the TO-BE information item and standard information item mapped here are assumed to already have a correspondence set between their respective parent TO-BE information model and standard information model.
2. Where a standard information item exists, further obtain the standard data attribute that is pre-linked to that standard information item within the standard specification.
3. Confirm whether a TO-BE data attribute corresponding to the standard data attribute exists, and if not, newly set one.
4. Map the newly set TO-BE data attribute to the TO-BE information item.

In 2. above, where no suitable information item exists in the target standard information model, the standard data attribute can be referenced directly, and the one closest to the content of the TO-BE information item can be set. In that case, the TO-BE data attribute is set following the procedure in 3. and 4. above.

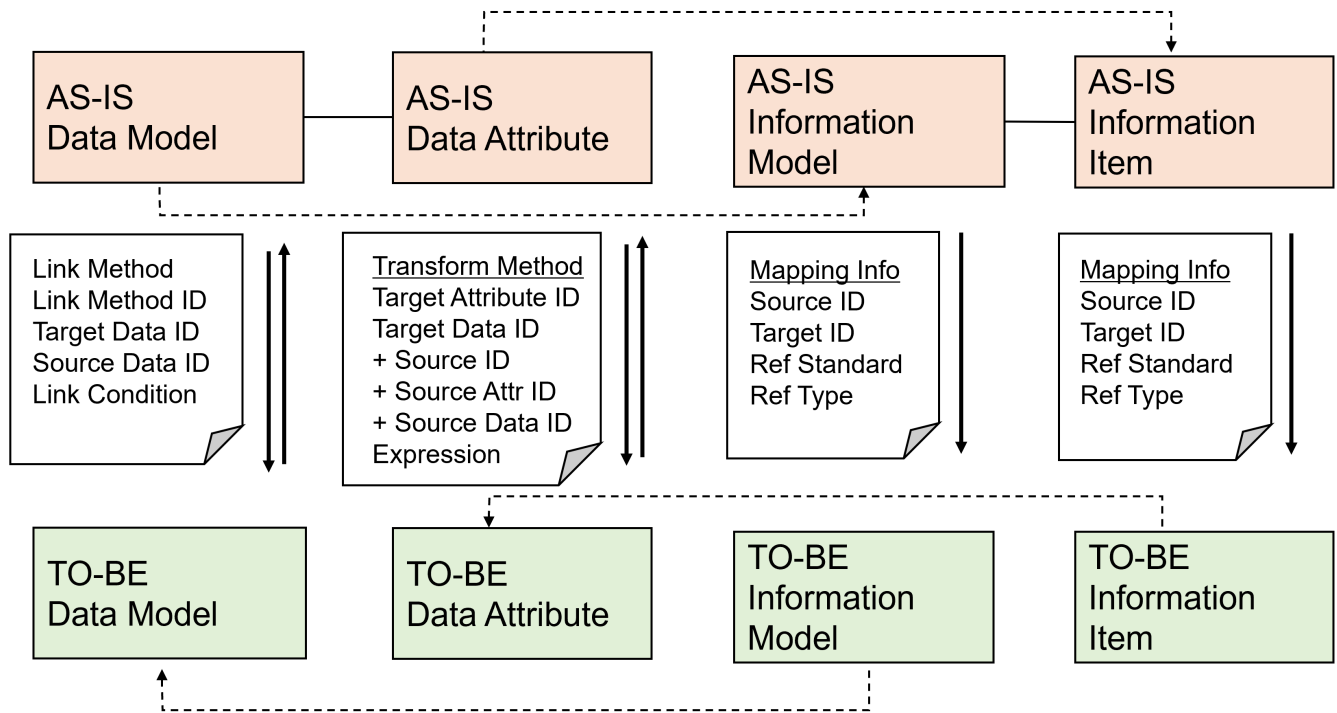
However, the standard data attribute that can be referenced here must belong to the same data model as another standard data attribute set for an information item of the same TO-BE information model, or a linking relationship via primary key and foreign key must be defined between the parent standard data models.

Note that where no corresponding standard data attribute is found, it is set as a new data attribute corresponding to the target TO-BE data model.

Defining the Data Conversion Method

The relationship ultimately obtained between the AS-IS data model and TO-BE data model is defined, in the form of a linking method and a conversion method, as the integration profile shown in Chapter 10.

Figure 71 Definition of the Data Conversion Method



The actual conversion of data content follows the definition of the data conversion method shown in Figure 71. Here, the data conversion method is performed between the AS-IS data model and TO-BE data model, but there is a direction depending on which is converted to which.

In this specification, the data model on the side that sets the content is called the target data model, and the data model on the side that references the content is called the referenced data model. When setting the content of the AS-IS data model into the TO-BE data model (PUSH), the referenced data model is the AS-IS data model. On the other hand, when setting the content of the TO-BE data model into the AS-IS data model (PULL), the referenced data model is the TO-BE model.

Where the target model is the TO-BE data model and the data needed for data conversion does not exist on the AS-IS data model side, an extended data model and extended data attributes can be newly added on the TO-BE data model side as an auxiliary data model, and these can be added to the referenced data model.

Using the data attributes in the referenced data model, a definitional formula and calculation procedure are described for setting the content of each data attribute of the target data model. The reference needed to calculate the value of one piece of data may be multiple pieces of data spanning multiple different data models. In defining the calculation procedure, in addition to general operations, there are also record-level filtering, sorting, copying, and aggregation.

10. Integration Profile (normative)

The integration profile is for each stakeholder to describe its own Lean PLM based on this standard. By explicitly describing each Lean PLM following the usage procedure of this technical specification shown in Section 9.1, interoperability is facilitated.

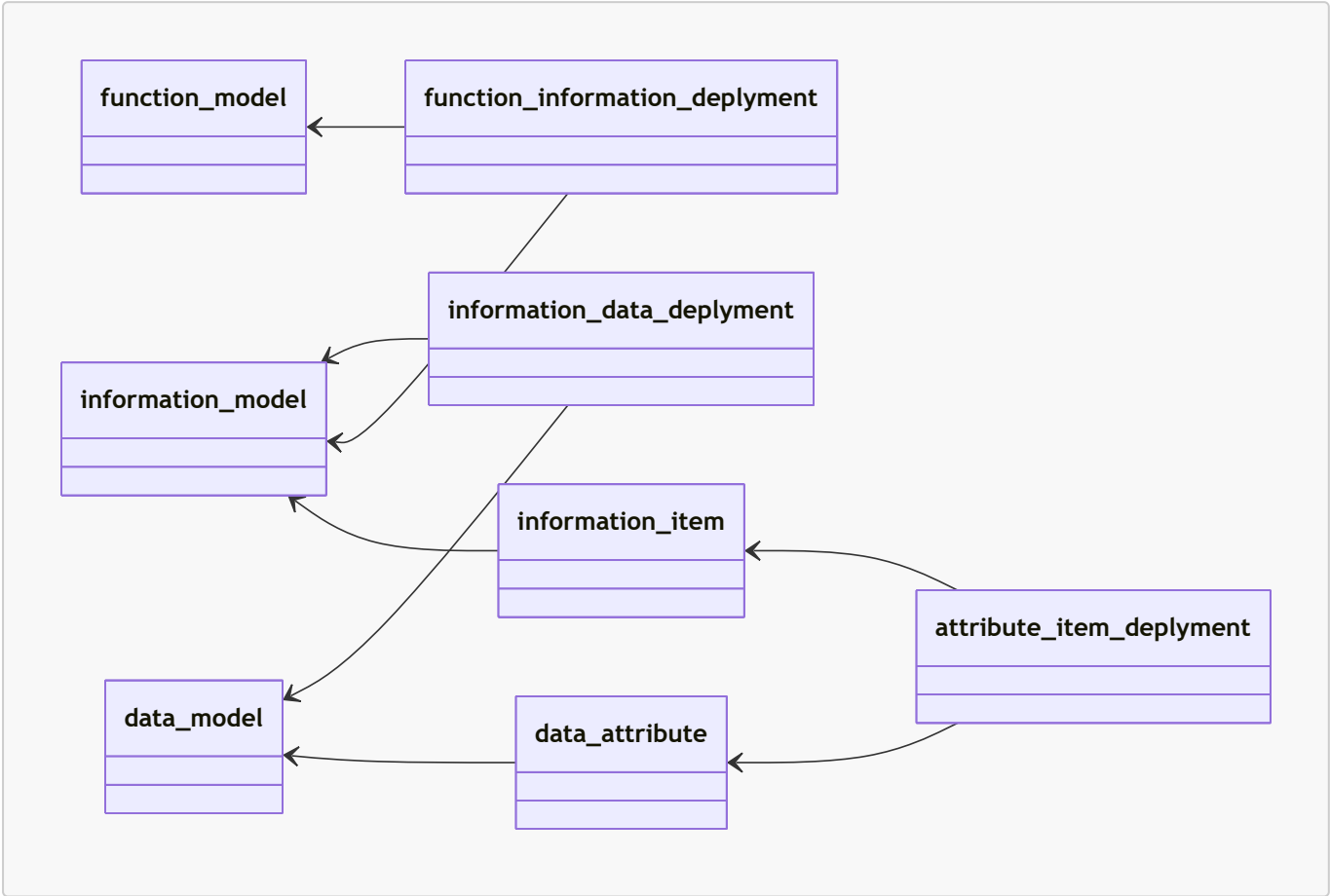
The content of the model for the integration profile shown here is defined for each stakeholder. For the standard model, only the eight models in the model definition can be defined. These are shown in a separate

document (lplm-schema.json).

10.1. Model Definition

The function model, information model, and data model, which are the basic models making up Lean PLM, have the following structure.

Figure 72 Entities for Model Definition



The function model is a compilation of the content of the various business operations held by the target stakeholder, organized as functions that resolve requirements corresponding to specific things or information. Any act performed by a business entity can be captured as a function. However, the range that can be explicitly described as a function model is limited.

Table 117 Function Model (function_model)

Name	Abbreviation	Data Type	Description
id	ID	PK	Model identification code
category	Function Category	varchar	Category name (selected from the content defined in this specification)
group_name	Function Group	varchar	Group name (selected from the content defined in this specification)
name	Function Name	varchar	Name representing the function

Name	Abbreviation	Data Type	Description
description	Description	varchar	Description of the function
stakeholder_id	Stakeholder ID	FK	ID of the stakeholder that entered it (a separate identification code is set)
created_at	Registration Date/Time	datetime	Date the content was first registered
updated_at	Update Date/Time	datetime	Date the content was last modified
status	Status	varchar	Current state (active/inactive, etc.)
note	Remarks	varchar	Management memo, etc.

The information model is the unit that represents the information targeted by the business operations defined in the function model. It expresses, in a compiled form, the set of information items necessary for decision-making in the activity carried out by the person in charge of the business, with the information items making up the information laid out spatially.

Table 118 Information Model (information_model)

Name	Abbreviation	Data Type	Description
id	ID	PK	Model identification code
category	Information Category	varchar	Category name (selected from the content defined in this specification)
group_name	Information Group	varchar	Group name (selected from the content defined in this specification)
name	Information Name	varchar	Name representing the information
description	Description	varchar	Description of the information
stakeholder_id	Stakeholder ID	FK	ID of the stakeholder that entered it (a separate identification code is set)
created_at	Registration Date/Time	datetime	Date the content was first registered
updated_at	Update Date/Time	datetime	Date the content was last modified
status	Status	varchar	Current state (active/inactive, etc.)
note	Remarks	varchar	Management memo, etc.

An information item is an element making up an information model; it expresses its content and is also the unit that accepts input from the person in charge of the business. The meaning of an information item must be inferable in advance from the information model and its corresponding function model.

Table 119 Information Item (information_item)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the information item
information_model_id	Information Model ID	FK	ID of the target information model
name	Item Name	varchar	Name of the information item
description	Description	varchar	Description of the information item
note	Remarks	varchar	Management memo, etc.

The data model is the unit of entity in the data schema for representing the information model on a computer. In an implementation using a relational database, it corresponds to a table. Data models are linked to one another. Due to normalization, they do not correspond one-to-one with the units of the information model, but can be mapped via data attributes.

Table 120 Data Model (data_model)

Name	Abbreviation	Data Type	Description
id	ID	PK	Model identification code
category	Data Category	varchar	Category name (selected from the content defined in this specification)
group_name	Data Group	varchar	Group name (selected from the content defined in this specification)
name	Data Name	varchar	Name representing the data
description	Description	varchar	Description of the data
stakeholder_id	Stakeholder ID	FK	ID of the stakeholder that entered it (a separate identification code is set)
created_at	Registration Date/Time	datetime	Date the content was first registered
updated_at	Update Date/Time	datetime	Date the content was last modified
status	Status	varchar	Current state (active/inactive, etc.)
note	Remarks	varchar	Management memo, etc.

A data attribute is an element making up a data model, and can have a concrete value expressed as a number, character string, etc. It is implemented as a field in a database or via a key-value. A data attribute is assumed to be intended for persistence, but a temporary one corresponding to an information item can also be defined.

Table 121 Data Attribute (data_attribute)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the data attribute
data_model_id	Data Model ID	FK	ID of the target data model
name	Data Attribute Name	varchar	Name of the data attribute
description	Description	varchar	Description of the data attribute
note	Remarks	varchar	Management memo, etc.

10.2. Model Deployment Definition

Function-Information Deployment shows the relationship between the function model and information model. Each function model uses information, or provides information, in carrying out its function. Where such information is defined as an information model, the relationship is shown here.

Table 122 Function-Information Deployment (function_info_deployment)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the function-information deployment
function_model_id	Function Model ID	FK	ID of the function model that is the deployment source
information_model_id	Information Model ID	FK	ID of the information model that is the deployment destination
note	Remarks	varchar	Management memo, etc.

Information-Data Deployment shows the correspondence when an information model is expressed by a data model. An information model is made up of one or more data models. On the other hand, a data model corresponds to zero, one, or more information models. Note that where one piece of information corresponds to multiple data models, a relationship via data model linkage must be set between those data models.

Table 123 Information-Data Deployment (info_data_deployment)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the information-data deployment
information_model_id	Information Model ID	FK	ID of the information model that is the deployment source
data_model_id	Data Model ID	FK	ID of the data model that is the deployment destination
note	Remarks	varchar	Management memo, etc.

Item-Attribute Deployment shows which data attribute, in which data model, an information item in an information model corresponds to. An information item always corresponds to one data attribute. On the other hand, a data attribute corresponds to zero, one, or more information items.

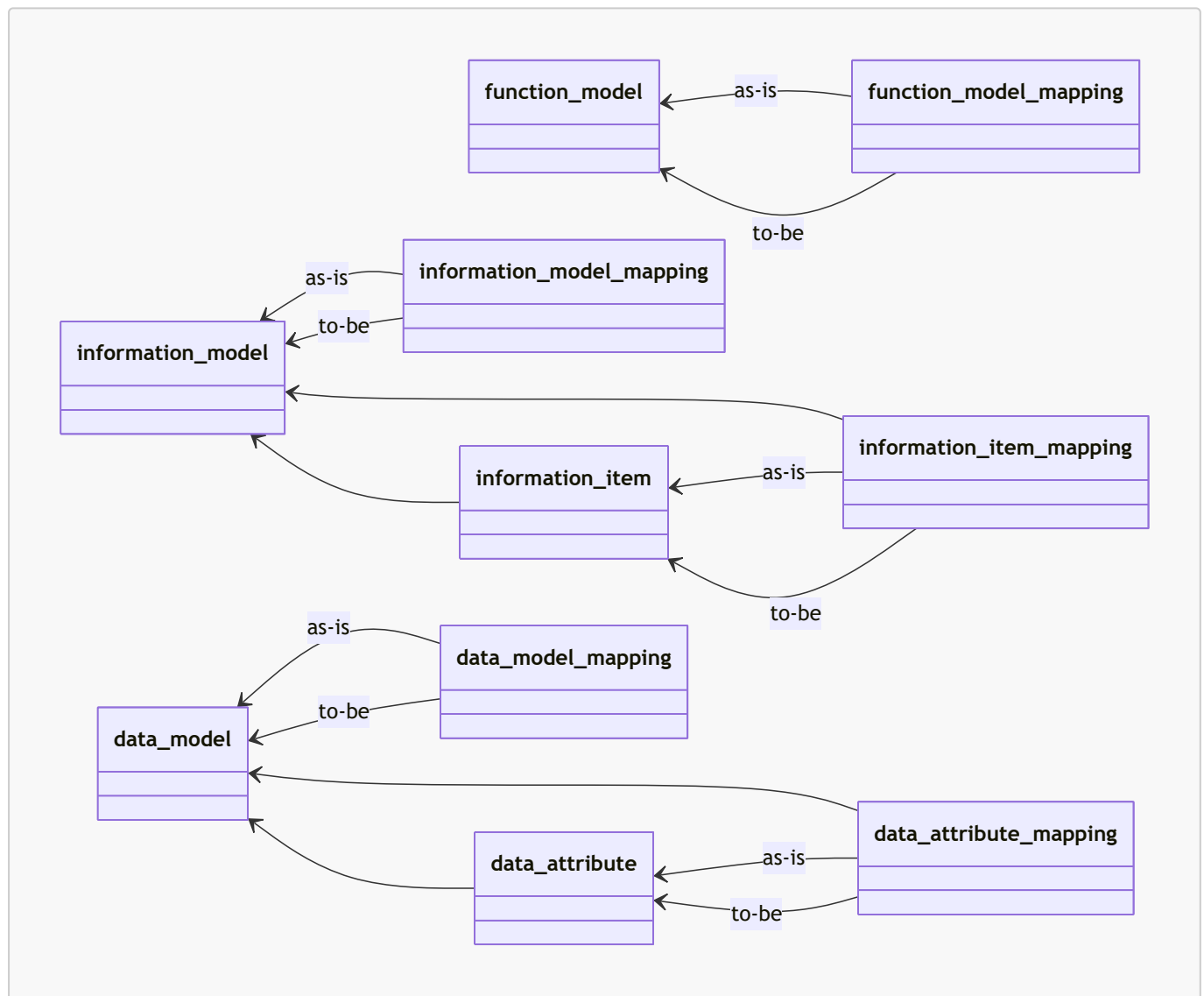
Table 124 Item-Attribute Deployment (attribute_item_deployment)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the item-attribute deployment
information_item_id	Information Item ID	FK	ID of the target information item
data_attribute_id	Data Attribute ID	FK	ID of the corresponding data attribute
note	Remarks	varchar	Management memo, etc.

10.3. Mapping Information

Mapping information shows how the unique function model, information model, and data model defined by each stakeholder correspond to the standard model shown in this specification. Here, each stakeholder's model is the AS-IS model, and the standard model shown in this technical specification is the TO-BE model.

Figure 73 Mapping Information



Function model mapping shows the correspondence between the AS-IS model and TO-BE model in the function model.

Table 125 Function Model Mapping (function_model_mapping)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID for the function model correspondence
as_is_model_id	AS-IS Function Model ID	FK	ID of the function model representing the current business
to_be_model_id	TO-BE Function Model ID	FK	ID of the function model representing the target business
standard_model_id	Standard Function Model ID	FK	ID of the corresponding standard function model
mapping_type	Mapping Type	varchar	Type of correspondence between models
note	Remarks	varchar	Management memo, etc.

Information model mapping shows the correspondence between the AS-IS model and TO-BE model in the information model.

Table 126 Information Model Mapping (information_model_mapping)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID for the information model correspondence
as_is_model_id	AS-IS Information Model ID	FK	ID of the information model used in the current business
to_be_model_id	TO-BE Information Model ID	FK	ID of the information model used in the target business
standard_model_id	Standard Information Model ID	FK	ID of the corresponding standard information model
mapping_type	Mapping Type	varchar	Type of correspondence between models
note	Remarks	varchar	Management memo, etc.

Data model mapping shows the correspondence between the AS-IS model and TO-BE model in the data model.

Table 127 Data Model Mapping (data_model_mapping)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID for the data model correspondence
as_is_model_id	AS-IS Data Model ID	FK	ID of the data model used in the current system
to_be_model_id	TO-BE Data Model ID	FK	ID of the data model used in the target system
standard_model_id	Standard Data Model ID	FK	ID of the corresponding standard data model
mapping_type	Mapping Type	varchar	Type of correspondence between models
note	Remarks	varchar	Management memo, etc.

Information item mapping shows the mapping between the AS-IS model and TO-BE model at the information item level. For information items belonging to the same information model, the information model to which the corresponding partner information item belongs, via the mapping, may be the same or different. Where the partner information model is different, a direct or indirect relationship must be defined between those information models.

Table 128 Information Item Mapping (information_item_mapping)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID for the information item correspondence
as_is_model_id	AS-IS Information Model	FK	ID of the information model used in the current business
as_is_element_id	AS-IS Information Item	FK	ID of the information item used in the current business
to_be_model_id	TO-BE Information Model	FK	ID of the information model used in the target business
to_be_element_id	TO-BE Information Item	FK	ID of the information item used in the target business
mapping_type	Mapping Type	varchar	Type of correspondence between items
note	Remarks	varchar	Management memo, etc.

Data attribute mapping shows the mapping between the AS-IS model and TO-BE model at the data attribute level. For data attributes belonging to the same data model, the data model to which the corresponding partner data attribute belongs, via the mapping, may be the same or different. Where the partner data model is different, a direct or indirect relationship must be defined between those data models.

Table 128 Data Attribute Mapping (data_attribute_mapping)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID for the information item correspondence
as_is_model_id	AS-IS Information Model	FK	ID of the information model used in the current business
as_is_element_id	AS-IS Information Item	FK	ID of the information item used in the current business
to_be_model_id	TO-BE Information Model	FK	ID of the information model used in the target business
to_be_element_id	TO-BE Information Item	FK	ID of the information item used in the target business
mapping_type	Mapping Type	varchar	Type of correspondence between items
note	Remarks	varchar	Management memo, etc.

10.4. Data Linking and Logic

Data model linkage and calculation logic show the structure of the data model and the structure of calculations using the values of data held by the data model. The basic configuration has the following three stages.

Definition Name	Configuration Name
data_link	Determines the referenced model
data_calculation	Defines the calculation method for the target attribute
data_calculation_link	Defines the attributes used within the calculation formula

Data Model Linkage

Data model linkage shows the relationship between data models. This corresponds to a JOIN statement in SQL. At most one data model can be related per definition in a data model linkage, but by combining these, values can be set from multiple data models.

Note that there are two forms of linkage: one where the target data model side holds no actual data and always references the value of the linked data model, and one where the target data model side duplicates and holds, as an actual value, the value of the linked data model.

Figure 74 Data Linking and Logic

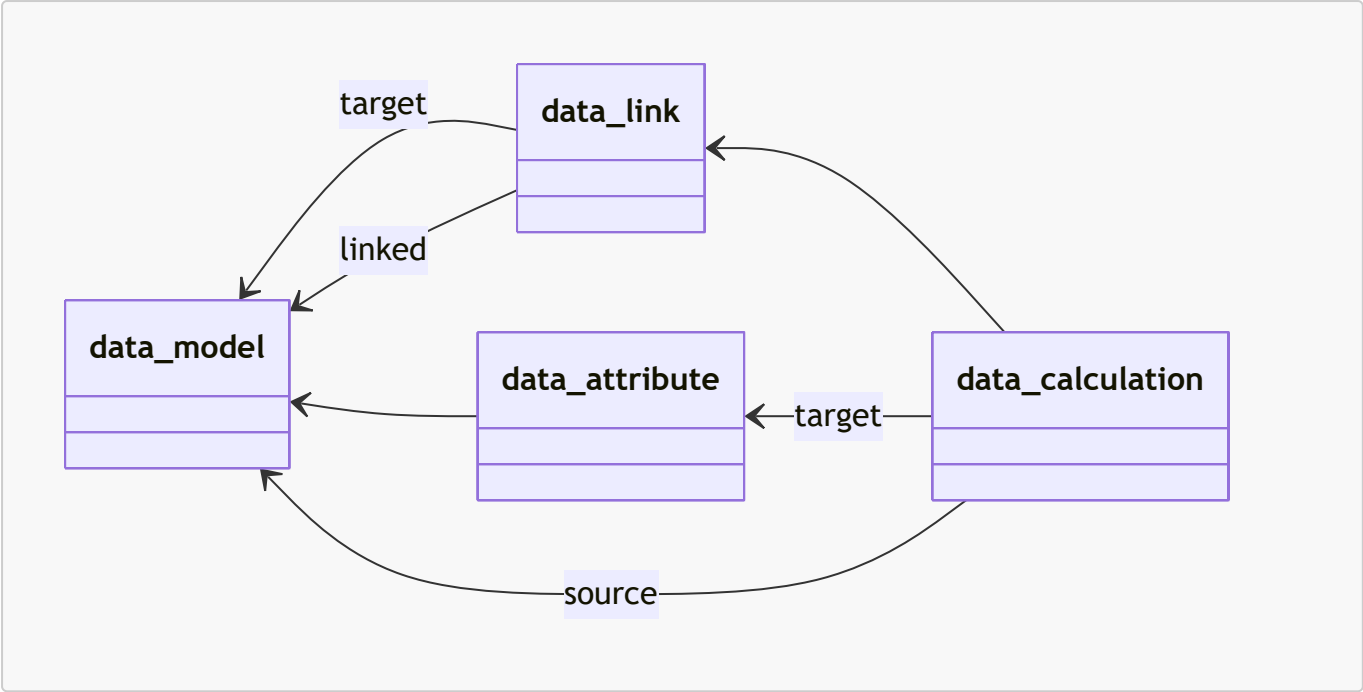


Table 130 Data Model Linkage (data_link)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the data linkage
target_model_id	Target Model ID	FK	The target model. Obtains a value from the linked source.
linked_model_id	Extension Model ID	FK	The linked model. In the case of copying, becomes the reference source.
link_type	Extension Type	varchar	There is the case of static linkage as a JOIN, and the case of dynamically copying a value.
join_key	Linkage Method	varchar	Specifies, as a character string, the linkage condition between the two models.
note	Remarks	varchar	Management memo, etc.

The candidate values of link_type are as follows.

Value	Meaning
JOIN	Statically links data models. A static model linkage like an SQL JOIN.
COPY	Dynamically copies a value. Obtains and copies the value at the time of data conversion.
LOOKUP	Used as a reference table. Master reference.
AGGREGATE	Used as an aggregation source. Aggregates multiple records.

Calculation Logic

Calculation logic shows the method of calculating the value of the target data attribute. The calculation method is shown by a calculation formula, and the constituent elements of the formula shown there are values of data attributes in the referenced data model. The referenced data model can have multiple data models set by using a calculation-logic link.

Table 131 Calculation Logic (data_calculation)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the data conversion method
target_attribute_id	Target Attribute ID	FK	ID of the target attribute whose value is set
source_model_id	Referenced Data Model ID	FK	ID of the referenced data model subject to the calculation
calculation_type	Calculation Type	varchar	One of: copy, aggregate, merge, filter, or calculate
condition	Condition Expression	varchar	Limits the target data that is the premise of the calculation
expression	Calculation Formula	varchar	Describes the calculation content or condition expression
note	Remarks	varchar	Management memo, etc.

The table below shows, as calculation_type, the types of calculation available for calculating an attribute value and their implementation method.

Value	Meaning	Description
COPY	Copy	Specifies a search condition, selects a specific record in the referenced data model, and sets the value of that data attribute.
AGGREGATE	Aggregate	Aggregates the specified data attribute value across multiple records in the referenced data model matching the search condition.
MERGE	Merge	Specifies a search condition, selects a specific record in the referenced data model, and adds it to the target data.
FILTER	Filter	Specifies a search condition to select the applicable data, for the referenced data model used in a calculation.
CALCULATE	Calculate	Executes a predefined calculation using the values of data attributes of records selected in the referenced data model.

When calculation_type is AGGREGATE, the value of the target attribute is calculated according to the calculation formula described in expression.

expression	Meaning
COPY	Copies the quantity as is

expression	Meaning
SUM	Aggregates the quantity
AVG	Average value
MAX	Maximum value

When calculation_type is CALCULATE, the value of the target attribute is calculated within the data of the target data model according to the calculation formula described in expression.

Example description

```
Production_Order.Cost = Production_Quantity × Standard_Unit_Price
```

Calculation Logic Link

A calculation logic link specifies, when calculating based on a calculation formula defined in calculation logic, where there are multiple target referenced data models, the data model linkage for linking them. For example, in performing a calculation $A = B + C$, if A, B, and C belong to Data-01, Data-01, and Data-02 respectively, then Data-01 and Data-02 must be related to each other in order to execute this calculation.

Table 132 Calculation Logic Link (data_calculation_link)

Name	Abbreviation	Data Type	Description
id	ID	PK	Unique ID of the data conversion reference
data_calculation_id	Calculation Logic ID	FK	ID of the corresponding calculation logic
data_link_id	Data Model Linkage ID	FK	ID of the data model linkage used in the calculation
note	Remarks	varchar	Management memo, etc.

10.5. KPI Model

To evaluate, from a neutral standpoint, the level of system-integration capability of individual Lean PLM solutions belonging to manufacturing establishments as stakeholders, or of solution companies, three KPIs are set: data conformance, data sufficiency, and data reliability. These indicators are calculated separately for the stakeholder on the receiving side (PULL) of the data model content, i.e. the user side, and for the stakeholder on the providing side (PUSH) of the data. So that one side does not affect the performance of the other, they are split at the layer of the standard model positioned in between, and each side is evaluated independently.

Data Conformance

Data conformance, when converting the target AS-IS data model A into the TO-BE data model A* following the procedure in Section 9.1, is shown by the proportion in which:

1. a TO-BE data model A* corresponding to AS-IS data model A exists, and
2. all of the mandatory data attributes possessed by AS-IS data model A are defined as data attributes of A*.

The same applies, as a PULL type, to the case of obtaining data on the AS-IS data model side from a TO-BE data model based on the standard model.

The counting target is counted on the AS-IS side from the perspective of the information model. That is, all data models making up the AS-IS information model are investigated as targets of data conformance, and anything else is excluded from the target.

Data Sufficiency

Data sufficiency is the proportion by which, for setting the content of an AS-IS data model implemented by a stakeholder, all of the data model content theoretically needed to reproduce equivalent content can be conveyed in a single transaction.

For example, in a PUSH-type use case, where the partner requires AS-IS data model A*, even when AS-IS data model A alone is insufficient in content, the content can be made 100% sufficient by linking data model B, as in Figure (a).

On the other hand, in practice, as in Figure (b), there are also cases where data model B, needed to constitute data model A*, is transmitted beforehand or afterward and reconstructed on the user side. In this case it is not subject to the calculation of sufficiency.

The calculation of data sufficiency is investigated at the level of the data attribute. The data sufficiency rate can be set for each data model, and at the same time the data sufficiency of the stakeholder as a whole is clarified by taking the weighted average of these data sufficiency rates.

Data Reliability

Data reliability shows, for each data model, at the level of the data attribute, the proportion for which the data attributes necessary and sufficient for setting its value exist. Data reliability can be determined per target data model or per business establishment. Even for the same target data model, the value of data reliability differs between the PUSH type and PULL type.

Where there is no applicable data attribute, it is necessary to add it as an attribute on the TO-BE data model side; if this is not done, it is treated as missing, and reliability decreases. Note that for attributes counted as missing, a default value defined for each is set.

For a data attribute under investigation, in order for the data attribute for setting its value to exist on the referenced data model and to be counted in the calculation of data reliability, the conditions are that:

1. an attribute of the referenced data model is assigned to each attribute of the target data model, and
2. calculation logic is defined for the value of the target data attribute, and furthermore, the value of the data attribute referenced within that logic can be obtained.

Note that an extended data model or extended data attribute established to calculate the value of a data attribute corresponds to a corresponding information model or information item; where it does not exist, one is to be newly generated, and where this is not possible, it is excluded from the reliability count.

11. Compliance and Certification (normative)

Manufacturing activities in general carried out by a company, organization, or individual, or a product or service they provide, can be said to comply with this technical specification if all of the following requirements are satisfied.

1. It is a product or service within the scope of this technical specification.
2. The terminology describing the product or service is mapped to the terminology defined in this technical specification.
3. The product or service is described in correspondence with the functions and information defined in this technical specification.
4. There is otherwise no violation of the rules explicitly prescribed in this technical specification, and there is no risk that its operation will cause significant damage to users.

Furthermore, where a product or service provided by a company or individual satisfies, in addition to the above four requirements, the following requirements, as recognized through the certification process provided by IVI, IVI certifies that it complies with this technical specification.

1. The function model, information model, and data model are described as an integration profile as shown in Chapter 9 of this technical specification, and the content is submitted to IVI, with no deficiencies in that content.
2. A test environment corresponding to the submitted integration profile is built, and it can be confirmed that results as specified are obtained in a certification test using three or more sample data sets. Note that the method and criteria of the certification test are prescribed in an attached document.
3. For bidirectional collaboration between design and manufacturing, the system implementation of the desired future state is periodically reviewed, and efforts to improve the business mechanism and the added value of the product or service provided are continued.